

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

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Role. Senior Supply Chain Manager, NACF Supply Chain — Global Capacity Planning team. Job ID 3195330. Bellevue, WA.

Team mission, in their words. Plan and deliver North America Customer Fulfillment capacity to meet customer demand at the desired speed and quality while improving the long-term cost structure of the network.

Their customers. S&OP, PPT, Finance, FC Operations, WW and Operations Engineering, Real Estate, WW Design, Launch, Safety/EHS. Partners in SCOT, Product, Engineering and Construction.

Format. One hour. Expect two to four Leadership Principle stories with hard follow-ups, and one functional block on capacity planning. If this is the hiring manager, the functional block gets longer.

Your position. You have designed and consolidated DC capacity, allocated demand across a 3-DC / 100-store network through peak, and turned measured lane data into planning parameters. You have not built an Amazon FC capacity plan. Say so once, early, then show the mechanism.

Read sections 2, 3, and 4 out loud once — section 3 has "tell me about yourself" in the Amazon shape and "why Amazon / why this team / why now." Section 5 is the story bank in STAR with the interviewer's own probes answered — know which four you are leading with. Section 6B answers all 128 questions in Amazon's question bank — story, STAR answer, probes, and the WBR/design translation. Section 7A is the deep model — how a unit flows, what it costs, and the WBR this role would run; read it once, slowly. Section 7 is the functional block; walk Case A on paper once before Friday.

1. What they are actually scoring

The L6 bar, in one table

L6 GUIDELINE SAYS	WHAT THE INTERVIEWER LISTENS FOR	WHERE YOU PROVE IT
Ambiguity. Strategy may not be defined. You pick the stakeholders, set the goals, design the long-term solution.	Did you start from a vague ask and define the problem yourself?	Elementum: the ask was a report; you built lane-level lead-time planning. Flexport: inherited an 18-month plan and re-costed the sequence.
Scope and influence. A team, sometimes an org or country. Influences 1–2 levels up.	Did you move a decision above your level with evidence, not seniority?	Roadmap flip ratified in Bellevue after private math with your manager. CIO/COO weekly pack across 60+ stakeholders.
Execution. Tactical and strategic. Decides where to simplify or extend. Mitigates long-term risk. Trade-offs short vs long.	Can you name the trade-off and the number that decided it?	Partner-first vs operator-first (CTS ~50% vs ~25%). Robots at 30% vs the launch date. Mock tolerance vs go-live.
Complex. Visible risks, roadblocks, constraints. You design an appropriately simple solution.	Was the solution simpler than the problem? Did a gate exist before the crisis?	0.5% tolerance published before mock one. 98% pick-accuracy gate published before the robots arrived.

L6 GUIDELINE SAYS	WHAT THE INTERVIEWER LISTENS FOR	WHERE YOU PROVE IT
Designing or modeling the supply chain, not working in it.	Are your stories about the network's shape and parameters, or about running a shift?	Allocation across 3 DCs + 100 stores. Two buildings into one omni DC. 200 lanes measured. Safety-stock parameters re-set from data.
Process improvement. Drives operational-excellence best practice.	Did you leave a mechanism that ran without you?	WBR grammar at Flexport; QA-authored release gate that survived your leaving; carrier escalation and drift flags.

How the interviewer is trained to probe — Amazon's STAR worksheet

Amazon's interviewer guide gives every interviewer the same worksheet: establish the story in STAR, then **probe** on the assigned Leadership Principle, then **challenge** the candidate's decisions. Results are scored against a legend — \$ cost savings or revenue, # volume/scale, % change, **time** saved, **customer/team impact**, **quality**. Every story in Section 5 is written so the probes below are already answered before they are asked, and every Result is written to that legend.

PHASE	WHAT THEY ARE TRAINED TO ASK	HOW YOUR ANSWER PRE-EMPTS IT
Situation / Task	Why is this important? What was the goal? What was the initial scope? What were the challenges? What were the risks if nothing happened?	Open every story with <i>problem</i> → <i>goal</i> → <i>the metric and its baseline</i> → <i>the cost of doing nothing</i> . Fifteen seconds. Then stop describing and start doing.
Action	Were you the key driver? Biggest contribution? Unique value? Most significant obstacles? What did you do versus the team? How did you set priorities / get buy-in? What decisions did you challenge?	Every action sentence starts with "I" and names an artifact — the gate, the table, the rule, the log, the query. Name the obstacle and the decision you overruled. Name what you deliberately did <i>not</i> do.
Results	How did you measure success? Why these results? What percentage change year over year? What trade-offs (quality, cost, time)? "I'm concerned about... tell me more."	Give the metric, grain, baseline, target, actual — in that order. Then the trade-off you paid. Then the mechanism that made it stick.
Challenge	Why this story? Can you give a more recent one? What were the lessons? What would you do differently? How would you implement this at Amazon? How did results compare to the goal you stated?	Every story below ends with "Did the metric move" against the stated goal, a real "differently," and an "At Amazon" sentence that lands in capacity planning.

Three words they must hear in every story: supply, demand, capacity. Each story in Section 5 carries a box naming what supply was, what demand was, what capacity was, and which one bound. Say the box out loud when the question is functional; keep it in your head when the question is behavioral — but make sure the story's logic follows it.

The JD, translated into what you say

- **"Take end-to-end ownership of building and manifesting our capacity plan."** Say: a plan is a set of numbers with named owners and triggers, not a deck. You have run that as a WBR.
- **"Balance capacity and demand."** Say: that is allocation at Manhattan and consolidation at Sephora UDC. Demand did not fit the building; you changed where units went, then changed the building.
- **"Define metrics, develop program management mechanisms, invent/simplify processes."** Say: input lever vs output metric, pre-wired reds, COE that changes a mechanism. Flexport Snowflake story.
- **"Craft concise written narratives to influence executive decisions."** Say: the two-column cost-to-serve table that flipped an 18-month roadmap. Offer to write it.

- **"Adherence to lean philosophy at scale."** Say: gates that can stop the line. Andon is not a poster.

Vocabulary you should use correctly, and what to ask about

TERM	USE IT AS
NACF	North America Customer Fulfillment — the FC network that picks, packs and ships customer orders.
Capacity plan	Units of demand the network must absorb, by week, by building type, against what each building can store and push out. Two constraints, not one: storage (cube) and throughput (units per hour through inbound, stow, pick, pack, ship). Say both every time.
S&OP	The weekly plan that reconciles demand forecast, inventory and capacity; the basis of labor planning and steering decisions. Your Elementum forecast-alert story lives here.
OP1 / OP2	Amazon's annual operating-plan cycles (fall and spring). Long-range capacity decisions — new builds, expansions, retrofits — are argued in these. You can say "the OP cycle" without pretending you have sat in one.
SCOT	Supply Chain Optimization Technologies — the science and product org that owns forecasting, buying, placement and planning systems. The interface you would manage: they own the model, you own whether the plan is honest.
PPT	Listed as a customer of the team. Do not guess the acronym. Ask: "what does PPT own relative to Global Capacity Planning?"
Sortable / non-sortable / AR	FC types by item size and automation. Capacity is planned per type; a sortable AR building and a non-sort building are not interchangeable. Say "by building type" rather than "by FC."
Launch / Real Estate / WW Design	The teams that put new capacity on the ground. Your UDC and dual-DC stories are launch stories; speak to them in launch gates and freeze windows.
Speed	Promised delivery speed to the customer. Capacity in the wrong place is a speed miss, not just a cost miss. Your promise-equation work maps straight onto this.

2. Say this about the gap. Once.

UNPROMPTED, INSIDE THE FIRST FIVE MINUTES

"I have not built an Amazon FC capacity plan. I have done the three things underneath one: I allocated demand across three mega-DCs and a hundred stores through peak when the network did not fit the plan; I consolidated two warehouses into one omni facility with robotics and ran the launch gates; and I replaced assumed planning parameters with measured ones across two hundred lanes and released the inventory the bad assumptions were holding. I know the scale here is different. The mechanism is not."

Then stop. Do not list what else you have not done. If they probe the gap later, the answer is in section 8 ("Why you over an internal transfer").

3. "Tell me about yourself" — Amazon format, 60 seconds

Present → past → future. One sentence each for present and future; three numbered chapters for past, one number each. Under 160 words. No adjectives about yourself. Say the three words — **supply, demand, capacity** — once, in the last line, where they land as a claim about the job.

SAY THIS, THEN STOP — ABOUT 60 SECONDS

Present. "I'm the technical program manager for retail fulfillment and inventory at Sephora — I own how inventory is defined, placed and promised across stores, DC and dotcom on S/4HANA, and I just consolidated two warehouses into one robotic omni DC: processing efficiency up fifteen percent, zero customer incidents through peak.

Past. Before that, three chapters, one job. Manhattan Associates: allocation and promising across three mega-DCs and a hundred stores — 99% fill through Cyber Monday, about eight million protected. Elementum: replaced brochure lead times with measured distributions on two hundred lanes — a million and a half of safety stock released at 99% accuracy. Flexport: flipped an eighteen-month roadmap with a two-column cost table — five million avoided, value in six months.

Future. I have not built an Amazon FC capacity plan. What I have done four times is plan supply, demand and capacity as one equation, deliver it through gates, and keep it honest in a weekly review. I want to do that at network scale, in the one place where it is the operating system."

If they want the long version, add one sentence per chapter from the table in Section 11. Do not add adjectives; add numbers. If they interrupt, the last two sentences are the only ones that must land.

Why Amazon. Why this team. Why now.

The generic version — "customer obsession, scale, Leadership Principles" — scores zero because every candidate says it. The specific version names one thing Amazon did, one thing you did, and the exact seam where they meet. Under 45 seconds each. Items marked **[public]** are from Amazon's public statements; re-read the source before Friday and do not quote a number you have not re-read.

WHY AMAZON — SPECIFIC

"Three specific reasons, none of them 'because it's Amazon.' First: regionalization **[public — 2023 shareholder letter]**. When Amazon reorganized the U.S. fulfillment network into regions so that most orders ship from inside the customer's own region, it got faster *and* cheaper at the same time — fewer touches, shorter distance. I read that as a capacity-planning decision, not a transportation one: where you put cube and hours decided both the speed line and the cost line. That is the exact trade I made at one-thousandth the scale when I consolidated two Sephora warehouses, and when I re-costed Flexport's roadmap on cost-to-serve. I want to make it at the scale where it is the company's P&L. Second: the review. For six years I have run weekly reviews in borrowed Amazon grammar — input metrics, output metrics, reds pre-decomposed, a correction of error that changes a mechanism. The Flexport roadmap decision I am proudest of was argued on a two-column cost table in a Bellevue onsite and on nothing else. Here that is the default, not something I have to install. Third: the demand signal. Everywhere I have planned capacity, the forecast was a spreadsheet I had to defend. Here it is a model with an error distribution. That changes what a capacity planner can promise — you can plan to a quantile instead of to an argument."

WHY THIS TEAM — GLOBAL CAPACITY PLANNING

"The charter has both halves of the only trade I care about: meet demand at the promised speed, *and* improve the long-term cost structure. Most roles own one half and lobby against the other; this one owns the dial. And the customer list — S&OP, Finance, FC Ops, Engineering, Real Estate, Launch — is the set of rooms where I have already had the hard version of the conversation: handed S&OP a forecast alert they trusted enough to act on; given Finance a cost table and let them attack it; held a launch gate against a go-live date. I know how each room argues. I want the seat that has to get them to one number — and the first thing I would want to understand is where the storage-versus-throughput call actually gets made today, because that is where the plan is either honest or not."

WHY NOW — AND WHY LEAVE SEPHORA

"I've delivered what I went to Sephora to do: the WMS migration is live, the omni DC is live and fifteen percent more efficient, the inventory operating model is the standing reference. If I stay I run the same play a fourth time at retail scale. I would rather run it once at this one, where supply, demand and capacity are planned for a network of hundreds of buildings and the forecast is a model. I'm not running from anything. I'm running toward the version of this problem where the network is the product."

Do not say: "most customer-centric company," "I love the Leadership Principles," "Day 1," or anything about compensation or Bellevue here. **If asked "why not stay in retail":** the last three sentences of "Why now."

4. The sixty-minute clock

MINUTE	WHAT IS HAPPENING	YOUR JOB
0–3	Intros. They describe the team.	Listen for the word they use for their pain: "peak," "utilization," "launch," "forecast," "cost per unit." That word picks your lead story.
3–7	"Tell me about yourself." Often followed by "why Amazon."	Section 3, Amazon format. Land the gap sentence. Keep "why Amazon" under 45 seconds.
6–40	Two to four LP questions. Each gets four or five follow-ups.	One story per question, 90 seconds, then stop and let them drill. Answer follow-ups in one or two sentences. Say "I" for decisions. Name the data pull.
40–52	Functional block: build a plan, handle a gap, define metrics.	Sections 7A and 7. Clarify out loud for 60 seconds before you solve. Say both currencies — cube and hours. Do a math checkpoint. End with owners and triggers.
52–58	Your questions.	Two from section 9. Not four.
58–60	Close.	Section 10.

Amazon interviewers write notes the whole time and score after. Pauses are fine. A tight 90 seconds followed by "happy to go deeper on any part" beats a four-minute story every time.

5. The story bank — STAR, probed to the worksheet

Sixteen stories. Every one has the same skeleton, in this order, because that is the order the interviewer's worksheet scores: **Problem** → **Goal** → **Key metric (baseline → target)** → **Situation** → **Task** → **Action** → **Result** → **Did the metric move?** Then the box naming **supply, demand and capacity** in that story. Then "what was innovative" — the usual approach versus what you did — because "actions" are scored on whether a competent peer would have done the same thing. Then the probes, grouped exactly as the worksheet groups them: Situation, Action, Results, Challenge. Then one sentence on how you would implement it at Amazon, because the worksheet tells them to ask.

Every number is from your resume or your war book. Where a probe answer describes *how* you computed a number, check the method against your memory before Friday — change the method if needed, keep the number. "AIQB" references are the question numbers in Amazon's interviewer question bank, so you can hear a question and know which story it was written for. The spoken answer is 90–120 seconds. Then stop and let them probe.

Two kinds of stories. *Domain stories* — where you designed the supply/demand/capacity logic: S10 (S/4HANA ATS and reconciliation), S3/S4/S5/S16 (Elementum), S6/S7/S8/S15 (Flexport), S2/S11 (Manhattan). *Program-management stories* — where you ran a large delivery: S1 (UDC commingle) and S9 (RP JDA). Lead with domain stories for capacity, inventory, data and metrics questions; use S1 and S9 for delivery, launch, deadline and obstacle questions. **Lead with these four** unless their pain word says otherwise: S2 (peak allocation), S3 (200 lanes), S10 (ATS and reconciliation), S7 (roadmap flip).

Every story now has "The math," a "Solution design / flow" diagram where there is a system, and "How it translated to a WBR." In the math blocks, numbers in [brackets] are illustrative inputs chosen so the arithmetic lands on your real result; replace them with your actual inputs before Friday. The bolded result numbers are yours.

S1 — Two warehouses into one omni DC; robots throttled to 30% on a gate I published before they arrived (Sephora UDC commingle · program-management story)

LPs: Insist on the Highest Standards · Customer Obsession · Bias for Action · Deliver Results. **AIQB:** Highest Standards 1, 4, 6 · Deliver Results 1, 2 · Bias for Action 1 · Customer Obsession 8.

Problem	Store replenishment and dotcom ran out of two warehouses: the same SKU stowed twice, two labor pools, two inbound docks for one vendor base, and no shared view of what was sellable to whom. Cost per unit above plan and a transfer every time one channel needed the other's stock.
Goal	One omnichannel DC on Blue Yonder with Locus robotics, live before peak, with zero customer-facing incidents and pick accuracy ≥ 98% before the robots carried full volume.
Key metric	Processing efficiency (units per labor hour) vs the two-building baseline; pick accuracy by zone by hour vs 98%; customer-facing incidents.
Did it move?	Yes. Processing efficiency +15%. Pick accuracy 94% → ≥ 98% within one week of the throttle. Zero customer-facing incidents through peak. Launched on date.

Situation. "Sephora served stores and dotcom from two separate warehouses. Same SKUs stowed twice, two labor pools, one vendor base shipping to two docks, and no single answer to 'how many can we sell.' The decision was to collapse both into one omni DC on Blue Yonder with Locus robots — inside a window that ended at peak.

Task. I owned the program: fifteen-plus tracks, sixty-plus stakeholders, about a dozen systems — WMS, ESB and Kafka, OMS, finance, the data platform. I set the goal in numbers before we started: live before peak, zero customer-

facing incidents, pick accuracy at or above 98% by zone before robots took full volume, processing efficiency above the two-building baseline.

Action. Three things that were not in the standard playbook. First, I published the pass/fail numbers before the robotics vendor was on site — 98% pick accuracy measured by zone by hour, not a building average — so the go/no-go would be read, not argued. Second, I designed the robot ramp as a dial, not a switch: the manual pick path stayed fully staffed and certified through the ramp, volume could move between bots and people within a shift, and the decision right for that dial sat with me. Third, one issues log — 836 rows over fourteen months, every row an owner and a date — behind dated public test gates, November 22, December 20, January 30, and an October-to-December change freeze so the building would survive peak. Week two after robots went live, pick accuracy printed 94%. I throttled robots to about 30% of volume the same day — only the zones that had cleared 98% in the debug sample — kept the launch green on the manual path, sat the zone-logic debug with the vendor, and restored full volume in week three when every zone cleared the bar.

Result. Processing efficiency finished plus fifteen percent against the two-building baseline. Pick accuracy from 94% to above 98% in a week. Zero customer-facing incidents through peak. Launched on the date. And structurally: one inventory, one dock, one labor pool — a unit can serve a store or a customer order without a transfer. The gate existed so the throttle needed no meeting."

Supply · Demand · Capacity. **Demand:** two profiles — store replenishment (lumpy, planned) and dotcom (daily, peaky) — now served from one pool. **Supply:** one vendor inbound stream to one dock instead of two. **Capacity:** two buildings' cube and hours collapsed into one, with robotics as the throughput multiplier. **What bound at launch:** pick throughput *at quality*, not cube — the building had the space; the question was whether the bots could pick at 98%. The throttle was a throughput-capacity decision made against a quality gate.

What was innovative. The usual approach is a cutover date plus a readiness checklist and a vendor who says "ready." I made the quality bar a number published before the technology existed in the building, built the ramp as a reversible dial with named decision rights, and kept a certified manual path as paid insurance. That turned a potential launch crisis into a routine Tuesday decision.

The math (program-management story — the math is throughput at quality)

Processing efficiency (UPH) baseline, two buildings:	= units processed ÷ direct labor hours, inbound + outbound, weekly [1.00M units ÷ 20,000 hrs] = 50.0 UPH	← bracketed inputs illustrative
after consolidation:	[1.05M units ÷ 18,260 hrs] = 57.5 UPH	← +15% is your number
where the 15% came from:	one stow instead of two (-dup handling), no inter-DC transfers, bots on dense zones	
Pick accuracy (zone, hour) week 2 building average 94% ;	= correct pick-confirms ÷ total picks by zone: dense multi-unit zones [88-91%], others [≥ 98%]	
Throttle share	= Σ volume of zones with accuracy ≥ 98% ÷ total volume	≈ 30% ← output, not target
Insurance price (manual path)	= 3 wks × [incremental hrs/wk] × [loaded rate]	vs
Cost of a 6-pt miss at peak	= [peak units/wk] × 6% × [cost per mis-ship: re-ship + concession + store OOS]	
the second number is an order of magnitude larger → manual path is cheap insurance		

Solution design / flow

```
graph LR
    Vendor[vendor] --> Dock[ONE inbound dock]
    Dock --> Receive[receive]
    Receive --> Stow[stow one inventory, one WMS]
    Stow --> ManualPick[manual pick]
    Stow --> Locus[Locus bots zones ≥ 98%]
    Stow --> ManualPickOther[manual pick other zones]
    ManualPick --> Pack[pack]
    Locus --> Pack
    ManualPickOther --> Pack
    Pack --> Ship[ship]
```

DIAL (owned by program): zone list bots may carry; re-evaluated daily on zone × hour accuracy
GATES: Nov 22 / Dec 20 / Jan 30 test gates · Oct-Dec change freeze
· 836-row issues log (owner + date per row)

How it translated to a WBR

Outputs: processing efficiency vs baseline (weekly); customer-facing incidents (daily, target 0). **Inputs:** pick accuracy by zone by hour vs 98%; bot share of volume; issues-log rows past due; freeze exceptions. **Red rule:** any zone < 98% for two consecutive hours drops out of the bot list the same shift — no meeting.

SITUATION PROBES

Why was this important? What was the goal? Every unit stowed twice is cube paid for twice and a transfer waiting to happen. The business case was lower cost per unit and one view of sellable inventory; my program goal was the three numbers — zero incidents, ≥98% by zone, efficiency above baseline — because a consolidation that ships wrong units has destroyed the reason it was built.

What would have happened if nothing was done? Two buildings through another peak with inter-DC transfers to cover channel imbalances, duplicated safety stock, and a cost per unit that stayed above plan. The lease and labor math did not work for two buildings at our volume.

What was the initial scope? Did it change? WMS, robotics, and inventory merge. Scope grew to the integration layer when we found store and dotcom order flows hit the WMS through different interfaces with different unit-of-measure conventions. I added the track and re-baselined the gate dates once, publicly, in writing.

Biggest challenge? Time. The window ended at peak and peak was non-negotiable. Everything was designed to be safe to stop: the manual path, the freeze, the gates.

ACTION PROBES

What did you personally do versus the team? I wrote the gate definition and published it; I made the throttle call the day 94% printed; I owned the issues log and the change freeze; I decided which zones the bots kept. The vendor fixed the zone logic; ops ran the manual path; I decided when volume moved.

Why 30%? How did you arrive at it? It was an output, not a target. I let the bots keep only the zones that had already cleared 98% in the debug sample. Those zones carried about 30% of volume. Anything else would have been a guess dressed as a plan.

Why not delay the robotics go-live until the vendor was confident? Vendor confidence is not a metric. I had a measured path to run both and a number that said when to move volume. Delaying would have spent calendar we did not have and taught us nothing about the zone logic under real volume.

Why not push through — the business had bought robots? A gate that cannot stop the program is decoration. The robots were an input; the customer-facing output was accuracy. I will not ship a customer-facing miss to protect an automation narrative.

What did the manual path cost? Who paid? About three weeks of labor hours for a certified path we were planning to shrink. I priced it against one week of accuracy misses at peak — mis-ships, re-ships, concessions, store out-of-stocks — and against pulling the launch. It was the cheapest insurance in the program and I said so in the steering pack before the bots arrived.

How did you get sixty stakeholders to one log? I made the log the only place a date lived. If a dependency was not a row with an owner and a date, it did not exist for the program. People stopped sending emails because the log was where decisions happened.

What decision did you challenge? The original plan had robots at 100% from day one with a two-week "stabilization." I challenged the word stabilization — it had no number — and replaced it with the gate and the dial.

How did you handle the freeze? Someone always has an exception. One exception path, written down: a change in freeze needed a customer-facing defect it fixed, a rollback, and my signature. Three got through in ten weeks. The freeze held because the exception path was real and narrow.

RESULTS PROBES

How did you measure +15%? Units processed per direct labor hour, inbound and outbound, versus the two buildings' combined trailing baseline for the same weeks, normalized for the dotcom-versus-store mix. Weekly, from WMS and the labor system, in the steering pack.

How did you measure pick accuracy? WMS pick-confirm scan versus gold-standard location, by zone, by hour. A building average would have hidden the bad zones inside the good ones — which is exactly how 94% was first noticed.

Compared to your goal? Zero incidents: met. 98% by zone before full volume: met, one week late versus the vendor's own plan. Efficiency above baseline: exceeded.

What trade-off did you make? Three weeks of robotics ROI, deliberately, for zero customer risk through peak. I would make it again.

I'm concerned a 94% print means your testing was weak. Tell me more. Fair. The zone logic passed test volume and failed real volume mix — the test data under-represented multi-unit orders in the dense zones. The miss was in the test design. The mechanism caught it in week two instead of at peak, which is what the gate was for; and the fix I made since is below.

CHALLENGE

What would you do differently? Shadow the bots on one zone for a week at full real volume before they touched any customer order. I now put "one zone, real volume, a week" into every automation ramp I write.

Why this story for Highest Standards? Because the standard was set before it was tested, and it cost us something to hold it. Standards that are free are not standards.

At Amazon. A launch or retrofit ramp as a dial with zone- or path-level gates: the capacity plan counts a new building's or retrofit's throughput only for the process paths that have cleared the rate and quality bar at real volume. You never plan on capacity that has not earned it.

S2 — Cyber Monday, 4 AM: rebalancing allocation across three DCs with 70% of the information (Manhattan Associates)

LPs: Bias for Action · Are Right, A Lot · Dive Deep · Customer Obsession. **AIQB:** Bias for Action 2, 4, 5 · Are Right, A Lot 1, 3 · Dive Deep 2 · Customer Obsession 8.

Problem Peak day, 4 AM: backorder lines piling up across a 3-DC / 100-store network. Millions of dollars of orders would miss cutoff. Root cause unknown; full diagnostic an hour away.

Goal Hold fill rate at 99% through the peak day — which meant stopping backorder growth inside the hour.

Key metric Fill rate by hour (target 99%); backorder lines; orders at risk of missing promised cutoff (in \$).

Did it move? **Yes.** Fill rate held at 99% through peak; ~\$8M of at-risk sales shipped to promise; structural stockouts ~20% over the engagement once the triage logic became standing policy.

Situation. "I owned order allocation and promising across three mega-DCs and about a hundred stores. The node we sourced from determined the lead time the customer saw — choose the node, and you have chosen the promise. Cyber Monday, 4 AM, a VP on the phone: backorder lines climbing, millions at risk, and the full diagnostic was an hour out.

Task. One number: hold fill rate at 99% through the day. That meant stopping the backorder growth inside the hour, with about seventy percent of the information I wanted.

Action. I split what had to be decided now from what could be learned in parallel. On the bridge I ranked every open order by *promise-at-risk* — dollar value times hours to its cutoff — instead of by age, and rebalanced allocation across the three DCs to the nodes that had hours, overriding the engine's assignments manually in that priority order. Every decision timestamped in one log, so the diagnostic team could later separate my effects from the fault's. I kept two actions explicitly off the table: no blanket promise downgrade, and no emergency inbound — both one-way doors

at 4 AM. The trace an hour later showed the allocation scheduler had died quietly at the daylight-saving change. Inventory was fine. DC capacity was fine. The engine had simply stopped assigning.

Result. Fill rate held at 99% through the peak. Roughly eight million dollars of orders that would have missed cutoff shipped to promise. The promise-at-risk ranking became the standing allocation priority, and structural stockouts were down twenty percent over the engagement. Afterward I wrote a scheduler heartbeat and a DST checklist into peak readiness. I never want a second 4 AM."

Supply · Demand · Capacity. Demand: peak-day orders by promise tier and cutoff. **Supply:** inventory by DC and store — adequate, as it turned out. **Capacity:** two kinds — the physical pick/pack hours by node, which were fine, and the *planning-system* capacity to assign orders to nodes, which had gone to zero. **What bound:** a false capacity signal. The lesson for a capacity planner: when the plan says "capacity is short," first prove the shortage is physical.

What was innovative. The standard move at 4 AM is FIFO — clear the oldest backorders first — and a blanket promise downgrade to stop the bleeding. I ranked by dollar-weighted hours-to-cutoff so the scarce decision bandwidth protected the most promise per minute, and I classified every available action as a one-way or two-way door before taking it. The timestamped log turned a firefight into a dataset.

The math

Promise-at-risk score (order)	= order value (\$) × urgency, urgency = $1 \div \max(\text{hours to cutoff}, 0.5)$
\$400 order, 2 h to cutoff	→ 200 ; \$900 order, 9 h to cutoff → 100 → the \$400 order ships first
Node feasibility check	= units assigned to node before cutoff ≤ node hours before cutoff × pick/pack rate
node B: 3 h × [4,000 units/h]	= 12,000 units of room → assign in score order until full, then next node
Cost of waiting one hour	≈ pool doubles every ~40 min at 4 AM arrival rate → pool(5 AM) ≈ 2.8 × pool(4 AM)
Sales protected	= Σ order value of orders in the 4 AM pool that shipped before cutoff ≈ \$8M
Fill rate (hourly)	= lines shipped before cutoff ÷ lines ordered → held ≥ 99%
Structural stockouts	= SKU-locations with zero ATS at order time ÷ SKU-locations ordered → ~20% over engagement

Solution design / flow

```
order → OMS promise (node? lead time?) → allocation scheduler → DC pick/pack → ship
      ▲                                     | (died at DST change:
      |                                     ▼ queue grew, nothing assigned)
      | promise-at-risk ranking (manual bridge, 4 AM) → node assignment by score,
      | one timestamped log → reconciled against      feasibility-checked
      |                                     scheduler restart (no double allocation)
AFTER: scheduler heartbeat (alarm if no run in 15 min) · DST/peak-regime checklist
      · ranking rule = standing policy
```

How it translated to a WBR

Outputs: fill rate by hour vs 99%; orders past cutoff (\$). **Inputs:** allocation queue depth by node by hour; scheduler last-run age; promise-at-risk \$ in pool; node hours vs units assigned. **Red rule:** queue depth rising two consecutive hours with scheduler age > 15 min pages the bridge before anyone sees a backorder.

SITUATION PROBES

Why was this important? Peak day is a disproportionate share of the year's revenue, and a missed cutoff on peak is a customer who does not come back. The VP's phrase was "millions by noon."

What if you had waited for the diagnostic? An hour of growing backorders at peak order rates. We modeled it after the fact: the at-risk pool roughly doubled every forty minutes at that hour's order arrival rate.

What did you know at 4 AM, and what didn't you? Knew: backorder lines, inventory by node, DC hours scheduled. Didn't know: why the engine was not assigning. I decided I did not need the "why" to move orders safely by hand, as long as every move was logged and reversible.

ACTION PROBES

What did you do versus the ops team? I took the bridge, set the ranking rule, made the node assignments in priority order, and kept the log. Ops executed the picks; the engineers ran the trace. The ranking rule was mine.

Why act at 70%? What would have made you wait? Waiting was a customer miss with a dollar sign; acting was a two-way door — manual assignments could be re-run when the engine came back. I would have waited if the action were a one-way door: a promise downgrade or an emergency PO. Those I kept off the table.

Why promise-at-risk and not FIFO? FIFO protects the oldest order; it does not protect the most promise per minute of my attention. With an hour of hand assignment, I wanted every minute to clear the most dollar-hours of risk. FIFO would have spent the hour on low-value orders with late cutoffs.

How did you decide which DC got which order? Hours available at the node against the order's cutoff, then inventory at the node, then distance to the customer — in that order. If a node had the units but not the hours before cutoff, it did not get the order.

What was the risk of the manual rebalance? Double allocation when the engine resumed. The timestamped log was the mitigation: when the scheduler came back, the engineers reconciled against the log before re-enabling auto-assignment.

What decision did you challenge? The first instinct on the call was to downgrade all two-day promises to standard. I refused — that is a one-way door that touches every customer to fix a problem that might be local. It was local.

RESULTS PROBES

How did you compute \$8M? Orders that were in the backorder pool at 4 AM and shipped before their cutoff after the rebalance, at order value. Conservative: it does not count orders that would have joined the pool over the next hour.

How did you measure fill rate? Lines shipped before cutoff ÷ lines ordered, by hour, against the 99% line. Hourly, because a daily average would have hidden a 4 AM hole.

How did the -20% in stockouts come about? That's not from one night. Correct. The ranking rule became the standing allocation priority — scarce units went to the orders with the highest promise-at-risk rather than first-come. Over the engagement, structural stockouts fell twenty percent against the prior-year baseline. The night was the proof of concept.

Compared to goal? Goal was 99% through the day. Met. We did not claw back the backorders that had already missed before 4 AM; I counted those as a miss and said so.

CHALLENGE

What would you do differently? The heartbeat and the DST check existed by the following week. The deeper fix: I now treat every peak as a different operating regime with its own readiness checklist, because the failure modes at peak are not the failure modes at average.

Why this story for Bias for Action and not something more recent? Because it is the cleanest case where speed had a price and I can show the math on both sides. The Sephora robot throttle (S1) is more recent and is the same muscle: a two-way door turned the same day.

At Amazon. When a region shows a capacity shortfall, I would first prove it is physical — hours or cube — before spending a lever. And every lever in the plan would be labeled one-way or two-way before peak, so the bridge at 4 AM already knows which ones it may pull without a meeting.

brochure was holding (Elementum)

LPs: Think Big · Ownership · Dive Deep · Customer Obsession · Frugality · Invent and Simplify. **AIQB:** Think Big 1, 3, 6 · Ownership 3 · Dive Deep 5, 6 · Frugality 1 · Invent and Simplify 1, 6.

Problem	The inventory plan ran on carrier-brochure lead times — one optimistic number per lane, never measured. Safety stock on every lane was sized against a number nobody had checked; planners padded it by feel.
Goal	Planned lead times that hold 99% of the time, <i>and</i> less safety stock, not more — measured variability should buy inventory out, not in.
Key metric	Lead-time accuracy (arrivals inside planned lead time, by lane, by week) → 99%; safety stock units and dollars released, auditable by lane.
Did it move?	Yes. 99% lead-time accuracy held. 580,000 units and ~\$1.5M of safety stock released. Drift flag per lane made it permanent.

Situation. "A planner asked me for a lead-time variability report. When I looked underneath, the inventory plan ran on carrier-brochure lead times — one optimistic number per lane, never measured against an actual arrival. Planners knew the number was soft, so they padded safety stock by feel. Nobody owned the seam between the brochure and the plan.

Task. The ask was a report. The goal I set was different: planning lead times that hold ninety-nine percent of the time, with *less* safety stock — because if you measure variability honestly, it should release inventory on most lanes, not add it.

Action. I pulled the shipment history across two hundred lanes myself — ship date, arrival date, carrier, and the source of the transit number, brochure or scan. I built the empirical distribution per lane: mean, spread, tail. Then the move that mattered: I set every planning lead time to the lane's empirical p99 — the smallest number of days by which 99 of 100 shipments had already arrived — not the mean, and not a normal approximation, because right-skewed lanes have fatter tails than a normal distribution admits. The counter-intuitive result: on most lanes the measured p99 was tighter than brochure-plus-gut, so stock came out; on the minority where it was wider, stock went in, deliberately. I wrote the release rule with finance before any dollar moved: stock released only where measured spread was tighter than assumed. Then I productized it — a drift flag per lane that fires when the rolling distribution widens past a threshold, so widening triggers a review instead of a silent re-plan.

Result. Ninety-nine percent lead-time accuracy held. Five hundred eighty thousand units and about a million and a half dollars of safety stock released, auditable lane by lane. The ask was a report. The product was lane-level planning as a system — and the drift flag is why it was still true a year later."

Supply · Demand · Capacity. **Supply:** inbound lead time per lane — the variable I measured. **Demand:** replenishment requirements by SKU-location, unchanged. **Capacity:** safety stock is storage capacity and working capital consumed by supply uncertainty. **What bound:** a bad supply parameter was consuming cube and cash. Fixing the parameter released both without touching a building. This is the cheapest capacity there is: capacity hidden inside a wrong assumption.

What was innovative. The standard answer is a variability report and a safety-stock formula with a normal z-score. I planned to the empirical tail per lane, wrote a release rule that required the measurement to be *better* than the assumption before money moved, and built a drift detector so the plan re-opened itself. The system replaced the report.

The math

Per lane: sort n transit times $t(1) \leq \dots \leq t(n)$; empirical p99 = $t([0.99\ n])$
lane example: $n = 400$, mean 21 d, σ 4 d, right-skewed

normal p99 = $21 + 2.33 \times 4 = 30.3$ d empirical p99 = 34 d → normal understates the tail by ~4 days
 brochure = 18 d (+ planner pad "about a week") → planned ≈ 25 d, held 93% of the time, not 99%

Why the mean is a coin flip: in a right-skewed lane the median < mean, and $P(t > \text{mean}) \approx 0.40 \rightarrow 2$ in 5 late

Safety stock per lane $SS = d \times (\text{planned LT} - \text{mean LT}) + z \times \sigma_d \times \sqrt{(\text{mean LT})}$ (d = units/day)
 before (brochure + pad): planned 25 d, but variability unknown → planners held extra "just in case" stock
 after (p99 = 34 d measured): on most lanes the measured spread was tighter than the pad implied → SS fell
 net across 200 lanes: $\sum (SS_{\text{before}} - SS_{\text{after}}) = 580\text{K units}$; $\times$ avg standard cost [\$2.59] ≈ \$1.5M

Release rule (finance-signed): release only where $\sigma_{\text{measured}} < \sigma_{\text{assumed}}$ (lane-level, auditable)

Drift flag: fire when rolling-90-day p99 > planned LT + [2 days] or $\sigma_{90} > 1.25 \times \sigma_{\text{baseline}}$

LT accuracy = arrivals within planned LT ÷ arrivals, by lane, by week → 99%

Solution design / flow

shipment events (ship scan, arrival scan, carrier, lane key, transit source)
 ↳ lane history store → per-lane distribution (mean, σ , p50, p90, p99, n)
 |
 | ↳ planning LT = p99 → ERP planning parameter (lane × SKU)
 | ↳ drift flag per lane → planner review
 | ↳ safety-stock recompute → release ledger (finance-signed rule) → \$ by lane

How it translated to a WBR

Outputs: LT accuracy vs 99% (lane × week); safety stock \$ vs plan. **Inputs:** lanes with drift flag open; lanes with n below confidence floor; % planning LTs sourced from scan vs brochure. **Red rule:** drift flag open > 2 weeks without a planner decision is a red on the planner's line, not the lane's.

SITUATION PROBES

Why important? What if nothing happened? Every lane carried either too much stock (cash and cube) or too little (stockouts), and nobody could say which. Doing nothing meant continuing to pay for an unmeasured number in inventory every week.

What was the initial scope and how did it change? One report for one planner. I expanded it to all two hundred lanes and to the planning parameter itself — I took that to leadership with mockups before I built it, because changing the planning parameter is a change to the plan, not to a report.

What was the goal before you set yours? There was none beyond "a report." That was the problem.

ACTION PROBES

Why p99 and not p95 or the mean? The mean is a coin flip — in a right-skewed lane, roughly two in five shipments arrive after it. p95 leaves one shipment in twenty late, which at our order frequency was a stockout a month on a busy lane. 99% was the service level the business had already committed to; I made the lead time honor it. On lanes where the cost of carrying justified less, the method allows a lower quantile per lane — the point is to choose it, not inherit it.

Why empirical quantile and not a normal distribution? Transit times are right-skewed — a shipment can be very late but not very early. A normal fit understates the tail, which is exactly where the stockouts live. I checked: on long ocean lanes the normal p99 was days short of the empirical p99.

How much history per lane? Seasonality? Enough shipments per lane to make a p99 meaningful — thin lanes were pooled with similar lanes (same origin region, mode, carrier) and flagged as lower confidence. Peak-season transits were kept as their own regime where the data supported it; the drift flag catches the seasonal widening on lanes where it did not.

What did you do versus the team? I pulled the data, built the distributions, chose the quantile rule, pitched leadership, wrote the release rule with finance, and designed the drift flag. Engineering productized the pipeline from my spec.

How did you get planners to trust a number that cut their safety stock? I showed each planner their own lanes: brochure number, measured p99, actual late arrivals under each. Where the data said "add stock," I added it first. Credibility came from the lanes where I gave inventory back.

What did you challenge? The assumption that the carrier's number was a fact. And the plan to ship a report — a report would have been read once.

RESULTS PROBES

How did you measure 99%? Actual arrivals inside the planned lead time, by lane, by week, from scan data — the same data the distributions came from, held out forward in time.

How did you compute \$1.5M? Safety-stock units before and after, by SKU-lane, valued at standard cost. Released only on lanes where the measured spread was tighter than the assumed spread — the finance-signed rule — so no lane's release depends on a judgment call.

Was the whole 580K units a release? Did any lanes get more stock? The 580K is net. A minority of lanes got more stock because their measured tail was wider than the brochure. I would rather not quote the lane count without the file in front of me.

Trade-off? Lanes with wider tails carried more inventory than before. That was the right trade: we had been running them short and calling it luck.

Compared to goal? 99% accuracy: met. Less safety stock, not more: met, net \$1.5M. Permanence: drift flag live.

CHALLENGE

What would you do differently? Start every new domain with a measurement audit in week one — "which planning parameters are measured, and which are inherited?" — instead of discovering the brochure problem mid-project. I have done that on every program since, including S/4HANA.

Why this story for Think Big? Because the opportunity was bigger than the ask, I took it, and I can show the gap: a report versus a planning system that still runs.

At Amazon. Every capacity planning parameter — dock-to-stow hours, process-path rates, vendor lead times, days of cover — is a distribution, not a number. I would ask on day one which of them are planned at a mean, plan the ones that bind at a chosen quantile, and put a drift flag on each so the plan re-opens when the world moves.

S4 — Forecast alerts that entered the weekly S&OP because planners trusted them (Elementum)

LPs: Invent and Simplify · Deliver Results · Earn Trust · Dive Deep. **AIQB:** Invent and Simplify 3, 5 · Earn Trust 1 · Dive Deep 6 · Customer Obsession 3, 4.

Problem	Planners learned about shortages when a production line stopped. Forecast accuracy ~70% at SKU-location-week. The SAP data any alert would run on was not trusted — the ERP's own reports disagreed with each other.
Goal	Alerts planners act on inside the weekly S&OP; forecast accuracy 70% → 85%; safety stock reduced — in that order, because the third is legitimate only after the first two.
Key metric	Forecast accuracy at SKU-location-week; daily reconciliation drift (MD04 vs MB52); alert false-alarm rate; safety stock \$ released.
Did it move?	Yes. 70% → 85%. Zero recon drift daily. ~\$2M safety stock released under a two-signal rule. Live in three months; adopted into S&OP without mandate.

Situation. "Planners found out about shortages when a line stopped. Forecast accuracy was about seventy percent at SKU-location-week, and the SAP data behind any alert was not trusted — the demand view and the stock view disagreed on any given day.

Task. Alerts planners would act on inside the weekly S&OP, forecast accuracy to eighty-five, and safety stock down. In that order.

Action. I refused to ship a threshold email on data nobody trusted. I led the SAP ingestion — inventory, production, POs, ASNs, forecasts — into one supply-and-demand view, with a daily MD04-to-MB52 reconciliation published and held at zero drift, and one rule: no alert fires on a day the recon is red. The alert itself was coverage math — projected supply against demand by week — with a thirty-second reason line: which input moved, by how much. Then a public false-alarm burn-down: every alert a planner dismissed was classified and the logic fixed in a weekly forum until the dismiss rate fell. The forecast-accuracy gain came mostly from the inputs the recon exposed — stale POs, duplicated demand, wrong lead times — not from a new model.

Result. Live in three months. Forecast accuracy from seventy to eighty-five percent at SKU-location-week. Recon at zero drift daily. About two million dollars of safety stock released — only where *both* forecast error and lead-time error had improved, a rule finance signed. The alert entered the weekly S&OP because planners trusted it, not because anyone mandated it."

Supply · Demand · Capacity. Demand: the forecast at SKU-location-week — the input I made 15 points more accurate. **Supply:** inventory, production, open POs, ASNs — reconciled into one view. **Capacity:** two kinds — the S&OP's capacity to act (alert lead time before a shortage), and safety stock as storage/working-capital consumed by forecast error. **What bound:** trust. Without a reconciled base, no amount of alerting moves a decision.

What was innovative. The standard approach is a dashboard with thresholds. I built the trust mechanism first — a daily reconciliation with a hard "no alert on a red day" rule — then alerts that explain themselves in one line, then a public burn-down of false alarms, and a two-signal release rule so no single improved number could justify a dollar. The alert was the least important part.

The math

Forecast accuracy (SKU-loc-week, lag-matched) = $1 - \frac{\sum |F - A|}{\sum A}$ 70% → 85%
Recon drift (daily) = $\frac{\sum_{\text{plant}} |\text{MD04 stock} - \text{MB52 stock}|}{\sum \text{MB52 stock}}$ held at 0.0%
Coverage (weeks) = $(\text{on-hand} + \text{scheduled receipts inside LT}) \div \text{avg weekly demand}$
Alert fires when coverage < (LT weeks + safety weeks) AND recon is green today
Reason line = the input whose Δ moved coverage most: "PO 4471 slipped 9 d → coverage 1.6 wk in wk 7"
False-alarm rate = $\text{alerts dismissed} \div \text{alerts fired, weekly, classified (data / logic / timing / ignored)}$
Two-signal release release SS on SKU-loc only if $\Delta \text{FA} > 0$ AND $\Delta \text{LT-accuracy} > 0$
 $\text{SS}_{\text{new}} = z \times \sqrt{(\text{LT} \times \sigma_d^2 + d^2 \times \sigma_{\text{LT}}^2)}$ with the improved σ_d and σ_{LT} ; $\sum (\text{SS}_{\text{old}} - \text{SS}_{\text{new}}) \times \text{cost} \approx \2M

Solution design / flow

SAP ECC: inventory (MB52) · requirements (MD04) · POs · ASNs · production · forecast
↳ daily ingestion → reconciliation (MD04 vs MB52 = 0?) —red→ no alerts today; fix data
| green
▼
supply-demand view by SKU-loc-week → coverage math → alert + reason line
| ↳ S&OP weekly
↳ false-alarm burn-down (weekly, public) → logic fixes
→ dismiss rate ↓

How it translated to a WBR

Outputs: forecast accuracy at planning grain; stockouts with no prior alert (target 0); SS \$ released. **Inputs:** recon drift (daily); alert false-alarm rate; alert lead time before shortage; % alerts actioned in S&OP. **Red rule:** any stockout without an alert $\geq$ LT weeks earlier gets a root-cause row the same week.

SITUATION PROBES

Why important? Cost of nothing? A stopped line is the most expensive shortage there is, and planners were compensating with safety stock they could not justify. Doing nothing was paying for 30% forecast error in inventory and in expedites.

Who was the customer? The planners, and through them the plants. The S&OP meeting was where their decisions were made, so that was where the alert had to earn a seat.

ACTION PROBES

Why reconciliation first? Planners wanted alerts. An alert on untrusted data trains people to ignore alerts. I would rather be three weeks late with something they believe than a day early with something they dismiss. I told them that in the first meeting.

What's the MD04-to-MB52 reconciliation in plain terms? The planning view of stock and requirements versus the physical stock view, by plant, by material, by day. If they disagree, one of them is wrong, and any alert built on either is wrong. We held the difference at zero and published it.

Why a reason line? A planner has thirty seconds per alert. "Coverage falls below two weeks in week 7 because PO 4471 slipped nine days" is actionable. "Coverage alert" is noise.

How did you run the false-alarm burn-down? Every dismissed alert got a category — wrong data, wrong logic, right alert wrong timing, right alert ignored — and we fixed the top category weekly in a public forum. The dismiss rate was on the wall. Planners saw their feedback change the logic within a week; that is where trust came from.

How did forecast accuracy improve 15 points if you did not change the model? Garbage in. Stale POs were being counted as supply; duplicated demand from two systems inflated requirements; lead times were brochure numbers. The recon exposed each one. Fixing inputs moved the number more than a new model would have.

What did you personally own? The ingestion scope and the recon design, the "no alert on a red day" rule, the reason-line requirement, the burn-down forum, and the two-signal release rule with finance.

RESULTS PROBES

How did you measure forecast accuracy? $1 - |\text{forecast} - \text{actual}| / \text{actual}$ at SKU-location-week, lag-matched to the planning horizon, weekly, in the S&OP pack.

Why two signals before releasing stock? Forecast accuracy alone could be mix; lead-time accuracy alone could be a quiet quarter. Requiring both to improve on the same SKU-location is how I kept the \$2M honest. Finance signed the rule before the first dollar.

Compared to goal? Adoption: met, un-mandated. 85%: met. Stock release: \$2M, met under the stricter rule.

Trade-off? Three weeks of calendar on the recon before anyone saw an alert. Worth it; the alternative was alerts nobody read.

CHALLENGE

Differently? Put the reason line in the first prototype — I added it after planners dismissed the first week's alerts, and I lost a week of trust I had to earn back.

At Amazon. Any capacity signal I bring to S&OP or Finance carries its reconciliation state and a one-line reason — "FC-A storage utilization crosses 90% in week 9 because inbound for category X pulled forward two weeks." And a lever fires on two signals, not one: forecast and placement, or utilization and dock-to-stow.

S5 — Austria: I arrived certain it was carriers; the data said we could not measure lateness

(Elementum)

LPs: Dive Deep · Are Right, A Lot · Earn Trust · Have Backbone. **AIQB:** Are Right, A Lot 1, 6 · Dive Deep 1, 2, 4 · Earn Trust 6 · Have Backbone 2.

Problem	A 25-location European freight program: reports said carriers were late everywhere and freight cost was over plan. The sponsor had committed to a routing engine to move volume to better carriers.
Goal	Auditable freight savings through better routing — the sponsor's number — without building an engine on data that could not support a recommendation.
Key metric	Data quality at the planning grain (reported 40%; true ~55%) → 99% by defect class; freight savings \$ auditable to a shipment.
Did it move?	Yes. Data quality 99% by defect class before any routing ran. ~\$8M freight savings across 25 locations, every dollar auditable to a shipment under a finance-signed method.

Situation. "I walked into a 25-location European freight program in Austria convinced we had a carrier problem. Every report said carriers were late everywhere, freight cost was over plan, and the sponsor had promised a routing engine to move volume to better carriers.

Task. Deliver auditable freight savings. The sponsor wanted the engine first.

Action. Before the engine kickoff I pulled three years of shipment history myself — not an analyst, me. The data said something more embarrassing than late carriers: we could not measure lateness. True data quality at the planning grain was about fifty-five percent, against the forty the reports claimed — wrong in both directions because the reports were measuring different things. Missing rates, lanes keyed at the wrong grain, the same carrier under five spellings, transit times copied from brochures. So I changed the program's first milestone from engine to measurement, in front of a sponsor who had promised the engine, with one page: a routing recommendation on 55% data is a coin flip with a confident interface. We fixed quality by defect class to ninety-nine percent before any routing ran. And I had finance sign the savings method — baseline lane rate versus executed rate, per shipment — before anyone claimed a dollar.

Result. About eight million dollars in freight savings across 25 locations, every dollar traceable to a shipment. Data quality ninety-nine percent by defect class. And a changed mind, on the record: I had been hired to fix carriers, and I told the people who hired me that carriers were not the problem."

Supply · Demand · Capacity. **Supply:** carrier capacity and rates by lane. **Demand:** shipment volume by lane across 25 locations. **Capacity:** the program's capacity to decide — you cannot route what you cannot measure. **What bound:** measurement, not carriers. For a capacity planner the lesson is literal: a utilization or on-time number at the wrong grain is wrong in both directions, and spending a lever on it is a coin flip.

What was innovative. The usual response to "the data is bad" is to clean it on the way into the engine. I made measurement the deliverable, scored by defect class with a 99% bar, and signed the savings method with finance before the first claim — so the result could not be argued later. I also changed my own thesis in public, which is what made the sponsor willing to change theirs.

The math

Data quality (planning grain) = records passing ALL defect-class checks ÷ records
defect classes: rate present · lane key at correct grain · carrier canonical · transit from
scan (not brochure)
reports said 40% (field-present only, counted brochure as data) ; true = 55% ; after = 99% by
class
Why a routing engine on 55% is a coin flip: recommendation needs rate AND lane AND transit

```
correct
if each is ~0.82 complete independently  $\rightarrow 0.82^3 \approx 0.55 \rightarrow P(\text{recommendation built on bad input}) \approx 45\%$ 
Savings method (finance-signed, per shipment):  $\Sigma_{\text{shipments}} (\text{baseline lane rate} - \text{executed rate}) \times \text{units}$ 
25 locations, 3 years of baseline, no lane averages  $\rightarrow \approx \$8\text{M}$ , each dollar joined to a shipment ID
```

Solution design / flow

```
3 yrs shipment history  $\rightarrow$  profile by defect class  $\rightarrow$  quality scorecard (by class, by location)
| (my first pull: "we cannot measure lateness")
▼
MILESTONE 1 (new): fix measurement  $\rightarrow$  99% by class
└─ MILESTONE 2: routing engine (lane  $\times$  carrier  $\times$  mode)
    └─ savings ledger per shipment (baseline - executed)
```

How it translated to a WBR

Outputs: freight \$ saved vs plan (auditable); on-time by lane (now measurable). **Inputs:** data quality by defect class; % shipments with scan-based transit; carrier canonicalization coverage. **Red rule:** savings claimed on any lane below 99% quality are held out of the ledger until the lane clears.

SITUATION PROBES

Why important? What if nothing happened? An engine on 55% data would have moved volume to carriers that looked good because their data was missing. We would have claimed savings nobody could audit and lost the sponsor's credibility with finance.

What was the initial scope? Routing engine, 25 locations, savings target. I kept the target and the locations and inserted a measurement phase in front.

ACTION PROBES

Why pull it yourself? Because I was the one who was going to tell the sponsor his first milestone was wrong. I needed to have seen the rows.

How can "40% reported" and "55% true" both be wrong? The reports counted a field as present if it had any value; some values were brochure numbers, so the reports overstated quality on those. Elsewhere they counted lanes as incomplete because of spelling variants that were actually resolvable, so they understated. Measured at the planning grain with defect classes, the true number was 55% — and the point is that neither report number could be used for anything.

How did you move a sponsor who had promised the engine? One page, privately first: here is what the engine would recommend on three real lanes using current data, and here is what actually happened on those lanes. Two of three recommendations were wrong. He took the page to his leadership himself.

What did you do versus the team? I pulled the history, defined the defect classes and the 99% bar, re-sequenced the program, and negotiated the savings method with finance. The team built the cleansing by class.

What decision did you challenge? My own first. Then the sponsor's milestone. Then the finance assumption that savings could be claimed on a lane average — we went per shipment.

RESULTS PROBES

How did you compute \$8M? Per shipment: baseline rate for that lane and mode before the program, minus executed rate after routing, summed across 25 locations, under the method finance signed. No lane averages.

Why 99% and not "good enough"? Because the engine's recommendations are only as reliable as the worst defect class. At 99% by class, a wrong recommendation was rare enough to catch by exception. At 90% it was a weekly argument.

Compared to goal? Savings target: met and auditable. The engine shipped later than the sponsor had promised. I owned that date change in writing.

CHALLENGE

Differently? Pull the data in week one, before the kickoff deck, not week four. It would have saved the sponsor an embarrassing re-plan.

Why this story for Are Right, A Lot? Because being right here meant changing my mind in public with data, which is the only version of that principle that is hard.

At Amazon. Before I plan to a utilization number, a design rate, or an on-time metric, I would ask what grain it is measured at and how it was last validated against actuals. A capacity ceiling that was set three years ago and never re-measured is a brochure number.

S6 — Seven to sixty-eight customers behind six graduation gates; cost-to-serve down 95% while the book grew ten times (Flexport)

LPs: Insist on the Highest Standards · Deliver Results · Frugality · Success and Scale Bring Broad

Responsibility. AIQB: Highest Standards 1, 6 · Deliver Results 3 · Frugality 1, 8 · Success and Scale 1, 8.

Problem A global 4PL shipment-visibility platform with seven pilot customers, cost-to-serve \$0.66 per container, a support team already stretched, and a sales pool of 60+ waiting to board. Scaled the usual way, support cost scales with customers.

Goal 68 customers by the committed date, with cost-to-serve *down* and ticket response *up* at volume.

Key metric Cost-to-serve per container (\$0.66 baseline); ticket response rate (72%); customers live vs plan; data quality and ETA accuracy per wave.

Did it move? **Yes.** 68 customers, two months early. Cost-to-serve \$0.66 → \$0.03 (~95%) while the base grew ~10×. Ticket response 72% → 90% at 5× volume.

Situation. "A global 4PL shipment-visibility platform: seven pilot customers, sixty-six cents a container to serve, a support team already stretched, and sixty-plus customers in the sales pipeline. Scaled the way pilots usually scale, support cost would have grown with the customer count.

Task. Sixty-eight customers by the date — with cost-to-serve going down and ticket response going up at volume. Both, or the scale was not worth having.

Action. I treated onboarding as a capacity problem. I wrote six graduation criteria a wave had to clear before it boarded, published to sales and leadership: data quality at or above ninety percent against source of truth, milestone and ETA accuracy at or above eighty, verification SLA, engagement floors, platform stability, CSAT. A wave was admitted only when the data and support layer had cleared the bar at the *current* load — so each wave paid for the next by forcing the fixes that made the next one cheaper. I held wave three in front of everyone on an ETA-accuracy miss traced to one bad data source, while sales wanted the pool boarded. We fixed the source, built a kill switch for that class of source, and boarded a week later.

Result. Sixty-eight customers, two months early. Cost-to-serve from sixty-six cents to three cents a container — a ninety-five percent cut — while the base grew about ten times. Ticket response from seventy-two to ninety percent at five times the volume. Unit cost falling while the book grows only happens if the gates keep bad data out."

Supply · Demand · Capacity. **Demand:** customer onboarding — the sales pool, arriving faster than the platform could absorb. **Supply:** carrier and port data feeds — the raw material; bad feeds were the defect. **Capacity:** the support and data layer's ability to serve customers, measured as cost-to-serve per container. **What bound:** capacity to serve at quality. The gates metered demand to capacity and forced capacity to grow cheaper than demand.

What was innovative. The usual scaling plan is a headcount curve. I inverted it: admission was gated on the platform's measured ability to serve at the current load, so scale was earned, wave by wave, and the cost curve bent down instead of up. The public hold on wave three was the mechanism teaching itself.

The math

Cost-to-serve (CTS) = (support + data-ops cost for the platform) ÷ containers tracked, monthly
start: \$0.66/container at 7 customers ; end: \$0.03 at 68 customers
decomposition: containers ×10 at flat cost → \$0.066 ; actual \$0.03 → cost base fell ~55% as well
Gate per wave (all six must pass at CURRENT load):
data quality ≥ 90% vs source · milestone/ETA accuracy ≥ 80% · verification SLA · engagement floor · stability · CSAT
Tickets per customer (wave w) ≈ f(data quality, ETA accuracy) → fell wave over wave → response 72% → 90% at 5× volume
Wave-3 hold: ETA accuracy 76% vs 80% gate ; one source contributed [70%] of misses → fix + kill switch → 1 week

Solution design / flow

```
carrier / port / forwarder feeds → ingestion → data-quality scoring → visibility → customer
                                   | (kill switch per source class)           |
                                   ▼                                           ▼
                                graduation scorecard (6 criteria, per wave) → admit / hold wave (public)
                                                                                   ↳ CTS per container,
                                                                                   tickets per customer
```

How it translated to a WBR

Outputs: CTS per container; customers live vs plan; ticket response %. **Inputs:** six gate criteria by wave; tickets per customer; bad-source kill-switch activations. **Red rule:** a wave boards only on a green scorecard; a hold is announced with the criterion and the fix date in the same line.

SITUATION PROBES

Why important? Cost of nothing? Boarding sixty customers onto a platform that cost sixty-six cents a container would have been a profitable product turned into a support organization. The pilot economics did not survive multiplication.

Who set the 68 and the date? Sales and leadership. I accepted both and added the two conditions — cost down, response up — so "68" could not be met by burning the team.

ACTION PROBES

Why those six criteria? Each was a leading indicator of a support ticket: bad data → a ticket; bad ETA → a ticket; low engagement → churn; instability → a ticket; CSAT → the lagging check. If a wave cleared all six, it generated fewer tickets per customer than the previous wave. That is the whole cost curve.

Why hold wave three publicly? Sales was furious. A public hold teaches the mechanism; a private hold teaches politics. Sales saw the criteria, saw the miss, saw the fix, saw the wave board a week later. Nobody argued a gate after that.

What did you do versus the team? I wrote and published the criteria; I made the hold call and named the data source; I decided the kill switch was a product requirement, not a support workaround. Engineering built the switch; support ran the waves.

What decision did you challenge? The plan to hire support ahead of the curve. I said: hire after the gate proves what the next wave costs. We hired less than planned.

How did you set priorities with sixty customers waiting? Waves were sequenced by data-source quality, not by deal size — the customers whose carriers had clean feeds boarded first and made the platform cheaper for the harder ones.

RESULTS PROBES

How did you compute cost-to-serve? Fully loaded support and data-ops cost for the platform ÷ containers tracked, monthly. \$0.66 at seven customers, \$0.03 at sixty-eight.

Isn't a 95% drop mostly denominator — ten times the containers? Partly, and I say so. Ten times the containers alone would have given about \$0.07 if cost had stayed flat. It did not stay flat — it fell, because tickets per customer fell wave over wave. The rest of the drop is the gates.

How did ticket response improve at 5× volume? Fewer tickets per customer (gates), plus the kill switch removing a whole class of noisy tickets, plus the verification SLA making response a measured commitment.

Compared to goal? 68: met, two months early. Cost down: −95%. Response up: 72 → 90. All three.

Trade-off? One week on wave three, and a hard conversation with sales. Cheap.

CHALLENGE

Differently? Build the kill switch into the platform before wave one. I knew bad sources existed; I let the first miss force the fix.

At Amazon. New capacity — a launch, a retrofit, an overflow site — graduates into the plan behind published criteria at real volume, and the plan counts only graduated capacity. And the cost side: I would want the capacity plan's cost per unit to fall as the network grows, which only happens if each wave of capacity is admitted at a quality bar rather than a date.

S7 — Flipping a socialized eighteen-month roadmap with a two-column cost-to-serve table

(Flexport)

LPs: Have Backbone; Disagree and Commit · Think Big · Frugality · Earn Trust. **AIQB:** Have Backbone 1, 2, 6, 8 · Think Big 2, 3 · Frugality 4 · Earn Trust 3.

Problem An inherited eighteen-month, operator-first launch sequence, already socialized with leadership. Engineering capacity was fixed; the sequence decided when value arrived and what it cost to serve.

Goal Value to customers sooner, at lower total cost-to-serve, with the same engineering headcount.

Key metric Engineering effort to first value; time to value (months); cost-to-serve reduction (%); cost avoided (\$).

Did it move? **Yes.** Engineering effort −66%. Time to value 6 months vs 18. Cost-to-serve cut ~50% vs ~25% under the original sequence. ~\$5M cost avoidance.

Situation. "I inherited an eighteen-month, operator-first launch plan that leadership had already seen and nodded at. Engineering capacity was fixed. The only real variable was sequence: which half of the problem we solved first.

Task. Get value to customers sooner at lower cost-to-serve, without new headcount — and do it without a public fight that would make my manager choose between me and the plan he had presented.

Action. I costed both sequences in the same currency — cost-to-serve, the number the business was actually paid on — rather than in features. Partner-first cut total cost-to-serve by about fifty percent and shipped value in six months; operator-first cut about twenty-five percent and shipped in eighteen. Two columns, same rows, same assumptions, every assumption footnoted. I did not flip it in a meeting. I took the table to my manager privately and asked him to break it. He pressure-tested it with engineering for two days; it held; and the Bellevue onsite ratified the new sequence. Then I committed completely — the operator work I had deprioritized, I personally protected a date for so the operators were not the losers.

Result. Sixty-six percent less engineering effort to first value. Value in six months instead of eighteen. About five million dollars of cost avoidance. I did not win the argument with seniority. I won it with a table — and the table is why it stayed won."

Supply · Demand · Capacity. **Capacity:** engineering hours — fixed, and the scarce resource. **Demand:** two customer populations, partners and operators, both wanting the platform. **Supply:** the sequence in which fixed capacity was

allocated to demand. **What bound:** capacity allocation. The flip was a capacity-planning decision in a different costume: same hours, re-sequenced against value per hour.

What was innovative. Roadmaps are usually argued in features and dates. I converted the argument into the business's own P&L currency, made the two options structurally identical so only the sequence differed, and chose the private route so the decision-maker could change his mind without losing face. Backbone is the private table; theater is the public ambush.

The math

Two columns, same rows (cost-to-serve over the planning horizon):

	operator-first	partner-first	
months to first value	18	6	
eng effort to first value	[E]	[0.34 E]	→ -66%
CTS reduction at horizon	~25%	~50%	
cost avoided vs baseline	—	≈ \$5M	$= \sum_{\text{months}} (\text{CTS}_{\text{baseline}} - \text{CTS}_{\text{partner-first}}) \times \text{volume}$
			$+ (E - 0.34 E) \times \text{loaded eng rate}$

Sensitivity: ratio held under engineering's two-day pressure test (effort $\pm 20\%$, volume $\pm 15\%$)

Solution design / flow

inherited: operators → screens → (18 mo) → partners
flipped: partners → partner path (6 mo, removes operator interactions at the source)
→ operators (protected date)
decision path: table → manager (private) → engineering pressure test (2 days)
→ Bellevue onsite ratifies → commit

How it translated to a WBR

Outputs: CTS vs plan; monthly operator interactions eliminated. **Inputs:** partner adoption by wave; engineering hours vs plan; operator-date milestone. **Red rule:** if partner adoption trails plan two months running, the table is re-run with actuals and re-presented — the decision stays data-owned.

SITUATION PROBES

Why important? Eighteen months of fixed engineering capacity is the most expensive thing the business owned. Spending it in the wrong order was a cost-to-serve decision nobody had priced.

Cost of nothing? Twelve extra months before customers saw value, and a cost-to-serve reduction half the size. The plan was not wrong; it was unpriced.

How socialized was it? Presented to leadership, dates in the planning system, teams staffed to it. Flipping it meant my manager going back to the people he had presented to.

ACTION PROBES

Why not raise it in the planning meeting? Because a public challenge forces the person who owns the plan to defend it, and then the argument is about him, not the number. Privately, he could attack the table and own the change. He did both.

What if he had said no? I would have asked what number would change his mind, gone to get it, and committed to operator-first in the meantime without relitigating. Disagree once, with data; then commit.

Why not do both in parallel? We did not have the engineers. Parallel would have shipped neither in six months. Sequencing was the whole bet.

How confident were you in the ~50% vs ~25%? Confident in the ratio, less in the absolutes — I said so in the footnotes. Engineering's two-day pressure test moved a few assumptions and the ratio held.

What did you do versus the team? I built the table, chose the currency, chose the private route, and protected the operators' date afterward. Engineering validated the effort estimates.

What decision did you challenge? The plan itself, and the implicit assumption that a socialized plan is a finished plan.

RESULTS PROBES

How did you compute \$5M? Difference in projected cost-to-serve between the two sequences over the planning horizon, plus the engineering effort not spent on the operator-first path to first value. Conservative assumptions, footnoted, reviewed by engineering and finance.

What did the operators lose? Twelve months of waiting for their screens. I owned that trade and protected a date for it. They got a worse year and a better platform.

Compared to goal? Sooner: 6 vs 18. Cheaper: -66% effort, ~50% CTS cut. Same headcount: yes.

CHALLENGE

Differently? Price the customer-side cost explicitly — speed, breach risk — not only cost-to-serve. The table was right; it would have been more right with a third row.

Why this story for Backbone? Because it was a real disagreement with the person who wrote my review, about a plan with his name on it, and I can show the artifact that resolved it.

At Amazon. When Finance says a building is not justified, I do not argue — I price both columns in the same currency: annualized cost of the build against overflow, transfers, premium transportation, speed downgrades and the risk-adjusted cost of a peak miss, with the break-even forecast shown. The six-page narrative this JD asks for is that table with sentences around it.

S8 — Four systems into one pipeline: the weekly review is only as good as the data under it (Flexport)

LPs: Invent and Simplify · Dive Deep · Frugality · Ownership. **AIQB:** Invent and Simplify 1 · Dive Deep 5 · Frugality 5, 6 · Ownership 2.

Problem	The Monday business review ran on four systems stitched together in Excel — by me. Two hours a week, ~60% system accuracy, one person as the single point of failure, and a VP steering on numbers that disagreed with their sources.
Goal	A review whose data is boring: accuracy $\geq 90\%$, zero manual assembly, no single point of failure — without a new team or a new tool budget.
Key metric	Metric accuracy vs source systems (60%); manual hours per week (2); cost of status quo ($>\$100\text{K}/\text{yr}$ priced); parallel-run delta ($<1\%$).
Did it move?	Yes. Accuracy 60% $\rightarrow$ 90%. Manual hours $\rightarrow$ 0. Parallel run within 1% for consecutive weeks before cutover. First automated catch: a 4% ticket-sync drop the manual era never saw.

Situation. "Flexport's Monday review ran on four systems stitched together in Excel by me — two hours every week, about sixty percent agreement with the source systems, and a VP making decisions on it. I was the single point of failure, and I knew the numbers were soft.

Task. Make the data under the review boring: ninety percent accuracy or better, no manual assembly, no dependence on me — and do it without a new team.

Action. I wrote the BRD myself and froze seven metric definitions exactly as the team already used them — no new metrics, no renamed ones, because changing the vocabulary during a migration lets everyone relitigate every red. I priced doing nothing at over a hundred thousand dollars a year in the infrastructure team's own labor rates, so the request competed on their numbers, not mine. I gated cutover on a parallel run: manual versus pipeline within one percent for consecutive weeks, published. Four systems became one Snowflake pipeline. I hit one wall I did not plan for — ticket data carries customer content, and the security review cost two weeks. I owned it upward the day I found it and re-sequenced; the parallel run meant Monday never missed.

Result. Accuracy from sixty to ninety percent. Two manual hours a week to zero. The first automated catch was a four percent ticket-sync drop the manual era had never seen, because the manual era could not see it. The weekly cadence only works if the data underneath it is boring."

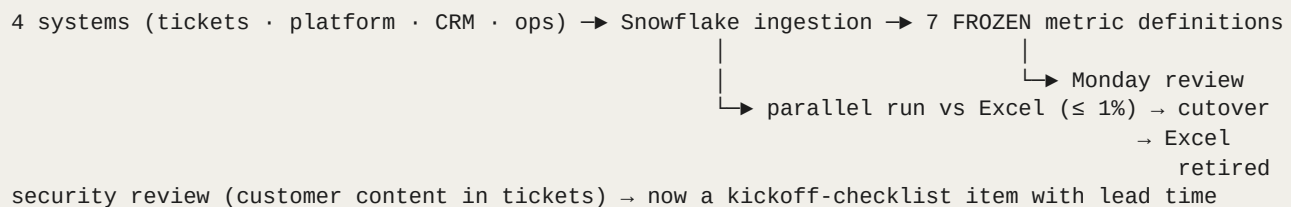
Supply · Demand · Capacity. Demand: decisions the review had to support every Monday. **Supply:** data from four systems. **Capacity:** the review's capacity to see — bounded by two manual hours and one person. **What bound:** the review itself. A WBR is a capacity system for decisions; this story is about expanding its throughput and removing its single point of failure.

What was innovative. Most "automate the dashboard" projects redesign the metrics at the same time and lose the room. I froze the vocabulary as a contract, made the status quo compete on the infrastructure team's own cost, and treated cutover like a WMS go-live — a numeric parallel-run gate. The pipeline was the easy part.

The math

Accuracy	= metric-weeks within tolerance of source-system recompute ÷ metric-weeks 60% → 90%
Status-quo cost	= 2 h/wk × 52 × [loaded rate] + infra team's ad-hoc fixes [h/yr × rate] > \$100K/yr
Parallel-run gate	= manual - pipeline ÷ manual ≤ 1% for each of 7 metrics, [3] consecutive weeks
First catch	= ticket-sync completeness fell 4% (records in source - records in pipeline) ÷ source

Solution design / flow



How it translated to a WBR

Outputs: review metric accuracy; decisions reversed due to data error (target 0). **Inputs:** pipeline freshness; source completeness by system; definition-change requests pending. **Red rule:** any metric whose source completeness drops > 2% is shown with a flag that week, not silently.

SITUATION PROBES

Why important? A VP was steering on numbers that were forty percent wrong somewhere. One Monday the wrong forty percent would be the one that mattered.

Cost of nothing? Two hours a week of my time, priced; the latent cost of a bad decision, unpriced; and a review that stopped the week I was out.

ACTION PROBES

Why freeze the definitions? Some were bad. Some were. I wrote the bad ones down for a later change with its own approval. Changing definitions mid-migration means nobody can tell a pipeline bug from a definition change, and every red becomes an argument about the chart instead of the business.

Why price the status quo in their rates? Because the infrastructure team had to build it, and "it saves me two hours" is not a priority for them. "It costs you \$100K a year of your own people's time" is.

Why a 1% parallel run — why not just switch? Because the review was live. The parallel run is the mock cutover: I would not go live on a WMS without one and I would not go live on the VP's numbers without one.

What went wrong and what did you do? I did not plan the data-security review; ticket text carries customer content. Two weeks. I told my manager the day I found it, with the new date and the reason, and the manual path kept Monday alive. Every data program since has "whose approval does this data need, and what is its lead time" on the kickoff checklist.

What did you do versus the team? BRD, frozen definitions, status-quo pricing, parallel-run gate, the escalation. Infrastructure built the pipeline.

RESULTS PROBES

How did you measure 60% and 90%? Each of the seven metrics, weekly, recomputed from source systems and compared to the review value. Share of metric-weeks within tolerance.

The 4% catch — why does it matter? Ticket sync had been silently dropping records, which made response rates look better than they were. The manual process could not detect a drop it was built on top of. The first week of the pipeline, it flagged. That is the difference between a dashboard and a mechanism.

Compared to goal? 90%: met. Zero manual: met. No SPOF: met. Two weeks late: owned.

CHALLENGE

Differently? Security review on the kickoff checklist. I have never missed it since.

At Amazon. Before I change a single metric in a capacity WBR I would freeze the existing definitions, run the new source in parallel against the old for a published number of weeks, and only then cut over. New metrics come after, through their own door.

S9 — Dual-DC WMS migration in eight months, tolerance published before mock one (Sephora RP JDA · program-management story)

LPs: Learn and Be Curious · Deliver Results · Ownership · Earn Trust. **AIQB:** Learn and Be Curious 1, 2 · Deliver Results 1, 2 · Ownership 2 · Earn Trust 4.

Problem Two DCs on a legacy ERP had to move to JDA WMS in eight months. A \$20M ROI pipeline of four programs was gated behind it. Nothing was documented, and I had never run a WMS migration.

Goal Both DCs live on time and on the \$1.8M budget, with inventory reconciliation within 0.5% at cutover and no disruption to store or dotcom replenishment.

Key metric Inventory reconciliation miss at each mock cutover vs the 0.5% tolerance; schedule; budget; hypercare exit criteria.

Did it move? **Yes.** Mocks: 7% → 1.8% → 0.3% vs 0.5%. On time. ~3% under the \$1.8M budget. \$20M pipeline unblocked. One exit criterion slipped 1.5 weeks — owned, below.

Situation. "Sephora handed me a dual-DC migration from a legacy ERP to JDA WMS on a hard eight-month clock. Four programs worth about twenty million in ROI were gated behind it. Nothing was documented, and I had never run a WMS migration.

Task. Both DCs live on time and on a \$1.8 million budget, with inventory reconciled within half a percent at cutover, and stores and dotcom never noticing.

Action. I did not fake experience. I sat the sponsor for objectives, assumptions and success criteria in week one, and held discovery's written exit until every flow had a documented current state — that was the gate I could run without domain expertise, and it forced the expertise into the room. Then entrance and exit criteria for every phase. The decision I am proudest of: I published the 0.5% inventory reconciliation tolerance, agreed with finance, before mock one — so the first mock would be a measurement, not a verdict. Three mocks: seven percent, then 1.8, then 0.3. The seven went up unvarnished with my name on it. The go/no-go read the scorecard, not the mood.

Result. On time. About three percent under budget. 0.3% against a 0.5% tolerance at cutover. Four-week hypercare with numeric exit criteria. Twenty million of downstream ROI unblocked. One miss: training sign-off, a go-live exit

criterion, slipped a week and a half — I had scored the RF-gun conversion as low effort, my own pre-training test caught the upgrade script failing on newer hardware, and COVID night shifts killed the recovery window. I told leadership it was mine. The program still landed on the date."

Supply · Demand · Capacity. **Demand:** daily store and dotcom replenishment that could not stop. **Supply:** inventory accuracy — the reconciliation tolerance is a supply-truth metric. **Capacity:** two DCs' ability to ship through a cutover weekend. **What bound:** the one-way door. Cutover is capacity you take offline on purpose; the mocks were how I bought it back with a number.

What was innovative. Publishing the tolerance before the first mock inverts the usual politics: a 7% first result is data, not failure, and the trend to 0.3% is the story. And running discovery as a gate I could hold without expertise turned my inexperience into a forcing function for documentation.

The math (program-management story — the math is tolerance and trend)

```
Recon miss (mock k) =  $\sum \text{locations} |\text{expected units} - \text{counted units}| \div \sum \text{expected units}$ 
mock 1: 7.0% (mapping 60% · timing 30% · physical 10%) → burn mapping first
mock 2: 1.8% (timing 70%) → freeze window + cutoff sequencing
mock 3: 0.3% < 0.5% tolerance (finance materiality) → GO
Budget: $1.8M plan, ~3% under ≈ $54K ; cost of a month's slip = 4 programs' ROI delay ≈ $20M ÷
[36 mo] ≈ $0.55M/mo
Training slip: 1.5 wks on a go-live exit criterion; program date held because it was not on the
critical path
```

Solution design / flow

```
discovery (written exit: every flow documented) → design → build
→ mock 1 → mock 2 → mock 3 → cutover → hypercare (4 wk, numeric exit)
tolerance 0.5% published BEFORE mock 1 · each mock classified by cause
· go/no-go reads the scorecard
```

How it translated to a WBR

Outputs: recon miss vs 0.5%; schedule and budget variance. **Inputs:** RAID ageing; exit criteria met per phase; single-point dependencies with a what-if plan. **Red rule:** a single-point dependency in front of an exit criterion without a what-if plan is a red regardless of its size estimate.

SITUATION PROBES

Why important? Four programs and \$20M of ROI were waiting. Every month of slip was a month of four other teams idle.

What was the biggest risk? A cutover that reconciled badly and had to roll back — a week of two DCs shipping from a system of record nobody trusted.

ACTION PROBES

You'd never done this. Why not hand it to a specialist? I used specialists for the WMS. I owned the gates, the tolerance, and the go/no-go — which is the part that needs an owner, not an expert.

How did you learn fast enough? Discovery was my classroom. Holding the exit until every flow was documented meant the experts had to teach the program what the building did. By mock one I could read a reconciliation report.

Why publish the tolerance before mock one? So the first number could not be argued. If 7% arrives and the tolerance is set afterward, the tolerance becomes 7%.

Why 0.5%? Finance's materiality for inventory at the two buildings. I did not pick it; I got them to sign it.

What did you do about the 7%? Read it as a scorecard: classified every miss by cause — mapping, timing, physical — and burned the top class before mock two. 1.8% was mostly timing; 0.3% was noise.

What did you do versus the team? I set the tolerance and criteria, held the gates, made the go/no-go, and owned the slip. The integrator built; ops counted.

RESULTS PROBES

Compared to goal? On time: met. Budget: 3% under. Tolerance: 0.3 vs 0.5. Training exit: 1.5 weeks late. Three of four, and the fourth owned in writing.

Tell me more about the slip — it's a go-live exit criterion you missed. A single-point dependency — RF guns — in front of an exit criterion, scored low effort because it was "just a script." The script failed on newer hardware. The rule since: any single-point dependency in front of an exit criterion gets a what-if plan regardless of estimated size. It has caught two since.

CHALLENGE

Differently? The what-if rule above, from day one.

Why this story for Learn and Be Curious? Because the learning had a mechanism — discovery as a gate — and a measured outcome, not a book list.

At Amazon. Any one-way-door capacity change — a building cutover, a system migration, a network re-architecture — gets its tolerance published before the first rehearsal, and the rehearsals are read as a trend, not a verdict.

S10 — S/4HANA: one definition of "available," returns as first-class stock, and a cutover with a numeric rollback (Sephora S/4HANA ↔ ISM)

LPs: Ownership · Invent and Simplify · Earn Trust · Dive Deep · Deliver Results · Are Right, A Lot (being wrong).

AIQB: Ownership 1, 3 · Invent and Simplify 1, 3 · Earn Trust 1, 6 · Are Right, A Lot 4, 5 · Deliver Results 2.

Problem	S/4HANA was becoming the inventory system of record for stores, retail DC and dotcom with a warehouse execution system beside it. "Available" meant something different in every system because each definition protected a different number. Customer returns, RTV stock, quality-hold and lost-and-found were being promised to customers because somewhere they counted as available.
Goal	One signed definition per stock state, with one owner and one promising rule; returns handled as first-class inventory; cutover with zero customer-facing incidents and a rollback that is a number, not a paragraph; the legacy inventory system (MCS) retired.
Key metric	Units promised against non-sellable states (returns, QI, blocked, in-transit) → zero; nightly out-of-balance (OOB) between SAP supply and WMS supply within tolerance; customer-facing incidents at go-live; interface sync and p95 latency vs rollback triggers; MCS retired.
Did it move?	Yes. Non-sellable promises → zero, verified in production. Nightly OOB between SAP and WMS inside tolerance with every exception bucketed and owned. Zero customer-facing incidents on staged go-lives. 343-task runbook; rollback triggers never fired. MCS retired. One rule I wrote was wrong and was corrected through the same signed policy.

Situation. "Sephora was standing up S/4HANA as the inventory system of record for stores, the retail DC and dotcom, with a warehouse execution system beside it. The problem was not the software. 'Available' meant something different in every system, because each definition protected a different number — fill rate for stores, labor for the DC, the promise for dotcom, the credit memo for finance. Customer returns, return-to-vendor stock, quality-hold and lost-and-found were being promised to customers because somewhere they counted as available.

Task. One definition per stock state, one owner, one rule about who may promise against it — signed by the business before anyone configured anything. Returns as first-class inventory. A cutover with zero customer-facing incidents. And retire the legacy inventory system.

Action. I redesigned the inventory operating model. Every stock state — available-to-sell, reserved, in-transit, quality-inspection, blocked, lost-and-found — got one definition, one owner, and one promising rule, and I took the whole set to steering with the failure mode attached to each: here is the customer who gets hurt if this state is

promised. Returns got a plant of their own. Return-to-vendor stock-transport orders were modeled as pick-reserve so credit could never run ahead of the physical unit: receive, grade, then restock, scrap or RTV — nothing promiseable until disposition. The daily proof was the available-to-sell computation. S/4HANA was the aggregated book of record; the Inventory Supply Management layer computed supply and demand across store, retail DC and dotcom; and every night it rebuilt ATS from the WMS supply basis — minus pack reserve, minus e-comm pick and pack reserves, minus shipped-not-billed, plus mech reserve and lost-and-found added back — then reconciled it to SAP supply and published the out-of-balance by SKU and DC with a tolerance and a bucket for every unit outside it. If the OOB was not explained by six in the morning, it carried an owner. I ran it as a program: RAID log, a weekly CIO/COO pack across sixty-plus stakeholders on frozen metrics so a red stayed red, and a 343-task weekend cutover runbook with rollback written as a number before the weekend — interface sync under 99% for four hours, or p95 latency over twice budget for one hour. One rule I wrote was wrong: I forbade every channel from promising against units on a truck. It starved dotcom of sellable units while in-transit inventory rose. The data showed the miss; the fix went through the same signed policy — a per-lane in-transit promise window from measured lane data.

Result. Could a unit in quality-hold or on a returns pallet still be promised to a customer? No — verified in production. Zero customer-facing incidents on the staged go-lives. The rollback never fired. MCS came out only after the new definition held in production. And the in-transit correction recovered the sellable units I had hidden, through policy, not a hotfix."

Supply · Demand · Capacity. **Supply:** inventory by state — the point is that only some states are supply. **Demand:** three channels promising against the same pool. **Capacity:** returns are inbound volume the plan did not forecast; quality-hold is storage you cannot sell; in-transit is supply you may or may not count. **What bound:** definitions. A capacity plan that counts non-sellable states as available overstates usable capacity, and the promise lies. Giving each state a definition and an owner is the same move as giving each building type its own ceiling.

What was innovative. Treating "available" as a policy document signed by the business before configuration, with the failure mode written next to each state; modeling RTV as pick-reserve so finance could not out-run the floor; and writing rollback as two numbers so a tired bridge at 2 AM could call it without a meeting. Also: correcting my own wrong rule through the policy, not around it.

The math — available-to-sell (ATS) and the end-of-day reconciliation

This is the computation the program lived on. S/4HANA is the **aggregated book of record** (plant / storage-location stock in MM-IM); the **Inventory Supply Management (ISM)** layer computes supply and demand across **store, retail DC and dotcom** and publishes ATS by channel; the WMS holds the physical and the reserves. Every night the three had to agree, and the number that proved it was the **out-of-balance (OOB)**.

```
STEP 1 – SUPPLY BASIS (from WMS, by SKU × DC, EOD snapshot)
  Supply basis = physical on-hand in sellable locations
                  (excludes QI, blocked, damaged, returns-plant stock)

STEP 2 – AVAILABLE TO SELL (ISM computes by SKU × DC, then splits by channel policy)
  ATS = Supply basis
    - Pack reserve      store/retail orders waved, picked, at pack (committed)
    - Ecomm pick reserve dotcom orders allocated, not yet picked
    - Ecomm pack reserve dotcom orders picked, at pack
    - Shipped-not-billed left the dock, goods issue not yet posted in SAP
                        (gone physically, still in SAP stock)
  + Mech reserve       units parked in automation buffers – physically
                        present and sellable; WMS shows "reserved"
  + Lost & found       found, not yet re-binned – present, sellable once binned

STEP 3 – RECONCILE TO SAP (the OOB)
  SAP supply = SAP unrestricted (MB52 / MARD) – SAP reservations, same SKU × plant
  WMS supply = Supply basis + Shipped-not-billed
```

(add SNB back: SAP still holds it until GI posts)

OOB (units) = SAP supply - WMS supply ; OOB (\$) = OOB × standard cost

Tolerance = $|OOB| \leq 0.5\%$ of SAP supply per SKU × DC

 AND $\sum |OOB \$| \leq [\text{materiality}]$ per DC per day

Buckets = timing (in-flight IDocs) · SNB > 24 h · adjustment posted one side

 · receipt posted one side · L&F ageing

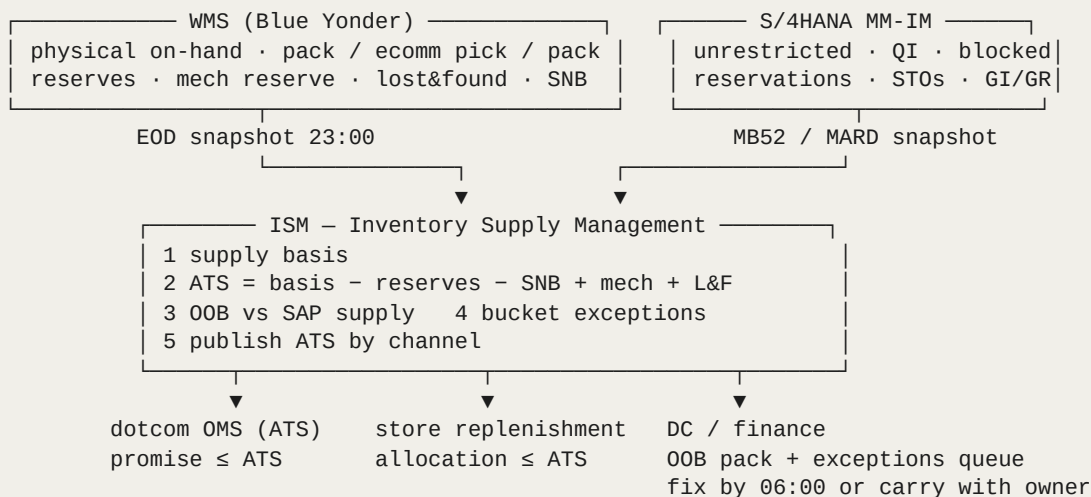
WORKED EXAMPLE — one SKU, one DC, one night (illustrative units)

Supply basis (WMS physical, sellable)	10,000	
- Pack reserve	-1,200	
- Ecomm pick reserve	-800	
- Ecomm pack reserve	-500	
- Shipped-not-billed	-300	
+ Mech reserve	+150	
+ Lost & found	+50	
= ATS	7,400	→ split by channel policy (dotcom floor, store allocation)
SAP unrestricted	10,420	
- SAP reservations (STO / delivery)	-120	
= SAP supply	10,300	
WMS supply = 10,000 + 300 (SNB)	10,300	
OOB = 10,300 - 10,300 = 0	✓	
(if SAP showed 10,420 after reservations → OOB = +120		
→ classify: 90 timing IDoc, 30 adjustment posted one side)		

WHY EACH TERM IS THERE

reserves	subtracted: already promised to someone — selling them twice is the oversell
SNB	subtracted from ATS (it is gone) but added back for the SAP comparison (SAP has not posted the goods issue yet)
mech + L&F	added back: physically here and sellable — WMS "reserved" ≠ "promised"
returns, QI, blocked	NOT in supply basis at all — nothing is promiseable until disposition (the returns plant)

Solution design / flow — EOD reconciliation and ATS publication



STOCK-STATE MODEL (signed before configuration)

state	owner	promiseable?	example
available	ISM	yes (by channel)	stowed, sellable location
reserved	WMS	no (already is)	pack / pick reserves
in-transit	logistics	inside lane window	STO on a truck (my wrong rule: "never" → fixed to per-lane window)
quality-inspect	QA	no	returns awaiting grade
blocked	QA/finance	no	damaged, recall
lost & found	DC ops	after re-bin	found on floor, not in a bin

returns plant returns no until disposition receive → grade → restock /
scrap / RTV (RTV STO = pick-reserve)

How it translated to a WBR

Outputs: oversells (orders promised > physical) — target 0; OOB \$ by DC by day within tolerance; ATS publication by 06:00. **Inputs:** OOB by bucket (timing / SNB ageing / one-sided adjustments / L&F ageing); SNB > 24 h units; L&F > 48 h units; IDoc failure count; in-transit units outside the lane window. **Red rule:** an OOB bucket that repeats three nights running becomes a defect with an owner, not a nightly fix. **Cutover rollback triggers:** interface sync < 99% for 4 h, or p95 latency > 2× budget for 1 h — never fired.

SITUATION PROBES

Why important? Cost of nothing? Every oversell was a cancelled order and a customer who did not trust the site; every return promised before grading was a damaged unit shipped to a second customer. Finance was issuing credit on units the floor had not touched.

What was the initial scope? Configure S/4HANA to match the legacy definitions. I changed it to: define first, configure second, because the legacy definitions were the problem.

ACTION PROBES

Why a signed policy before configuration? Most oversell bugs are vocabulary bugs. A config change nobody signed becomes next quarter's argument. Once the sentence was signed, the design arguments got small — the plant and the STOs were just the implementation of a sentence everyone had agreed to.

Why a separate returns plant? That's heavy. Because a returned unit is not the same supply as a new one until someone has graded it, and the only way to make the system enforce that was to make it physically and logically elsewhere. The heaviness is the point: nothing in that plant is promiseable until it leaves.

Why pick-reserve for RTV? So finance's credit memo could not run ahead of the physical unit. The reservation ties the money to the pick. Before, credit was issued on paper and the unit was sometimes still on a store shelf.

Why rollback as a number? A rollback that is a paragraph gets debated at 2 AM by tired people. A number gets called by whoever is on the bridge. The triggers existed so the weekend could start.

How did you keep sixty stakeholders from re-litigating reds? Frozen metrics in the weekly pack. A red could not be made green by changing the chart; it could only be made green by moving the number.

What did you do versus the team? Wrote the stock-state definitions and the failure modes; designed the returns plant and the pick-reserve RTV model; owned the RAID, the pack, the runbook, and the rollback triggers; made the MCS retirement call. SAP and the integrator configured.

RESULTS PROBES

Walk me through the ATS computation. Supply basis is WMS physical on-hand in sellable locations. Subtract what is already promised — pack reserve for stores, e-comm pick reserve and pack reserve for dotcom — and subtract shipped-not-billed because it has physically left. Add back mech reserve, units parked in automation buffers that WMS flags as reserved but nobody has promised, and lost-and-found, units physically present that will be sellable once re-binned. That is ATS, split by channel policy. Then reconcile: SAP supply is unrestricted minus SAP-side reservations; WMS supply is basis plus shipped-not-billed, because SAP still carries it until the goods issue posts. The difference is the OOB.

Why add back shipped-not-billed for the SAP comparison but subtract it from ATS? Two different questions. ATS asks "what can I promise" — SNB is gone, so it is out. The OOB asks "do the two books agree" — SAP has not posted the goods issue, so SAP still holds it; to compare like with like I add it back on the WMS side. If SNB ages past 24 hours it stops being a timing difference and becomes a defect with an owner.

What did the OOB usually come from? Timing — IDocs in flight at snapshot — was the biggest bucket and self-cleared by morning. Then one-sided adjustments, SNB ageing, and lost-and-found not re-binned. The nightly pack showed the buckets; a bucket that repeated three nights became a defect, not a nightly fix.

How did you verify "non-sellable promises → zero"? Promise log joined to stock state at promise time, daily in hypercare. The count of promises against QI, blocked, returns-plant or lost-and-found stock: zero.

How did you know the in-transit rule was wrong? Sellable units by channel fell while units in transit by lane rose. Two lines diverging is the signal. Fix: a per-lane in-transit promise window from measured lane transit — the 200-lane idea applied to promising instead of planning.

What did the wrong rule cost? Weeks of dotcom running with fewer sellable units than it physically had. I will not quote a dollar I did not measure; I can say the recovery was visible in sellable units the week the window went live.

Compared to goal? Definitions signed: yes. Returns first-class: yes. Zero incidents: yes. Rollback never fired: yes. MCS off: yes. One rule wrong and corrected: owned.

CHALLENGE

Differently? Set the in-transit window from lane data on day one instead of starting from "never." Conservatism has a price, paid in the same currency as risk. And sample the OMS-to-WMS handshake daily in hypercare from day one.

At Amazon. Returns are inbound volume the plan did not forecast; quality-hold is cube you cannot sell; in-transit is supply you may count only inside a measured window. I would want the capacity plan's "available" to be a signed definition by stock state — and I would want to know which states today's utilization number includes.

S11 — The \$2.5M discrepancy I refused to fix quietly (Manhattan Associates)

LPs: Earn Trust · Customer Obsession · Have Backbone. **AIQB:** Earn Trust 6 · Customer Obsession 1, 8 · Have Backbone 1.

Problem An OMS-to-WMS handshake defect had mis-stated a couple hundred customer orders — roughly \$2.5M of order value — and the manager's instinct was to correct the data quietly.

Goal Protect the customer's real priorities and the account, with zero undisclosed corrections.

Key metric Orders corrected to the customer's priority; disclosure before fix (yes/no); account outcome.

Did it move? **Yes.** Full disclosure; the customer prioritized the orders that mattered to their business; the class was bulk-patched; we later won a larger initiative with them.

Situation. "On an OMS-to-WMS handshake we found a discrepancy across a couple hundred orders, roughly two and a half million dollars of order value. **Task.** The manager's instinct was to correct it quietly and move on. I thought the customer had to decide the order of repair, because they knew which orders mattered. **Action.** I argued for full transparency and won it: we showed the customer the list the same day, let them prioritize, and bulk-patched the defect class rather than the instances — so the fix was structural, not cosmetic. It was uncomfortable; I was the junior voice in that room. **Result.** They protected the orders that mattered to their business, the class never recurred, and we later won a larger initiative with them. Silent corrections are loans against trust, and the interest rate is brutal. Same instinct at Flexport later: a truncated week printed 99.4% automation and I discarded it, because a VP steering on three data points is a decision-quality problem."

Supply · Demand · Capacity. **Demand:** the customer's orders, mis-stated. **Supply:** the truth about them. **Capacity:** the account's trust — the capacity to be believed next time. **What bound:** the willingness to show a bad number.

The math

Affected orders = orders where OMS allocation $\neq$ WMS pick confirmation for the same order line (handshake defect class)

count $\approx$ [200] orders ; value = Σ order value $\approx$ \$2.5M

Repair order = customer-ranked (their business priority), not system-ranked (order age)

Fix scope = the defect CLASS (handshake mapping), verified by zero recurrences over [4] weeks — not the 200 instances

How it translated to a WBR

Output: OMS ↔ WMS handshake mismatches per 10K lines (target 0, published to the customer weekly). **Input:** interface error queue age. **Red rule:** any customer-impacting data defect is disclosed with the list before the fix, every time.

Why not fix it and tell them after? Same outcome. Not the same outcome: they would have learned we fix things without telling them. Also not the same repair — they re-ordered the fix by business priority, which we could not have known.

What did the manager say? That it would create noise. I agreed it would, and said the noise now was cheaper than the noise later. He let me make the call and own it.

Why bulk-patch the class, not the instances? Two hundred orders were the symptom. The handshake was the cause. Fixing instances would have made the next two hundred our fault twice.

At Amazon. If the capacity plan is wrong, the WBR hears it from me first with the decomposition attached, before anyone finds it in a speed miss. Reds that arrive pre-decomposed are the version of this story that scales.

S12 — Giving the skeptic the gate (Sephora agile conversion)

LPs: Hire and Develop the Best · Earn Trust · Strive to be Earth's Best Employer. **AIQB:** Hire and Develop 1, 3, 4 · Earth's Best Employer 6, 9 · Earn Trust 3.

Problem A VP sponsored converting a twenty-person product/engineering/QA group from a six-week to a four-week release. The QA lead — the most experienced person in the room — was the most skeptical, and QA was where a faster cycle would break first.

Goal Four-week releases without a quality regression; the mechanism survives my leaving.

Key metric Release cycle (6 → 4 weeks); critical defects per release; whether the gate was still in use after I rolled off.

Did it move? **Yes.** Four-week cycle reached in six weeks. Critical defects –50%. The QA-authored gate was adopted by neighboring teams and outlived my engagement.

Situation. "A VP sponsored moving a twenty-person product, engineering and QA group from six-week to four-week releases. The QA lead was the most skeptical person in the room — and QA was where a faster cycle breaks first.

Task. Four-week releases, no quality regression, and a mechanism that would survive me leaving. **Action.** Instead of managing around her, I gave her the load-bearing artifact: the definition of done with entry criteria, and the authority to hold a release that did not meet it. I coached her on writing a gate that can say no — numeric, published, owned by her — and on how to defend it to engineering without me in the room. Within two cycles she was the one defending it. **Result.** Four-week cycle in six weeks. Critical defects dropped by half, because the gate was stricter than what had existed at six weeks. Neighboring teams copied the model and the ceremony survived my leaving. Development that depends on me staying is not development."

Supply · Demand · Capacity. **Demand:** releases per quarter, up 50%. **Capacity:** the team's throughput at quality.

Supply: the gate — the mechanism that let throughput rise without quality falling. **What bound:** trust in the person who would hold the line.

The math

Release cadence 6 wk → 4 wk = +50% releases per year (8.7 → 13)
Critical defects per release, trailing 3 releases: [4.0] → [2.0] = -50% (gate stricter than
the old 6-week process)
Gate = definition of done with numeric entry criteria (test pass %, open criticals
= 0, perf within budget), owned by QA lead

Durability test = gate still in use [N] quarters after I rolled off; copied by [2] neighboring teams

How it translated to a WBR

Outputs: releases on cadence; critical defects per release. **Inputs:** gate criteria pass rate; exceptions granted (with reason). **Red rule:** an exception without a documented reason code is a red on the release, not on QA.

Why the skeptic? Why not the person most bought in? The skeptic knew where it would break. Giving her the gate converted her objections into criteria, and her credibility made the gate real to engineering.

What specifically did you coach? How to write a criterion as a number, not an adjective; how to hold a release in a room full of engineers who want to ship; and when to let something through with a documented exception rather than block. She was good at the first two already; the third was the coaching.

What's the honest shape of your people leadership? Direct reports in the classic sense at MetricStream — onshore/offshore delivery teams, SOWs, the go-live chair. Everywhere since, matrix: this twenty-person conversion, UDC with 15+ tracks and 60+ stakeholders, about a hundred people across nine teams sequenced at Flexport. I do not dress matrix leadership up as a headcount I did not hold.

Did an investment in someone not work out? At Flexport I spent a quarter coaching an analyst toward owning the Monday review; he did not want the visibility and said so late. I should have asked earlier what he wanted rather than what I needed. I ask that question in the first month now.

At Amazon. The person who most doubts the capacity plan — in Ops, in Finance — is the person I would ask to write the gate the plan has to clear. Their objections become the criteria, and the plan is stronger for it.

S13 — Two go-lives, one engineering team, no 24-hour days (Sephora)

LPs: Strive to be Earth's Best Employer · Deliver Results · Invent and Simplify. **AIQB:** Earth's Best Employer 3, 10 · Deliver Results 1, 7.

Problem Two programs compressed onto one engineering team with overlapping go-lives; roughly \$200K per month per DC at stake in delay, and a team that would otherwise work around the clock.

Goal Both go-lives on date, with issue resolution moving around the clock and no individual working around the clock.

Key metric Go-live dates met; issue age in the RAID log; hours worked per person per week (held to normal shifts).

Did it move? Yes. Both landed on date; ~\$1M per DC protected; no 24-hour days.

Situation. "Two programs compressed onto one engineering team, with roughly two hundred thousand dollars a month per DC at stake in any slip. The default was heroics — the same six people on every call, day and night. **Task.** Both go-lives on date without burning the team. **Action.** One RAID log across both programs, so an issue existed once and had one owner. One 45-minute daily forum, not two. And a two-pass war room: a day shift that triaged and assigned, a night shift in another time zone that executed and handed back — so issues moved around the clock without anyone working around the clock. I set the rule publicly: no one on both passes. **Result.** Both landed on date, about a million dollars per DC protected, and the issue-age metric in the log fell because work never waited for morning. Heroics are a process miss."

Supply · Demand · Capacity. **Demand:** two programs' issues. **Capacity:** one team's hours — fixed. **Supply:** the two-pass structure that made the same hours cover twice the clock. **What bound:** hours per day, not people.

The math

Cost of delay ≈ \$200K / month / DC ; single-pass plan projected [~5] months of slip → ≈ \$1M per DC protected

Issue age (RAID)	= hours from open to resolved ; two-pass war room: work moves 16–20 h/day with 8-h shifts → age ↓
Rule	= no person on both passes (log shows who touched what, when) ; night decision rights = reversible actions only

How it translated to a WBR

Outputs: both go-live dates; issue age p50/p90. **Inputs:** issues per pass; people appearing on both passes (target 0); decisions escalated to morning. **Red rule:** anyone on both passes in a week is a red on me, not on them.

How did you enforce "no one on both passes"? The RAID log recorded who touched what and when. Anyone appearing on both passes got a conversation with me, not a thank-you.

How did you measure the \$1M per DC? Monthly cost of delay per DC — roughly \$200K — times the months of slip the original single-pass plan projected. Conservative: it ignores the morale cost of the alternative.

What if the night pass hit something only the day pass could decide? Published escalation: one named person per night with decision rights for anything reversible; anything irreversible waited for morning by design. That boundary was written down before the first night.

At Amazon. Peak readiness is the same shape: a 24-hour bridge staffed in passes with written decision rights, so the 4 AM call has an owner who is awake and a boundary they do not cross alone.

S14 — Teaching myself a stack: the LLM triage agent (independent build)

LPs: Learn and Be Curious · Frugality · Insist on the Highest Standards. **AIQB:** Learn and Be Curious 2, 5, 7 · Frugality 5 · Highest Standards 2.

Problem Incident triage cost ~\$25 per incident in analyst time; I did not know the stack to automate it.

Goal Automate triage at a fraction of the cost with accuracy held and every conclusion cited to a source.

Key metric Cost per incident; triage accuracy vs the evaluation set; citation coverage.

Did it move? Yes. \$25 → \$0.40 per incident (–98%); accuracy held against the pre-written evaluation set; 100% citation coverage.

Situation. "Incident triage cost about twenty-five dollars an incident in analyst time. I wanted to automate it and did not know the stack. **Task.** Build it, learn the stack, and hold accuracy — not ship a demo. **Action.** Python, FastAPI, tool use. The discipline I brought from supply chain: I wrote the evaluation set before the agent — a hundred-plus real incidents with known-correct triage — and every prompt or model change was gated on the known past still passing. Every output had to cite its source or it did not ship. **Result.** Cost per incident from twenty-five dollars to forty cents — a ninety-eight percent cut — accuracy held against the evaluation set, one hundred percent citation coverage. Same operating system as everything else I run: instrument first, then let the scorecard teach."

Use only if asked about learning something new or about AI. It is a side build, not a program you owned at a company. Say that in the first sentence.

The math

Cost per incident	= (LLM tokens × price + compute + retrieval) ÷ incidents ≈ \$0.40 vs analyst ≈ 20 min × [\$75/h] = \$25 → –98%
Accuracy	= correct triage ÷ evaluation set (written BEFORE the agent, [100+] real incidents) – held
Citation coverage	= outputs with ≥ 1 source citation ÷ outputs = 100%
Gate	= no prompt/model change ships unless the evaluation set still passes at ≥ baseline

How did you learn the stack? By building against a scorecard. Documentation plus a failing evaluation set is the fastest teacher I know.

Why the evaluation set first? Because otherwise "it works" means "it worked on the three I tried." The set is the gate; the agent is the candidate.

At Amazon. Any model or tool I bring to capacity planning gets a backtest against last year's weeks before it touches the plan. The evaluation set is the mock cutover.

S15 — A scope cut that was priced, published, and scored (Flexport FF App export launch)

LPs: Deliver Results · Bias for Action · Customer Obsession · Frugality. **AIQB:** Deliver Results 1, 4 · Bias for Action 1 · Frugality 2, 4.

Problem An export-side launch for forwarding partners with a hard date and a scope that could not all ship: Air and LCL, operator screens, and the partner path competed for the same engineers.

Goal Ship the MVP on date; eliminate the most monthly operator interactions per engineering hour; migrate existing import partners without churn.

Key metric Monthly operator interactions eliminated (the success metric set before scope); partners live; migration rate of import partners.

Did it move? Yes. MVP April 2022; 11 partners, 10 countries; 48 of 53 import partners migrated within a month. Air/LCL and operator screens deferred — openly, with dates.

Situation. "The export launch was minimum-scope by design: partner path first; Air and LCL waited a year; operators kept their old screens two more quarters. Real cuts, real complaints. **Task.** Ship on date and make every cut defensible. **Action.** I set the success metric before scope — monthly operator interactions eliminated, not features shipped — so every proposed cut was scored against that number, in a published table, not in a hallway. Then a burn-up tripwire: any new work was estimated and swapped for something of equal size, never absorbed. **Result.** MVP landed April 2022 — eleven partners, ten countries — and forty-eight of fifty-three import partners migrated within a month. The people who lost scope could read why in the table."

Supply · Demand · Capacity. **Capacity:** engineering hours to the date — fixed. **Demand:** three constituencies' features. **Supply:** the scored scope. **What bound:** hours; the metric decided who got them.

The math

Success metric (set before scope)	= monthly operator interactions eliminated
Scope score (feature)	= interactions eliminated ÷ engineering weeks → rank; cut below the line
Migration	= 48 ÷ 53 import partners in a month = 90.6%
Burn-up tripwire	= any added work estimated and swapped for equal size; absorbed work = 0 by rule

How it translated to a WBR

Outputs: operator interactions eliminated vs plan; partners live. **Inputs:** scope added vs swapped; partner migration rate. **Red rule:** scope absorbed without a swap is a red on the program, shown on the burn-up.

Why interactions eliminated and not revenue? Revenue lagged by quarters and was attributable to many things; operator interactions were measurable monthly and were the cost we were paid to remove.

How did you keep the door shut after launch? The burn-up chart was public; any line added had to show the line removed. Twice someone tried to absorb work; both times the chart made it visible within a week.

At Amazon. A capacity plan's scope is its lever list. Every lever added after the plan is published gets a cost and swaps for one of equal cost — or the plan is a wish list.

S16 — A thousand routes, one gate (Elementum EDI)

LPs: Insist on the Highest Standards · Deliver Results · Dive Deep. **AIQB:** Highest Standards 1, 3 · Deliver Results 3 · Dive Deep 5.

Problem ~1,000 supplier and carrier routes to onboard onto X12 EDI with SAP PO and ASN integration; shipment completeness at 60%, so planners could not see what was coming.

Goal All routes live with completeness $\geq 95\%$; no cohort graduates on bad data.

Key metric Valid ASN rate per cohort (gate $\geq 95\%$); shipment completeness (60% baseline).

Did it move? **Yes.** Completeness 60% $\rightarrow$ 95%. Every cohort graduated at $\geq 95\%$ valid ASNs. Routing-guide overrides carried reason codes.

Situation. "About a thousand supplier and carrier routes had to go onto X12 EDI with SAP PO and ASN integration; shipment completeness sat at sixty percent, so the plan was blind to forty percent of what was in motion. **Task.** All routes live, completeness to ninety-five, and no cohort allowed through on bad data. **Action.** Cohorts graduated only at ninety-five percent or better valid ASNs — one gate, published, no exceptions by relationship. And a detail that mattered: the routing-guide skeptics were often right about specific lanes, so I made overrides legitimate — every override carried a reason code, which turned sabotage into data and gave me the lane list to fix. **Result.** Shipment completeness from sixty to ninety-five percent. Every cohort graduated at the bar. The reason codes became the backlog for the routing guide's next version."

Supply · Demand · Capacity. **Supply:** inbound shipments — visible only if the ASN is valid. **Demand:** planners' need to see supply in motion. **Capacity:** inbound planning capacity is bounded by visibility — you cannot schedule a dock for a shipment you cannot see. **What bound:** data completeness.

The math

Valid ASN rate (cohort)	= ASNs passing X12 validation + PO match $\div$ shipments ; gate $\geq 95\%$ to graduate
Shipment completeness	= shipments with a valid ASN before arrival $\div$ shipments 60% $\rightarrow$ 95%
Override reason codes	= overrides with code $\div$ overrides = 100% ; top codes $\rightarrow$ routing-guide backlog
Residual	= 5% edge formats, documented with owner (cost to fix > visibility gained)

Solution design / flow

```
supplier / carrier  $\rightarrow$  X12 EDI (856 ASN, 850/855 PO)  $\rightarrow$  SAP PO  $\rightarrow$  validation (format · PO · qty)
├─ pass  $\rightarrow$  ASN accepted
└─ fail  $\rightarrow$  supplier scorecard
    └─  $\rightarrow$  cohort gate ( $\geq 95\%$ )

routing guide  $\rightarrow$  override + reason code  $\rightarrow$  lane fix backlog
```

How it translated to a WBR

Outputs: shipment completeness; receiving dock surprises (shipments with no ASN). **Inputs:** valid ASN rate by cohort; override reason-code mix. **Red rule:** a cohort below 95% does not graduate regardless of the relationship.

Why 95% and not 100%? The last five percent were edge formats that cost more to fix than the visibility they bought; I documented them as a known residual with an owner instead of pretending the gate was 100%.

Why let people override the routing guide? Because they were going to anyway. A reason code made the override visible and often correct; forbidding it would have made it invisible and always suspicious.

At Amazon. Inbound capacity planning depends on ASN accuracy and appointment adherence; I would gate vendor cohorts on measured ASN validity and treat every dock-appointment override as a reason-coded data point about where the plan is wrong.

6. Leadership Principles — which story, which hook

They will not say the LP's name. They will say "tell me about a time." Hear the LP, pick the row, speak the hook, tell the story, stop.

LP	LEAD / BACKUP	THE HOOK SENTENCE
Customer Obsession	S3 / S1	"I treated the brochure as a lie the customer was paying for in inventory." / "I will not ship a customer-facing miss to protect a robotics narrative."
Ownership	S3 / S10	"Nobody owned the seam between brochure lead times and the plan. I took the seam, not the ticket." / "Nobody owned what 'available' meant. I wrote the definitions, took them to steering with the failure mode attached, and owned the cutover."
Invent and Simplify	S8 / S10	"New metrics without killing old ones is complexity wearing a badge. I froze seven definitions and killed the spreadsheet." / "Most oversell bugs are vocabulary bugs. One definition, one owner, one rule per stock state — then the design arguments got small."
Are Right, A Lot	S2 / S10	"70% is enough when waiting is a customer miss." / "I forbade promising against in-transit units. I was wrong, the data said so, and the fix went through the same signed policy."
Learn and Be Curious	S9 / S14	"I had never run a WMS migration. I did not fake it. I built the gate and used the specialists."
Hire and Develop the Best	S12	"I gave the skeptic the gate. Development that depends on me staying is not development."
Insist on the Highest Standards	S6 / S1	"I held wave three in front of everyone. Standards without a stop-the-line are posters."
Think Big	S3 / S7	"The ask was a report. The product was lane-level planning as a system." / "Think Big without sequencing is a slogan."
Bias for Action	S2 / S1	"Timestamped decision, then a post-mortem that makes silence an alarm." / "Throttling the robots was a two-way door we had built. I turned it the same day." <i>One-way doors (S/4HANA cutover, JDA cutover) get rehearsals and a numeric rollback instead.</i>
Frugality	S6 / S7	"Cost-to-serve down 95% while the book grew ten times. Headcount would have scaled the waste."
Earn Trust	S11 / S10	"Silent corrections are loans against trust." / "Sixty stakeholders, one weekly pack, metrics frozen so nobody could re-litigate a red by changing the chart."
Dive Deep	S5 / S8	"I pull until the unit is a shipment, an order, a lane or a building — not a national average."
Have Backbone; Disagree and Commit	S7	"Backbone is the private table. Theater is the public ambush."

LP	LEAD / BACKUP	THE HOOK SENTENCE
Deliver Results	S9 / S10	"Results with a rollback are not results. The gates are why it counted." / "343-task runbook, rollback as a number, never fired, legacy system off, zero customer incidents."
Strive to be Earth's Best Employer	S13 / S12	"Heroics are a process miss."
Success and Scale Bring Broad Responsibility	S6	"We scaled ten times and ticket response improved. The cost of scale was paid in gates, not in someone else's backlog."

The follow-up discipline

- **"What did you do, specifically?"** One sentence that starts with "I." Name the artifact: the gate, the table, the log, the SQL.
- **"Why not [the obvious alternative]?"** Name the cost of the alternative in the same currency as the result.
- **"How did you measure?"** Metric, grain, baseline, target. Never "we saw improvement."
- **"What would you do differently?"** A real change you have since made. Not "nothing" and not "communicate more."
- **"Walk me through the data pull."** Tables joined, grain, filters. If you did not pull it yourself, say who did and what you checked.

6B. Every question in the bank — answered

All 128 questions from Amazon's interviewer question bank, in the bank's order. Each gets: the story, a four-to-six-sentence STAR answer you can say as-is, the two probes most likely to follow with one-line answers, and the line that matters for this seat — **how it translated to a WBR, a solution design, or a flow**. Full STAR, math and diagrams are in Section 5 under the story number. Questions marked (*Manager*) in the bank are answered with matrix leadership — say "matrix" out loud; do not invent a headcount.

Format of each line: **Story** → Answer → **Probes** → **WBR / design**.

Are Right, A Lot

AR-1. Didn't have enough data to make the right decision. What path did you take? Did it turn out correct?

S2. Cyber Monday 4 AM: backorders climbing across three DCs and a hundred stores, root cause an hour away. I separated what had to be decided now from what could be learned in parallel: ranked open orders by promise-at-risk — dollar times hours to cutoff — and hand-assigned them to nodes with hours, every move timestamped; I kept one-way doors (promise downgrade, emergency inbound) off the table. The fault was a dead scheduler, not capacity. Fill held at 99%, about \$8M shipped to promise. Correct, and reversible if it had not been.

Probe "What would have made you wait?" — If the only available actions were one-way doors. **Probe** "How did you know inventory was fine?" — Stock by node was on the dashboard; the queue was growing with stock sitting still, which pointed at the engine.

WBR / design: scheduler heartbeat as an input metric; "queue rising + stock flat" as a pre-wired red; every lever labeled one-way/two-way before peak.

AR-2. A strategic decision without clear data or benchmarks. Alternatives, trade-offs, risk mitigation?

S7. No benchmark existed for partner-first vs operator-first at Flexport. I built the benchmark: both sequences costed in cost-to-serve with identical rows and footnoted assumptions. Alternatives: operator-first (18 months, ~25% CTS cut), partner-first (6 months, ~50%), parallel (no engineers). Trade-off: operators waited a year. Mitigation: engineering pressure-tested the table for two days before anyone saw it; I protected a dated operator milestone. Result ~66% effort, ~\$5M avoided.

Probe "How confident in 50 vs 25?" — Confident in the ratio, less in the absolutes; I said so. **Probe** "What if the pressure test had broken it?" — Then I would have committed to operator-first with the table as the reason.

WBR / design: the two-column CTS table is the artifact; it re-runs with actuals when partner adoption trails plan two months running.

AR-3. A difficult decision with input from many sources. How did you arrive at it? Was it correct?

S10. "Available" meant five things to five groups — stores wanted fill rate, DC wanted labor, dotcom wanted the promise, finance wanted the credit memo, returns wanted throughput. I wrote one definition per stock state with the failure mode attached — the customer who gets hurt if this state is promised — and took the set to steering for signature before configuration. Returns got a plant; RTV became pick-reserve. Correct on every state but one: my in-transit rule was too conservative and was fixed through the same policy from lane data.

Probe "Who lost?" — Finance lost the ability to credit ahead of the physical unit; they agreed once they saw the duplicate-credit exposure. **Probe** "How did you know it was correct?" — Promise log joined to stock state: zero promises against non-sellable states.

WBR / design: stock-state model (owner, promiseable?) + nightly ATS/OOB reconciliation; oversells as the output metric, OOB buckets as inputs.

AR-4. A bad decision. Impact? What did you learn and apply?

S10 (in-transit). I forbade every channel from promising against units on a truck. It felt safe. It starved dotcom of sellable units while in-transit inventory rose — two lines diverging. I replaced "never" with a per-lane in-transit promise

window from measured transit, through the same signed policy. Applied since: conservatism is priced in the same currency as risk, and every policy default gets a measured number on day one, not "never."

Probe "How long did it take to notice?" — Weeks; the signal was two metrics diverging, which is now a standing chart.

Probe "Cost?" — I will not quote a dollar I did not measure; sellable units recovered the week the window went live.

WBR / design: "sellable by channel vs in-transit by lane" as a paired chart; in-transit units outside the lane window as an input metric.

AR-5. An error in judgment. Impact? Learned? Applied?

S9 (RF guns). I scored the RF-gun conversion as low effort because it was "just a script." The script failed on newer hardware; COVID night shifts killed the recovery window; training sign-off — a go-live exit criterion — slipped 1.5 weeks. Program still landed on date because it was off the critical path, and I told leadership it was mine. Applied: any single-point dependency in front of an exit criterion gets a what-if plan regardless of size. It has caught two since.

Probe "Why did you under-score it?" — Familiarity: I had seen it done before and did not test it on our hardware. **Probe** "How did you communicate?" — Same day, with the new date and the reason, in the steering pack.

WBR / design: "single-point dependencies without a what-if" as a RAID input metric, red regardless of estimate.

AR-6. Discovered your idea was not the best course of action. How did you find out? What was? Who provided it?

S5. My idea: carriers were the problem; build a routing engine. I pulled three years of shipment history myself and found we could not measure lateness — 55% true data quality at the planning grain, reported as 40%. The best course was measurement first, 99% by defect class, then routing. The data provided it; a finance partner provided the savings method. ~\$8M across 25 locations, auditable per shipment. Learned: audit the measurement before the thesis.

Probe "Who did you have to tell?" — The sponsor who had promised the engine; one page, privately, with three real lanes where the engine would have been wrong. **Probe** "Would you have found it later anyway?" — Yes, as wrong recommendations claimed as savings. Much more expensive.

WBR / design: data quality by defect class as a gating input; savings ledger per shipment; any lane under 99% held out of claimed savings.

AR-7. Brought different perspectives together to solve a problem. Which perspectives? How did you seek them? Outcome? Learnings?

S10. Stores, DC ops, dotcom, finance, returns, SAP and the WMS integrator — each with its own meaning of "available." I ran stock-state workshops where each group had to name the customer harmed if its definition won, then signed one definition per state. Outcome: zero non-sellable promises, zero go-live incidents, legacy system retired. Learning: get the failure mode on the table before the design; it makes the arguments small. Differently: lane data for in-transit from day one.

Probe "Who was hardest?" — Finance, on credit timing; the duplicate-credit exposure converted them. **Probe** "How did you keep it from being a committee?" — Signature by a date; silence counted as consent in writing.

WBR / design: the stock-state table (state · owner · promiseable · example) is the design; OOB and oversells are the review.

Bias for Action

BA-1. A calculated risk where speed was critical. Mitigation? Outcome? Differently?

S1. Robots went live into a window that ended at peak. Week two, pick accuracy printed 94% against a 98% gate I had published before the bots arrived. I throttled the bots to the zones that had cleared 98% — about 30% of volume — the same day, kept the launch green on a certified manual path, and restored full volume in week three. Mitigation was designed in advance: the gate, the dial, the manual path. +15% efficiency, zero incidents. Differently: one zone at real volume for a week first.

Probe "What was the risk you took?" — Going live with bots at all inside a peak window; the dial made it recoverable.

Probe "Who decided?" — Me; the decision right sat with the program.

WBR / design: bot share of volume as an input, pick accuracy by zone by hour as the gate; any zone below 98% two hours running drops out the same shift.

BA-2. Tight deadline, no time to consider all options. How much time? Approach? Learned?

S2. About an hour before the backorder pool doubled again. Approach: decide the reversible things now, learn the irreversible things in parallel. Promise-at-risk ranking, manual node assignment, timestamped log; no promise downgrade, no emergency PO. 99% fill held, ~\$8M protected. Learned: a peak is its own regime — it gets its own readiness checklist, including a scheduler heartbeat and a DST check.

Probe "What option did you not consider that you should have?" — Pausing order intake for ten minutes to let the queue drain; I would weigh it next time. **Probe** "What did the log buy you?" — Clean reconciliation when the scheduler restarted; no double allocation.

WBR / design: allocation queue depth and scheduler age by hour; peak-regime checklist as a dated input.

BA-3. An important business decision without consulting your manager. How did it turn out? Differently?

S1. The throttle. 94% printed on a Tuesday; the gate said what to do; I did it before the steering call and told my sponsor afterward with the zone data. It turned out right — accuracy above 98% in a week, no customer impact. I did not consult because the decision right had been given to the program in writing when the gate was published. Differently: nothing on the decision; I would have shadowed one zone at real volume first.

Probe "What if your sponsor had disagreed?" — The gate was theirs too — they had signed it. The conversation would have been about the gate, not the call. **Probe** "Where is the line?" — Reversible and inside a published rule: act, then inform. Irreversible or outside the rule: ask.

WBR / design: decision rights written next to each gate; "decisions taken under gate authority" reported weekly.

BA-4. Had to gather information and respond immediately. Outcome? Differently?

S2 is the lead; **S11** if they want a customer one: a couple hundred orders mis-stated between OMS and WMS, ~\$2.5M. I pulled the affected list that morning, argued for disclosure over a quiet fix, and had the customer re-rank the repair by their priorities the same day; we bulk-patched the defect class. Outcome: trust kept, class never recurred, larger initiative won later. Differently: nothing on disclosure; faster on the class fix.

Probe "What information did you need before telling the customer?" — The complete list and the cause class; not the fix. **Probe** "Why not fix first?" — They would have re-ordered the fix anyway, and they would have learned we fix quietly.

WBR / design: OMS ↔ WMS mismatches per 10K lines published to the customer weekly — the disclosure built into the cadence.

BA-5. Decide between moving forward or gathering more information. What did you do? What information is necessary before acting?

S4. Planners wanted alerts now; the SAP data under them was not trusted. I chose to gather — three weeks on a daily MD04-to-MB52 reconciliation held at zero — before shipping a single alert, because an alert on untrusted data trains people to ignore alerts. Then I moved fast: live in three months, forecast accuracy 70 → 85%, ~\$2M released. Necessary before acting: the data must reconcile and the action must be reversible. If both hold, act.

Probe "How did you hold planners off for three weeks?" — I showed them the two SAP views disagreeing on the same SKU; they stopped asking. **Probe** "What did you do in the three weeks that was visible?" — Published the drift number daily; falling drift was the progress.

WBR / design: "no alert on a red recon day" rule; recon drift as the gating input metric.

BA-6. Saw an issue that would impact your team and took a proactive approach. Issue? Outcome? Learned?

S8. I was the single point of failure for the Monday review — four systems in Excel, two hours, 60% accuracy — and nobody had asked me to fix it. I wrote the BRD, froze seven definitions, priced the status quo at >\$100K/yr in the infra team's rates, and gated cutover on a 1% parallel run. Accuracy 60 → 90%, manual hours to zero, first catch a 4% sync drop. Learned: price the status quo in the builder's currency and the request prioritizes itself.

Probe "What went wrong?" — Security review on ticket content: two weeks; owned upward the day I found it. **Probe** "Why not a better spreadsheet?" — Excel was the SPOF, not the formula.

WBR / design: the pipeline is the WBR's plumbing; frozen definitions and parallel-run delta are the design gates.

BA-7. (Manager) Your team was not moving to action quickly enough. What did you do? Outcome?

S13. Two go-lives on one engineering team; issues waited overnight because the same six people were on every call and exhausted. I did not ask for more urgency. I changed the structure: one RAID log, one 45-minute daily, and a two-pass war room — a day pass that triages, a night pass in another time zone that executes with written decision rights for reversible actions. Issue age fell, both go-lives landed, ~\$1M per DC protected, no 24-hour days. Matrix team; I owned the structure, not the headcount.

Probe "Why was the team slow?" — Not motivation — a single queue with eight working hours a day. **Probe** "How did you know it worked?" — Issue age p50 and p90 in the log, week over week.

WBR / design: issue age by pass; people on both passes (target 0); decisions escalated to morning.

BA-8. (Manager) Removed a serious roadblock preventing progress. Barrier? How? Outcome? Differently?

S8 (security review) or S5 (sponsor's milestone). Austria: the sponsor had promised a routing engine as milestone one, and the team was being pushed to build on 55% data. The barrier was the commitment. I took one page to the sponsor privately — three lanes where the engine would have been wrong on current data — and he moved the milestone himself. Team spent its time on measurement; 99% by class; ~\$8M auditable. Differently: pull the data before the kickoff deck, not after.

Probe "Why private?" — So he could own the change instead of defending the promise. **Probe** "What did the team see?" — The scorecard by defect class going up weekly; that was their progress metric.

WBR / design: data-quality scorecard by class as the team's weekly output until the engine milestone unlocked.

Customer Obsession

CO-1. A difficult interaction with a customer. How did you deal with it? Outcome? Differently?

S11. Telling a customer we had mis-stated ~200 of their orders, ~\$2.5M, before we had fixed it. Uncomfortable, and my own manager preferred a quiet fix. I showed the full list the same day, let them set the repair order, and fixed the class. They protected what mattered to them; we later won a larger initiative. Differently: nothing on disclosure — I would get to the class fix faster.

Probe "What did the customer say?" — Anger first, then "thank you for telling us" once they saw they could re-prioritize.

Probe "Would you do this at \$25K?" — Yes; the amount changes the meeting, not the rule.

WBR / design: customer-visible defect metric published weekly; disclosure-before-fix as a standing rule.

CO-2. Went above and beyond for a customer. Why? Response? Outcome?

S3. A planner asked for a lead-time variability report. I gave her — and two hundred lanes — a planning parameter that held 99% of the time and released 580K units and ~\$1.5M of safety stock, with a drift flag so it stayed true. Why: the report would have been read once; the problem was the brochure number in the plan. Response: planners adopted it because I gave stock back on lanes that needed it before I took it on lanes that did not.

Probe "Who authorized the bigger scope?" — I pitched leadership with mockups before building; the ask was mine to make. **Probe** "What did it cost?" — My time, and engineering time to productize the drift flag.

WBR / design: lane LT distribution pipeline → planning parameter → drift flag → S&OP review line.

CO-3. Anticipated a customer need with a solution they didn't know they needed. How did you know? Response?

S4. Planners asked for shortage alerts. What they needed was data they could trust — the ERP's demand and stock views disagreed daily — and a reason line they could act on in thirty seconds. I built the reconciliation first, then alerts that explained themselves, then a public false-alarm burn-down. I knew because I watched them dismiss the first prototype's alerts without reading them. Response: the alert entered the weekly S&OP un-mandated.

Probe "How did you measure adoption?" — % of alerts actioned in S&OP, weekly. **Probe** "What did they push back on?" — The three-week wait; the drift number falling daily kept them.

WBR / design: recon drift (input) → alert with reason line → % actioned (input) → stockouts without prior alert (output).

CO-4. Asked for customer feedback and used it to drive improvement. Response?

S4 (false-alarm burn-down). Every alert a planner dismissed got a reason category, and the top category was fixed in a public weekly forum. The dismiss rate was on the wall. Planners saw their feedback change the logic within a week,

which is where trust came from. Also **S16**: routing-guide overrides carried reason codes, so skeptics' feedback became the lane-fix backlog.

Probe "What was the top false-alarm category?" — Timing: right alert, fired a week early; we lag-matched the forecast.

Probe "How did you close the loop?" — The fix and the planner who flagged it were named in the forum.

WBR / design: false-alarm rate and override reason-code mix as input metrics with a weekly burn-down.

CO-5. Evaluated the customer experience of your product or service. What did you do? Result?

S6. The visibility platform's experience was its data: a bad ETA is a bad experience. I made ETA accuracy $\geq 80\%$ and data quality $\geq 90\%$ against source of truth two of six graduation criteria, measured per wave, and held wave three publicly on an ETA miss from one bad source. Result: 68 customers two months early, CSAT held as a gate, ticket response 72 $\rightarrow$ 90% at 5 $\times$ volume.

Probe "How did you measure ETA accuracy?" — Predicted vs actual milestone within a tolerance window, per shipment, per data source.

Probe "What did customers notice?" — Fewer tickets; that is the experience metric that moved.

WBR / design: graduation scorecard by wave; tickets per customer as the experience output.

CO-6. A customer came to you for something that wouldn't address their need. How did you approach it? Result?

S5. The sponsor wanted a routing engine. The need was auditable freight savings, and an engine on 55% data would have produced unauditable claims and wrong routings. I showed three real lanes where the engine would have been wrong, re-sequenced to measurement first, and delivered the engine after 99% quality. Result: ~\$8M across 25 locations, every dollar to a shipment. The sponsor got what he needed, later than he wanted the thing he asked for.

Probe "How did you keep him on side?" — I kept his savings target and his locations; I only moved the first milestone.

Probe "Was he right about anything?" — Yes: some carriers were bad; we could prove which ones only after the data was fixed.

WBR / design: measurement scorecard as milestone one; savings ledger per shipment as the output.

CO-7. Pushed back or said no to a customer request. What did you say or do?

S6 (wave three). Sales — my internal customer — wanted the pool boarded. ETA accuracy was 76% against an 80% gate. I said no, publicly, with the criterion, the source causing it, and the date we would re-test; we boarded a week later with a kill switch built. The no was a date, not a refusal. Also **S15**: operators asked for their screens in the MVP; the scored scope said no for two quarters, in a published table.

Probe "How did Sales react?" — Badly for a week; nobody argued a gate after they saw it work. **Probe** "When would you have said yes?" — At 80%, or with a signed exception naming who eats the tickets.

WBR / design: gate scorecard public to the requester; every "no" carries the criterion and the re-test date.

CO-8. Balanced the needs of the customer with the needs of the business. What did you do? Result?

S1. Business: robots at full volume on the plan date, the ROI the program was sold on. Customer: accurate orders through peak. I throttled bots to the 30% of volume in zones that had cleared 98%, spent three weeks of robotics ROI, and kept zero customer-facing incidents. Then restored full volume once every zone cleared. +15% efficiency on the year. Both got what they needed; the business paid three weeks.

Probe "How did you price the three weeks?" — Incremental manual hours vs one week of a 6-point accuracy miss at peak; an order of magnitude apart. **Probe** "Who signed?" — The gate was signed before the bots arrived; the throttle needed no new signature.

WBR / design: bot share and zone accuracy as inputs; efficiency and incidents as the two outputs that must both be green.

CO-9. (Manager) A change you implemented in your team or organization to meet customer needs. Result?

S10. I changed what "available" meant for the whole inventory organization — one signed definition per stock state, a returns plant, nightly ATS rebuilt from the WMS supply basis and reconciled to SAP with a bucketed out-of-balance. Result: zero promises against non-sellable stock in production, zero go-live incidents, legacy system retired. Matrix: sixty-plus stakeholders, one weekly pack, frozen metrics.

Probe "What did customers see?" — Fewer cancelled orders and no returned units shipped to a second customer.
Probe "How is it still running?" — The nightly OOB pack with owners; a bucket repeating three nights becomes a defect.
WBR / design: $ATS = basis - reserves - SNB + mech + L\&F$; OOB vs SAP supply; oversells as output, buckets as inputs.

Deliver Results

DR-1. Delivered an important project under a tight deadline. Sacrifices? Impact on the deliverable? Outcome?

S9. Dual-DC WMS migration in eight months, \$20M of ROI behind it, nothing documented, first time for me. Sacrifices: I held discovery's exit until every flow was documented, which cost early weeks and bought every later one; training scope was sequenced late and slipped 1.5 weeks. Deliverable was not cut: 0.3% recon vs 0.5% tolerance, on time, ~3% under \$1.8M. The sacrifice was my comfort, not the standard.

Probe "What did you cut?" — Nothing from the tolerance or the mocks; I compressed hypercare to four weeks with numeric exit criteria. **Probe** "What would you not do again?" — Score a dependency as small because it was familiar.

WBR / design: mock trend $7 \rightarrow 1.8 \rightarrow 0.3$ against a pre-published tolerance; exit criteria per phase.

DR-2. Significant unanticipated obstacles to a key goal. Successful? Differently?

S1. 94% pick accuracy in week two of robotics, inside a peak window. The obstacle was not in the plan; the mechanism was. Throttle to cleared zones, manual path carries the rest, zone-logic debug, full volume in week three. Successful: +15%, zero incidents, on date. Differently: one zone at real volume for a week before any customer order.

Probe "Why did testing miss it?" — Test mix under-represented multi-unit orders in dense zones; the miss was in test design. **Probe** "What if the bots had not cleared in week three?" — Manual path through peak; bots after. The plan said so.

WBR / design: zone $\times$ hour accuracy as the gate; test-mix coverage as a readiness input next time.

DR-3. Not only met a goal but considerably exceeded it. How? Challenges?

S6. Goal: 68 customers by date. Delivered two months early with cost-to-serve down 95% (\$0.66 $\rightarrow$ \$0.03) and ticket response up 72 $\rightarrow$ 90% at 5 $\times$ volume — the goal was the number; the exceeding was the direction of the cost curve. How: onboarding gated on six published criteria at current load, so each wave made the next cheaper. Challenge: holding wave three in front of Sales.

Probe "Isn't ~95% mostly denominator?" — Partly; 10 $\times$ containers at flat cost gives \$0.066. Cost also fell ~55% because tickets per customer fell. **Probe** "What did you hire?" — Less than planned; I hired after each wave proved the next one's cost.

WBR / design: CTS per container and tickets per customer by wave; six criteria as inputs.

DR-4. More than halfway to a goal when you realized it may not be the right goal. What did you do? Outcome?

S5. Four weeks into a routing-engine program, the data said the goal — "fix carriers" — was wrong; the problem was measurement. I re-sequenced: measurement to 99% by class first, engine second, savings method signed with finance before any claim. Outcome: ~\$8M auditable. Also **S3:** the goal was a report; halfway through I changed it to a planning parameter. Differently: measurement audit in week one.

Probe "How did you tell the sponsor?" — One page, three lanes, privately. **Probe** "What was the cost of the pivot?" — Engine shipped later than promised; I owned the date in writing.

WBR / design: data-quality scorecard replaced the engine milestone as the first output.

DR-5. A goal you didn't think was achievable. How did you help the team try? Successful? Differently?

S9. Eight months, two DCs, no documentation, no WMS experience on my side — I did not think it was achievable as handed to me. I made it achievable by making it measurable: discovery exit, phase criteria, a tolerance published before mock one so a 7% first result was data, not failure. Three mocks, 0.3%, on time, under budget. Differently: the what-if rule for single-point dependencies from day one.

Probe "What did you tell the team at 7%?" — "This is the scorecard; here are the three cause buckets; we burn the biggest one first." **Probe** "When did you believe it?" — At 1.8%; the trend was the evidence.

WBR / design: recon miss by cause bucket per mock — the trend chart was the team's morale.

DR-6. Did not effectively manage your projects and something was late. Impact? How do you focus on the right deliverables with competing priorities?

S8 (security review). I did not plan the data-security review for ticket content; two weeks late. Impact: none to the business because the parallel run kept Monday alive on the manual path; impact to my credibility, which I repaired by owning it the day I found it. How I focus: one RAID log across programs, every item with an owner and a date; the critical path published; dependencies in front of exit criteria get what-if plans. Lead time of approvals is now a kickoff question.

Probe "Was it really no business impact?" — The pipeline's first catch — a 4% sync drop — arrived two weeks later than it could have. **Probe** "What else did you miss?" — S9's RF guns; same lesson from a different angle.

WBR / design: approvals and their lead times as RAID rows; critical-path items flagged.

DR-7. (Manager) Setting goals that are challenging yet achievable. How did you approach it? Outcome?

S6. The 68 and the date came from Sales. I added two conditions — cost-to-serve down, ticket response up — which made the goal hard in a way the team could win: by building gates, not by burning out. Each wave's criteria were achievable at the current load and challenging because the load grew. Outcome: all three met, two months early. Matrix team across support, data ops and engineering.

Probe "How did you know 90% data quality was the right bar?" — It was the level below which tickets per customer rose sharply in the pilot. **Probe** "Did anyone think it was impossible?" — Support did, at \$0.66; the first two waves' cost curve changed their mind.

WBR / design: per-wave scorecard — goals set as gates the team can see itself clearing.

Dive Deep

DD-1. Understanding a complex problem by digging into details. Who did you talk to, where did you look? How did you use it?

S5. Three years of shipment history, pulled by me — not an analyst — plus the planners who keyed lanes and the finance team that would audit savings. The rows showed carriers under five spellings, lanes at the wrong grain, brochure transit times. I used it to flip the first milestone to measurement and to define the defect classes the 99% bar was scored on. ~\$8M, auditable.

Probe "Why you and not an analyst?" — I was about to tell a sponsor his milestone was wrong; I needed to have seen the rows. **Probe** "What was the most valuable source?" — The field that said where the transit number came from — brochure or scan.

WBR / design: defect-class scorecard; "% transit from scan" as a standing input.

DD-2. Dug deep to a root cause. How did you know you were focusing on the right things? Outcome? Differently?

S2. Backorders climbing; the noise said capacity. I checked stock by node (fine), DC hours (fine), and the allocation queue (growing with stock sitting still) — which pointed at the engine. The trace confirmed a scheduler death at the DST change. I knew I was on the right thing because the queue-vs-stock divergence was the only signal inconsistent with "capacity." Outcome: 99% fill; heartbeat and DST checklist since. Differently: the heartbeat before peak, not after.

Probe "How long to the root cause?" — About an hour for the trace; ten minutes to rule out inventory and hours. **Probe** "What did the post-mortem change?" — Peak readiness checklist with system-level heartbeats, not just physical capacity checks.

WBR / design: "queue depth rising while stock flat" as a pre-wired red; scheduler age as an input.

DD-3. A problem requiring in-depth thought and analysis. How did you know you were focusing on the right things? Outcome?

S3. Two hundred lanes of transit history: per-lane empirical distributions, the choice of the p99 over the mean and over a normal fit, and a finance-signed release rule. I knew I was right because the held-out arrivals landed inside the

planned lead time 99% of the time, lane by lane. 580K units, ~\$1.5M released; drift flag keeps it true. Differently: a measurement audit in week one.

Probe "Why not a normal distribution?" — Right-skewed transits; normal p99 was ~4 days short on long lanes. **Probe** "Thin lanes?" — Pooled by origin region, mode and carrier, flagged lower confidence.

WBR / design: LT accuracy by lane by week; drift flags open; % parameters from scan data.

DD-4. A big problem in your organization you helped solve. How did you become aware? What was missing and how did you fill gaps? Reflection?

S10. Aware through oversells and returned units shipped to second customers — and five groups each certain their "available" was right. Missing: a single definition of each stock state and a nightly proof that SAP and WMS agreed. I filled it with the signed stock-state model and the ATS/OOB reconciliation — basis minus reserves minus shipped-not-billed plus mech reserve and lost-and-found, reconciled to SAP supply, bucketed. Reflection: conservatism has a price; my in-transit rule proved it.

Probe "What was the hardest gap?" — Shipped-not-billed: physically gone, still in SAP; treated differently for ATS and for the OOB. **Probe** "What did the reflection change?" — Measured defaults on day one, never "never."

WBR / design: OOB by bucket nightly; oversells as the output; a bucket repeating three nights becomes a defect.

DD-5. A specific metric you used to identify a need for change. Created or existing? How did it influence the change? Outcome?

S8. Metric: review accuracy — share of metric-weeks where the Monday number matched a recompute from source — 60%. I created it, because nobody had asked how often the VP's numbers were right. It made the case for one pipeline with frozen definitions and a 1% parallel run. Outcome: 90%, zero manual hours, a 4% sync drop caught in week one.

Probe "Why was it 60%?" — Manual joins across four systems with different keys; some weeks the ticket export was truncated. **Probe** "Did anyone dispute the 60%?" — Infra did until they saw the recompute method; then they built the pipeline.

WBR / design: accuracy and source completeness as standing inputs under the review itself.

DD-6. Created a metric that identified a need for change. Why? How did it influence change? Outcome?

S4. Recon drift: $\Sigma[\text{MD04} - \text{MB52}] \div \Sigma \text{MB52}$, daily, by plant. I created it because planners were being asked to trust alerts built on two SAP views that disagreed. Drift at 8–12% made the case to build reconciliation before alerts; drift at zero made the alerts credible. Outcome: forecast accuracy 70 → 85%, ~\$2M released, alerts adopted un-mandated. Also

S3: the per-lane p99 as a planning metric.

Probe "What caused the drift?" — Stale POs counted as supply; duplicated demand from two feeds; timing of goods movements. **Probe** "Is it still measured?" — Daily; "no alert on a red day" depends on it.

WBR / design: drift as the gating input; alert false-alarm rate as the quality input; stockouts-without-alert as the output.

DD-7. (Manager) Validated the assumptions underlying a report's project. How did you decide follow-up was necessary? Steps? Result?

S5 with the team's analyst, matrix. The routing-engine plan assumed the carrier data was good enough; the analyst's quality report said 40%. I followed up because 40% and "good enough" could not both be true. I pulled the rows myself and re-scored at the planning grain by defect class — true 55%, wrong in both directions. Result: the assumption was replaced with a measured scorecard, and the program re-sequenced.

Probe "How did the analyst take it?" — Well; the defect classes became their scorecard and their work was visible weekly. **Probe** "What tells you to validate?" — A number nobody can say how it was computed.

WBR / design: every metric on page one has a computation note; metrics without one are flagged.

Earn Trust

ET-1. Communicated a change in direction people would have concerns with. How did you understand and mitigate concerns? Changes along the way? Did they get comfortable?

S10. Telling finance they could no longer issue credit ahead of the physical unit, and telling the DC that returns would live in a separate plant. I ran the stock-state workshops so each group named the customer harmed by its own definition; finance saw the duplicate-credit exposure, the DC saw returned units shipped to second customers. Changes along the way: the in-transit rule, corrected from lane data when my own version proved wrong. They got comfortable because the policy was signed before configuration and the nightly OOB proved it held.

Probe "What resistance did you not expect?" — Dotcom, who lost in-transit units to my conservative rule; they were right. **Probe** "How did you know they were comfortable?" — Nobody re-litigated a red in the weekly pack; metrics were frozen so they could not.

WBR / design: frozen metrics in the weekly pack; OOB buckets with owners; policy changes through signature, not config.

ET-2. A tough or critical piece of feedback you received. What did you do about it?

Flexport, from my manager after the roadmap flip (**S7**): "You were right, and you nearly made it impossible for me to agree with you" — I had circulated an early draft of the table to a peer before bringing it to him. I apologized, and the rule since is the one I now describe as the principle: backbone is the private table first. Every disagreement since has gone to the decision owner before anyone else. Also from Elementum (**S4**): planners said the first alerts were noise; I added the reason line and the burn-down within a week.

Probe "How did it feel?" — Fair. He was right about the sequence and I had been right about the plan; both were true.

Probe "How do you get feedback now?" — A standing "what would you have done differently" at the end of every steering, including about me.

WBR / design: a "what we got wrong" line in every weekly pack — mine first.

ET-3. Influenced a peer with a differing opinion about a shared goal. What did you do? Outcome?

S12. The QA lead believed four-week releases meant lower quality. We shared the goal — ship, safely — and disagreed on the path. I did not argue; I gave her the definition of done and the authority to hold a release, and coached her on writing a gate that says no with a number. Within two cycles she was defending it to engineering. Critical defects -50%; the gate outlived my engagement.

Probe "What if she had used the gate to block everything?" — Then the exception path with reason codes would have shown it, and we would have fixed the criteria, not her. **Probe** "Why her?" — She knew where it would break.

WBR / design: gate pass rate and exceptions with reason codes as inputs; defects per release as output.

ET-4. Not able to meet a commitment. Obstacles? Impact? Learned?

S9 (training sign-off, 1.5 weeks) or S8 (two weeks on security review). S9: RF-gun conversion failed on newer hardware; COVID night shifts closed the recovery window. Impact: a go-live exit criterion late; stores and dotcom saw nothing because it was off the critical path. I told leadership the day I knew, with the new date and whose miss it was. Learned: what-if plans for single-point dependencies regardless of size.

Probe "Why did you not catch it earlier?" — I scored it as familiar; my own pre-training test is what caught it, just late.

Probe "How did peers react?" — Better than if they had found it themselves.

WBR / design: single-point dependencies in RAID with what-if status; exit-criteria slippage reported with owner.

ET-5. Your team's goals out of alignment with a team you relied on. How did you work with them? Did you achieve your goals?

S8. I needed the infrastructure team to build a pipeline; their goal was platform work, not my Monday review. I priced the status quo in their own labor rates — >\$100K/yr of their ad-hoc fixes — so the pipeline competed on their numbers, and I froze definitions so their build had a fixed target. They built it; accuracy 60 → 90%. Also **S6**: Sales's goal was boarding; mine was cost-to-serve; the gates were the shared instrument.

Probe "What did you give them?" — A BRD with frozen definitions, so scope could not creep on them. **Probe** "What did they give you?" — Two weeks of pain on security review, and a pipeline that caught a 4% drop in week one.

WBR / design: status-quo cost in the builder's currency as the business case; frozen definitions as the contract.

ET-6. Uncovered a significant problem in your team. How did you communicate it? How did you address it? Manage the impact?

S11 (customer-facing) or **S5** (program-facing). **S5**: the program I had joined to fix carriers could not measure lateness — a problem in our own data, not theirs. I communicated it with one page, three lanes, privately to the sponsor, then to the team as a scorecard they could move. Addressed: measurement to 99% by class before any routing. Impact on the team: the engine milestone moved; their weekly progress became the scorecard, so nobody felt idle.

Probe "Did anyone get blamed?" — No; the defect classes were about fields, not people. **Probe** "How did you keep the team motivated on cleanup?" — Visible weekly movement by class, and the engine date tied to it.

WBR / design: defect-class scorecard as the team's weekly output during the pivot.

ET-7. Improved morale and productivity on your team. Underlying problems? How did you prevent recurrence?

S13. Two go-lives on one team; the same six people on every call, nights included; issues waiting for morning. Cause: one queue, eight working hours. I built the two-pass war room with written night decision rights and a rule that nobody works both passes — enforced from the RAID log. Issue age fell; both landed; ~\$1M per DC protected; no 24-hour days. Prevention: the structure is now how I run every compressed program.

Probe "How did you measure morale?" — Hours per person per week in the log, and who showed up on both passes — zero after week one. **Probe** "What did the night pass decide alone?" — Reversible actions; irreversible waited for morning by design.

WBR / design: people on both passes (target 0) and issue age by pass as inputs.

ET-8. A team member struggling to keep up or fit in; you stepped in. Why? What did you do? Impact? Outcome?

S12 (QA lead), reframed: she was not struggling with skill — she was being managed around because she said no. I stepped in by giving her the gate and coaching her to write and defend criteria as numbers. Impact on my work: more time in the first two cycles, far less afterward. Outcome: she owned the mechanism; defects -50%; neighboring teams copied it.

Probe "Why did you think she was struggling?" — She had stopped speaking in design reviews; I asked her why over coffee. **Probe** "What did you learn?" — The person being routed around is usually the one who knows where it breaks.

WBR / design: gate ownership named on the release scorecard — her name, not mine.

ET-9. A team member not performing well and impacting your work. How did you handle it? Why? Outcome? Learned?

Flexport, matrix: an analyst feeding the Monday review (**S8** context) was delivering late and inconsistent numbers. I separated the system problem from the person problem: 60% accuracy was a four-system Excel problem, not his. For the rest, I set a weekly delivery time and a definition of "done" for each metric, reviewed privately for three weeks, and asked what he wanted to own. He did not want the visibility and said so late; he moved to a role that fit. Learned: ask what someone wants in month one, not month four.

Probe "Was it fair to him?" — Fairer once I stopped blaming him for the system. **Probe** "Impact on your work?" — Two extra hours a week for a month; cheaper than a wrong number in front of a VP.

WBR / design: metric-level "done" definitions; delivery time as an input until the pipeline removed the dependency.

Frugality

FR-1. Helped save costs or eliminate waste. Situation? Impact?

S3. Safety stock sized against carrier-brochure lead times — paying for an unmeasured number in inventory on two hundred lanes. I replaced the brochure with the empirical p99 per lane and released stock only where measured spread was tighter than assumed. 580K units and ~\$1.5M out, 99% accuracy held, drift flag keeps it. Also **S4**: ~\$2M under a two-signal rule; **S5**: ~\$8M freight.

Probe "Where did the units go?" — Not re-ordered; the release was in the planning parameter, so the next replenishment cycle simply bought less. **Probe** "Any lane got worse?" — A minority got more stock, deliberately; they had been running short.

WBR / design: safety stock \$ vs plan as output; lanes with open drift flags as input.

FR-2. Completed a project with limited resources. Approach? Impact? Differently?

S13. Two go-lives, one engineering team, no budget for a second. Approach: make the same hours cover more of the clock — two-pass war room across time zones, one RAID log, one daily. Impact: both on date, ~\$1M per DC protected. Differently: write the night decision rights before the first night — I wrote them after the first escalation.

Probe "Why not ask for more people?" — New people on a compressed go-live are a cost for the first month; the structure was faster. **Probe** "What did the time zone give you?" — Sixteen to twenty working hours a day on eight-hour shifts.

WBR / design: issue age by pass; hours per person held to shift.

FR-3. Not enough resources for something important; found a creative way. Options? How did you decide? Outcome?

S8. No budget for a BI team; the Monday review depended on two hours of my Excel. Options: keep stitching (SPOF), buy a tool (budget I did not have), or get the infrastructure team to build a pipeline in their platform. I decided by pricing the status quo in their own rates — >\$100K/yr — so the build was their saving, not my ask. Outcome: Snowflake pipeline, 60 → 90% accuracy, zero manual hours.

Probe "How did you get engineering time?" — By competing on their number, with a frozen scope so the build was bounded. **Probe** "What did it cost?" — Two weeks of a security review I had not planned.

WBR / design: the pipeline itself; status-quo pricing as the business-case template.

FR-4. Trade-offs between quality and cost. How did you weigh them? Result? Differently?

S7. Partner-first was cheaper and faster; operator-first was "complete." I weighed in one currency — cost-to-serve — with the customer-side cost footnoted rather than priced. Partner-first won: -66% effort, ~\$5M avoided, value in six months; operators waited a year for screens. Differently: price the customer-side cost — speed, breach — as a third row, not a footnote.

Probe "What quality did you give up?" — Operator experience for four quarters; I protected a dated milestone so it was a delay, not a cut. **Probe** "Would partner-first have won with the third row?" — Yes, by less; the ratio was robust.

WBR / design: CTS vs plan and operator milestone on the same page so the trade stays visible.

FR-5. A creative solution without additional resources. Problem? Solution? How did you come up with it? Outcome?

S2 (promise-at-risk ranking) or S16 (reason-coded overrides). S16: a thousand routes, a routing guide people ignored, no budget to police it. Solution: make overrides legitimate with a reason code — the skeptics were often right about specific lanes — and turn the codes into the lane-fix backlog. Came from asking why people overrode instead of forbidding it. Outcome: completeness 60 → 95%, every cohort graduated at ≥ 95% valid ASNs.

Probe "Didn't that legitimize ignoring the guide?" — It legitimized telling us why; the guide got better because of it.

Probe "What was the top reason code?" — Lane-specific carrier capacity the guide did not know about.

WBR / design: override reason-code mix as the input that feeds the routing-guide backlog.

FR-6. Requested additional funding or budget. Why needed? Did you try another approach? Did you get it?

S1. I asked to fund a fully staffed manual pick path for three weeks through the robotics ramp — labor the plan had already assumed we would shrink. Other approach considered: delay the bots past peak, which cost calendar and taught us nothing. I priced it against one week of a six-point accuracy miss at peak; an order of magnitude apart. I got it. It paid for itself the day 94% printed.

Probe "What if you had not gotten it?" — Bots at full volume or bots after peak; I would have chosen after peak and said why. **Probe** "Did you use all three weeks?" — Just under; full volume restored in week three.

WBR / design: manual-path hours as a dated input with a sunset; bot share as the lever.

FR-7. (Manager) Decide whether to award or ask for additional resources. Criteria? Outcome? Differently?

S6. Support wanted to hire ahead of the 68-customer curve. My criteria: does the next wave's measured cost per customer justify it, and is there a cheaper fix in the data layer? I held hiring until each wave's scorecard showed the ticket rate; the gates and a kill switch for bad sources cut tickets per customer, and we hired less than planned. CTS \$0.66 → \$0.03. Differently: build the kill switch before wave one.

Probe "How did support feel about being told no?" — Skeptical until wave two's ticket rate fell; then relieved they were not hiring into a mess. **Probe** "When do you say yes?" — When the scorecard shows the cost is structural, not a data defect.

WBR / design: tickets per customer by wave as the hiring trigger — a number, not a request.

FR-8. (Manager) Challenged your team to find a more efficient solution. What drove it? How did you help? Challenges? Results?

S4. The team's first design was a threshold-email alert. I challenged it: an alert on untrusted data is waste — it trains people to ignore alerts. I helped by specifying the reconciliation and the reason line, and by running the false-alarm burn-down forum myself. Challenge: three weeks with nothing visible to planners. Results: 70 → 85% forecast accuracy, ~\$2M released, alerts adopted into S&OP un-mandated.

Probe "What did the team push back on?" — The reconciliation looked like scope creep; the drift number at 8–12% ended that. **Probe** "What did they invent?" — The reason-line logic — naming the input whose delta moved coverage most.

WBR / design: the alert flow with recon gate; false-alarm burn-down as the team's weekly metric.

Have Backbone; Disagree and Commit

HB-1. Strongly disagreed with your manager or peer on something important. How did you handle it? Differently?

S7. My manager had socialized an 18-month operator-first plan. I disagreed on sequence. I built both options in cost-to-serve — identical rows, footnoted assumptions — and took the table to him privately and asked him to break it. He pressure-tested it with engineering for two days; the Bellevue onsite ratified partner-first. ~66% effort, ~\$5M avoided, value in six months. Differently: nothing on the method; I would have priced the customer-side cost as a row.

Probe "What if he'd said no?" — Ask what number would change his mind, go get it, commit to his plan meanwhile.

Probe "Why not in the meeting?" — Public challenge makes it about him, not the number.

WBR / design: the two-column table — the artifact that makes a disagreement data-owned.

HB-2. An unpopular stance in a meeting with peers and your leader. Why strongly? What did you do? Outcome?

S5. Kickoff for a routing engine the sponsor had promised; I said the first milestone should be measurement, because we could not measure lateness. Unpopular with the sponsor and with the team that wanted to build. I felt strongly because a recommendation on 55% data is a coin flip with a confident interface. I had already shown the sponsor privately; in the room I showed the three lanes. Outcome: re-sequenced; 99% by class; ~\$8M auditable.

Probe "Did you lose anything?" — Some credibility with the build team for a month, until the scorecard moved. **Probe** "Would you have committed if overruled?" — Yes, with the savings-method signature as my condition so claims stayed auditable.

WBR / design: quality by defect class as a gate; savings ledger per shipment.

HB-3. Committed to a group decision you disagreed with. What led you to commit? Same decision now?

S15. I argued for a thinner MVP — partner path only, no operator screens for a year. The group kept a minimal operator path in. I committed because the success metric — operator interactions eliminated — had been set before scope, and the group's version scored acceptably against it; my disagreement was taste, not data. MVP landed April 2022, 48 of 53 import partners migrated in a month. Same decision now: yes; the metric was the right arbiter.

Probe "How did you commit visibly?" — I owned the operator-path milestone personally. **Probe** "When would you not commit?" — When the decision contradicts the agreed metric and nobody will change the metric.

WBR / design: success metric set before scope; burn-up tripwire so the compromise did not grow.

HB-4. Felt strongly but the team went another direction. How hard did you press? How did you approach the project afterward?

S10 (in-transit), reversed: I was the one who pressed for "never promise against in-transit," and the team's instinct for a window was right. I pressed hard enough to get it signed. When the data showed dotcom starved, I brought the

divergence chart to the same policy forum and asked for the window from lane data. Afterward: same process, measured default. The lesson is the one I now press for first.

Probe "Give one where you lost and were right." — S15's operator path: I thought it was waste; it scored acceptably and I committed. The metric decides, not me. **Probe** "How hard is too hard?" — Past the point where the other side has heard the number.

WBR / design: sellable-by-channel vs in-transit-by-lane as the divergence chart that reopens a policy.

HB-5. Pushed back against a decision that negatively impacted your team. Issue? Outcome? Differently?

S6 (wave three). Leadership's inclination was to board the pool on Sales's timeline, which would have dumped tickets on the support team. I pushed back with the scorecard: ETA accuracy 76% vs 80%, one source, fix in a week. Outcome: boarded a week later with a kill switch; ticket response rose to 90% at 5× volume. Differently: kill switch before wave one.

Probe "What if leadership had overruled?" — Board with a signed exception naming who absorbs the tickets, and the metric would show it. **Probe** "Did it hurt you?" — With Sales, for a week.

WBR / design: gate scorecard visible to Sales; exceptions signed, not implied.

HB-6. Submitted a great idea your manager did not support. Idea? How did you handle it?

S3. The idea: replace the planning lead-time parameter itself, not just report on variability. My manager's view: scope — deliver the report. I handled it by building the smallest proof — three lanes, brochure vs measured p99 vs actual late arrivals — and pitching leadership with mockups while delivering the report on time. The proof moved him. Result: 200 lanes, 99%, ~\$1.5M released.

Probe "Did you go around him?" — No; the leadership pitch had him in the room. **Probe** "What if the proof had failed?" — Then the report was right and I would have shipped it.

WBR / design: three-lane proof → planning parameter → drift flag — proof before scope.

HB-7. Supported a business initiative you didn't agree with. How did you handle it? How did you deliver the message to your team?

S15. Air and LCL deferred a year, operators kept old screens two quarters — cuts I had argued against in part. Once scored against the pre-set metric, I owned them. To the team: the published table — every cut next to its score — and "the metric decided; here is how we re-open it," with the burn-up tripwire as the door. Nobody was asked to pretend the cuts did not hurt.

Probe "How did the operators take it?" — Badly, then better when they saw their date held. **Probe** "Did anyone re-open it?" — Twice; both times the swap rule made the cost visible and the request withdrew.

WBR / design: scope table with scores; burn-up with swaps — the message and the mechanism are the same page.

HB-8. The business gained because you persisted for a length of time. Why determined? How did it turn out?

S10. Getting one definition of "available" signed by five groups took months of workshops and a returns plant nobody initially wanted. Determined because every oversell was a cancelled order and every ungraded return was a damaged unit to a second customer. Turned out: zero non-sellable promises, nightly OOB in tolerance, legacy system off. Also **S3**: the drift flag took a quarter of persistence after the release.

Probe "When did you nearly stop?" — When finance asked for a credit exception; the duplicate-credit exposure number ended it. **Probe** "What kept the pressure productive?" — A signature date; silence counted as consent in writing.

WBR / design: stock-state model signed; ATS/OOB nightly — persistence encoded as a mechanism.

Hire and Develop the Best

HD-1. Helped a team member develop their career. How? Result?

S12. The QA lead: I gave her the definition of done, the authority to hold a release, and coaching on writing criteria as numbers and defending them in a room of engineers. Result: she owned a mechanism neighboring teams copied; the gate outlived my engagement. Development that depends on me staying is not development.

Probe "What did she get out of it?" — Visibility as the person who made four-week releases safe; a seat in design reviews she had stopped attending. **Probe** "How did you measure development?" — She defended the gate without me in the room by cycle two.

WBR / design: her name on the release scorecard as gate owner.

HD-2. Invested in an employee's development. What and why? Outcome? One that didn't work out?

Worked: **S12.** Did not: Flexport analyst (**S8** context) — I spent a quarter coaching him toward owning the Monday review; he did not want the visibility and said so late. I had invested in what I needed, not what he wanted. He moved to a role that fit. Since then I ask "what do you want to own in a year" in month one.

Probe "What would you have done differently with him?" — Asked earlier; and separated the system problem — 60% accuracy — from his performance sooner. **Probe** "Honest shape of your people leadership?" — Direct reports at MetricStream; matrix since. I say so.

WBR / design: development goals written with the person, reviewed quarterly — same discipline as a metric.

HD-3. Feedback to develop the strengths of someone on your team. Impact on performance?

S12. Her strength was knowing where releases break; her feedback from engineering was "blocker." I told her the strength was right and the form was wrong: turn objections into criteria with numbers, published in advance, so a "no" is a reading, not an opinion. Impact: the same objections became the gate; defects -50%; engineering stopped arguing with her and started arguing with the number.

Probe "How did she take the feedback?" — Relieved; nobody had told her the instinct was right. **Probe** "Did you give feedback to engineering too?" — Yes: stop routing around QA; argue the criteria.

WBR / design: criteria as numbers on the scorecard; exceptions with reason codes.

HD-4. Coaching for a team member. What started it? Outcome?

S16. An EDI onboarding lead was graduating cohorts on relationships — "they're a good supplier" — rather than valid-ASN rate. What started it: a cohort graduated at 88% and the dock found out. I coached on holding the 95% gate regardless of the relationship, and on using reason codes so a hold was a conversation about data. Outcome: every later cohort graduated at $\geq 95\%$; completeness 60 $\rightarrow$ 95%.

Probe "Was the supplier relationship damaged?" — Improved; the supplier got a scorecard instead of a vague complaint. **Probe** "What did the lead learn?" — A gate protects the relationship by making the no impersonal.

WBR / design: valid ASN rate by cohort as the graduation input.

HD-5. (Manager) Someone you hired who complemented your skills. How did you coach where you have a weakness?

MetricStream, where I held direct reports: I hired a delivery lead stronger than me in offshore team operations and hands-on configuration. My weakness was depth in the configuration layer; I coached on what I was strong at — sponsor management, gate design, go-live decision — and let him lead where he was stronger, including in front of the customer. Say this plainly: "I have hired into my weaknesses when I held headcount, and in matrix roles I have given the gate to the person who knew more than me — the QA lead, the WMS specialists."

Probe "How did you know it was a weakness?" — Mock-cutover questions I could not answer without him. **Probe** "How did you avoid becoming redundant?" — I owned the decision; he owned the knowledge; both were needed.

WBR / design: named owners per gate on the scorecard — the owner is whoever knows most, not whoever ranks highest.

HD-6. (Manager) Helped a remote team member develop their career across distance.

S13 context: the night-pass lead in another time zone. I gave the night pass written decision rights for reversible actions, which made the lead a decision-maker rather than an executor, and reviewed the night log with him weekly — what he decided, what he escalated, what he would decide next time. Outcome: by week three he was escalating less and deciding well; the night pass became the model for the next program. Matrix.

Probe "How did you build trust remotely?" — The log: every decision visible, reviewed without blame. **Probe** "What did he get?" — A decision role on his résumé and a reference I have given since.

WBR / design: night-pass decisions vs escalations as a weekly development metric.

HD-7. (Manager) Someone who struggled to get promoted; how did you facilitate their success?

S12: the QA lead's reputation was "blocker," which is not a promotion case. The gate made her the owner of a visible mechanism with a measured result — defects -50%, cadence +50%. I wrote that into the review language and made sure her name, not mine, was on the scorecard neighboring teams copied. Honest framing: I influenced her case; I did not hold the promotion decision.

Probe "What was the barrier?" — Perception; the mechanism changed the evidence. **Probe** "What would you do with headcount?" — The same, with the decision too.

WBR / design: ownership visible on artifacts — the promotion case writes itself from the scorecard.

HD-8. (Manager) Most challenging talent review and promotion process. Factors? Bias tools? Perceptions of fairness? Learned?

Honest answer: MetricStream, onshore/offshore delivery teams. The challenge was comparing people the sponsor saw daily with people they never saw. Factors: delivered outcomes against written criteria, not visibility. Bias tool: I scored against the same written criteria for everyone before reading names, and I made the offshore leads present their own work to the sponsor. Perceptions: published criteria and a calibration conversation, not a private ranking. Learned: visibility is a bias; mechanisms correct it. "Since then my roles have been matrix; I apply the same criteria discipline to gate ownership and recognition."

Probe "Did anyone feel it was unfair?" — One onshore lead, until the criteria were shown to be the same. **Probe** "What would you change?" — Calibrate with a peer manager, not alone.

WBR / design: written criteria first, names second — same as frozen metric definitions.

HD-9. (Manager) Constructed a team to accomplish a goal. Factors? Diversity? Work allocation? Effectiveness? Differently?

S1: fifteen-plus tracks, sixty-plus stakeholders — a matrix team I constructed by track lead. Factors: each track lead owned a gate they could read without me; I paired a skeptic with each optimist; vendor, ops and IT each had a seat on the issues log, not just in meetings. Diversity: of function and of disposition — the people who thought it would fail wrote the criteria. Allocation: by who could say no credibly. Effectiveness: one log, dated gates. Differently: name the zone-logic owner earlier.

Probe "How did you handle a weak track lead?" — Paired them with the gate's co-owner; moved the gate if it did not improve in two cycles. **Probe** "What made it work together?" — The log was the only place a date lived.

WBR / design: owner per gate per track on one page; rows past due as the team-health input.

HD-10. (Manager) A low-performing individual. How did you deliver feedback? Did performance improve?

MetricStream direct report, delivery: missed internal dates repeatedly on a configuration track. Feedback: private, specific — the three dates, the impact on the go-live chain — and a question about what was in the way. The answer was an unclear spec and fear of asking. I fixed the spec and set a weekly check-in with a written "done." Performance improved within a month; he led the next track. If it had not, the next conversation would have been about fit, with HR.

Probe "How did you know it was the spec?" — I asked; and the dates he hit were the ones with clear specs. **Probe** "What if it had been effort?" — Then the written "done" would have shown it in two weeks, and the conversation changes.

WBR / design: written "done" per deliverable, reviewed weekly — the same instrument for a person as for a metric.

Insist on the Highest Standards

HS-1. Refused to compromise your standards around quality or customer service. Who was the customer? Result?

S1. Customer: Sephora's stores and dotcom customers through peak. Standard: 98% pick accuracy by zone before robots carried full volume, published before the bots arrived. At 94% I throttled to cleared zones rather than ride the plan date. Result: above 98% in a week, zero incidents, +15% efficiency. Also **S6:** wave three held on an ETA miss.

Probe "What did it cost?" — Three weeks of robotics ROI. **Probe** "Who pushed you to compromise?" — The robotics timeline; nobody by name, which is why the gate had to be written down.

WBR / design: accuracy by zone by hour vs the gate; bot share as the lever.

HS-2. Unsatisfied with the status quo. What did you do? Impact? Differently?

S3. The status quo was a brochure number in the plan that everyone knew was soft and padded by feel. I measured two hundred lanes and replaced the parameter with the empirical p99, then built a drift flag so the status quo could not quietly return. Impact: 99% accuracy, ~\$1.5M released. Differently: measurement audit in week one of any new domain.

Probe "Why had nobody done it?" — The brochure was free and the pad was invisible; nobody owned the seam. **Probe** "What did the drift flag catch first?" — A lane whose carrier changed mid-year; p99 widened by five days.

WBR / design: % planning parameters measured vs inherited — the standing status-quo check.

HS-3. Improved the quality of something already getting good feedback. Why? How did customers react?

S8. The Monday review got good feedback — it was on time and looked right. I measured it: 60% of metric-weeks matched a recompute from source. I rebuilt it as a pipeline with frozen definitions and a 1% parallel run. Customers — the VP and team — reacted first with surprise at the 60%, then with a 4% ticket-sync drop caught in week one that the old review had never seen. Also **S16:** 95% completeness when 60% "worked."

Probe "How did you know it needed improvement if nobody complained?" — Nobody had checked; I checked. **Probe** "Did the VP trust it less after learning 60%?" — Briefly; the parallel run rebuilt it.

WBR / design: review accuracy as a metric about the review itself.

HS-4. A decision between standards and delivery. Trade-offs? Outcome? Differently?

S9. Mock one at 7% against a 0.5% tolerance, with a hard date. Delivery said "we'll fix in hypercare"; the standard said the tolerance was published for a reason. I kept the tolerance, burned the cause buckets — mapping, then timing — and went live at 0.3% on the date. Trade-off: training scope was sequenced late and slipped. Differently: what-if plan for RF guns.

Probe "What if mock three had been 0.8%?" — No-go, and I had said so before mock one; the sponsor had signed it.

Probe "Where did delivery win?" — Hypercare was four weeks, not eight; numeric exit criteria made it safe to compress.

WBR / design: tolerance published before rehearsal; go/no-go reads the scorecard.

HS-5. A goal you wish you had done better on. How could you have improved?

S10 (in-transit). Goal: every stock state promiseable only under a measured rule. I met it for every state but one — in-transit got "never," not a number, and dotcom paid for weeks. Better: the per-lane window from lane data on day one; I had the method from Elementum and did not apply it. Also **S9** training: 1.5 weeks.

Probe "Why did you default to never?" — It felt safe during a cutover; safety that is not priced is not safety. **Probe** "What tells you a default is wrong now?" — Two lines diverging; I chart them from day one.

WBR / design: in-transit units outside the lane window as an input from day one.

HS-6. Most significant continuous-improvement project you led. Catalyst? Approach? Outcome?

S6. Catalyst: \$0.66 a container at seven customers and sixty waiting — the pilot economics did not survive multiplication. Approach: six graduation criteria per wave, measured at current load, so every wave forced the fixes that made the next cheaper; a public hold when a wave missed. Outcome: \$0.03 a container at 68 customers, response 72 → 90% at 5× volume, two months early.

Probe "What improved wave over wave?" — Tickets per customer, because bad sources were killed and ETA accuracy rose. **Probe** "What was the catalyst for the kill switch?" — Wave three's hold; I should have built it first.

WBR / design: CTS per container, tickets per customer, six criteria by wave.

HS-7. (Manager) Used feedback about your team to drive a change. How gathered? Outcome?

S4. Feedback about the alerting team's product came as dismissed alerts — planners voting with their mouse. I made it formal: every dismissal categorized, top category fixed weekly in public, dismiss rate on the wall. Outcome: false alarms

fell, the alert entered S&OP un-mandated, and the team's design changed — the reason line came from this loop.

Probe "How did the team feel about public dismiss rates?" — Exposed for two weeks; proud by week four when the line fell. **Probe** "What else did you gather?" — S&OP attendance of our alert — whether it was on the agenda.

WBR / design: false-alarm rate by category as the team's quality input.

Invent and Simplify

IS-1. A complex problem solved with a simple solution. What made it complex? How do you know it worked?

S10. Complex: five systems, five meanings of "available," returns and quality-hold leaking into promises, finance crediting ahead of the floor. Simple: one sentence per stock state — definition, owner, promiseable? — signed before configuration; and one nightly computation, $ATS = basis - reserves - SNB + mech + L\&F$, reconciled to SAP with a bucketed OOB. I know it worked because the promise log joined to stock state shows zero non-sellable promises, and the OOB sits in tolerance nightly.

Probe "What did you remove?" — Every channel-specific exception; the plant and STOs implemented one sentence.

Probe "What was hardest to make simple?" — Shipped-not-billed: one fact, two treatments.

WBR / design: stock-state table + ATS/OOB flow (Section 5, S10).

IS-2. The most innovative thing you've done and why. One or two more examples — is it a pattern?

S3: planning lead time at the empirical p99 per lane with a drift flag — the plan reopens itself. Pattern: **S6**, onboarding as capacity gated on measured ability to serve; **S10**, rollback as two numbers so a tired bridge can call it; **S2**, promise-at-risk ranking. The pattern is the same move: replace a judgment call with a measured rule that fires on its own.

Probe "Why is p99 innovative — it's a percentile?" — The innovation is the system: per lane, empirical, with a release rule and a drift detector. The percentile is the easy part. **Probe** "What innovation failed?" — "Never promise in-transit" — a rule without a measurement.

WBR / design: rules with numbers and triggers — the WBR grammar itself.

IS-3. Made something simpler for customers. What drove it? Impact?

S4. Thirty-second reason line on every alert — "coverage falls to 1.6 weeks in week 7 because PO 4471 slipped 9 days" — instead of "coverage alert." Driven by watching planners dismiss alerts unread. Impact: adoption into S&OP, false alarms down, 70 → 85% forecast accuracy, ~\$2M released.

Probe "How did you compute the reason?" — The input whose delta moved coverage most since yesterday. **Probe** "What did it replace?" — A planner opening four SAP transactions.

WBR / design: every red in a WBR carries its reason line — which input moved, whose lever.

IS-4. Drove new thinking and innovation out of your team. A specific innovation?

S16. I asked the EDI team why suppliers overrode the routing guide instead of how to stop them. The team's answer became the innovation: reason-coded overrides, which turned skeptics into a data source and the codes into the routing-guide backlog. Completeness 60 → 95%. Also **S4**: the reason-line logic was the team's design to my requirement.

Probe "What did you do versus the team?" — I asked the question and protected the time; they designed the codes.

Probe "How did you make it safe to propose?" — The first idea that failed was mine, in public.

WBR / design: override reason-code mix as the input feeding the routing-guide backlog.

IS-5. A problem the usual approach wouldn't address. How did you select an alternative? Alternatives considered? Result?

S6. Usual: scale support headcount with customers. Alternatives: hire ahead (cost curve up), slow sales (revenue), or gate onboarding on measured ability to serve at current load. I chose the gates because they made cost fall with scale rather than rise. Result: \$0.66 → \$0.03 per container at 10× the base.

Probe "Why not just slow sales?" — Same outcome for the customer, no improvement in the platform. **Probe** "What did the gates force?" — Bad-source fixes and a kill switch — the actual cost reducers.

WBR / design: graduation scorecard by wave.

IS-6. A novel idea or decision with big business impact. What was novel?

S7. Pricing a roadmap in cost-to-serve rather than features, and making the two options structurally identical so only sequence differed. Novel because roadmaps were argued in dates and demos. Impact: -66% effort, ~\$5M, value in six months instead of eighteen.

Probe "Would anyone else have done it?" — Someone would have argued for partner-first; nobody had priced it. **Probe** "Pattern?" — Same as S3 and S5: convert the argument into a measured currency.

WBR / design: two-column CTS table; re-run with actuals.

IS-7. (Manager) Enabled your team or a team member to implement a significant change. Problem? Success measure? Result?

S12. Problem: six-week releases and a QA function managed around. I enabled the QA lead to implement the gate — her artifact, her authority. Success measure set first: cadence to four weeks, critical defects per release not up. Result: four weeks in six weeks, defects -50%, copied by neighbors.

Probe "What did you remove from her way?" — Engineering's habit of routing around QA; I made the gate the only door. **Probe** "What if defects had risen?" — The gate would have shown which criterion was missing; fix the criterion.

WBR / design: defects per release and gate pass rate on one scorecard with her name.

Learn and Be Curious

LC-1. Realized you needed deeper expertise to do your job well. What did you do? Outcome? Differently?

S9. A WMS migration I had never run. I made discovery a gate — written exit, every flow documented — which forced the experts to teach the program what the building did while I learned enough to read a reconciliation report by mock one. Outcome: 0.3% vs 0.5%, on time, under budget. Differently: learn the RF-device layer too; I left one corner to assumption.

Probe "What did you read?" — Our own flows, the integrator's cutover playbook, and every mock's reconciliation line by line. **Probe** "How did you know you knew enough?" — When I could classify a reconciliation miss by cause without asking.

WBR / design: discovery exit criteria as the learning mechanism.

LC-2. Work outside your comfort area. How did you identify what to learn? How did you build expertise? Did you meet the goal?

S14. Building an LLM triage agent in a stack I did not know. Identified what to learn by writing the evaluation set first — a hundred-plus real incidents — so the gap between what I built and what passed told me what to learn next. Built expertise against a failing scorecard. Met the goal: \$25 → \$0.40 per incident, accuracy held, 100% citations. Side build; I say that.

Probe "Why that project?" — Because it would fail visibly if I was wrong. **Probe** "What transferred to work?" — Evaluation set first is a mock cutover; same discipline.

WBR / design: evaluation set as the gate for every change — backtest before the plan.

LC-3. Didn't know what to do next on a hard problem. How do you learn what you don't know? Options? Decision? Outcome?

S5. Four weeks in, the data said the thesis was wrong and I did not yet know what the right program was. I learn by pulling the rows myself and by asking the people who key them — the planners — what each field meant. Options: clean on the way into the engine, pause, or make measurement the deliverable. I chose measurement because the other two hid the problem. Outcome: 99% by class, ~\$8M auditable.

Probe "How long were you lost?" — About a week between "carriers aren't it" and "measurement is it." **Probe** "Who helped?" — A finance partner who insisted on a signed savings method; that defined the deliverable.

WBR / design: defect-class scorecard.

LC-4. Something you've worked on to improve your work effectiveness. Resources? Impact?

Written narratives over decks. After the roadmap flip (**S7**) I noticed the table did the work and the slides did not, so I moved every steering pack to a one-page written argument with the table on top — the Flexport and Sephora weekly packs (**S10**) are written this way. Resources: Amazon's own public writing on narratives, and feedback from my manager on the first three. Impact: shorter meetings and decisions that stayed decided.

Probe "What was hard?" — Writing "we were wrong" lines; now a standing section. **Probe** "Example of a decision that stayed decided?" — The stock-state policy: signed once, never re-litigated.

WBR / design: weekly pack as a written narrative with frozen metrics.

LC-5. Explored a new or unexpected area of an existing space. Why hadn't it been explored? Why did you proceed? Results?

S3. Lead-time variability was a known space; nobody had looked at the tail per lane because the brochure was free and the pad was invisible. I proceeded because the planner's question implied the plan was running on an unmeasured number. Results: 99%, ~\$1.5M, drift flag.

Probe "What surprised you?" — Most lanes released stock; I expected most to need more. **Probe** "What else was unexplored there?" — Seasonal regimes per lane; the drift flag catches them now.

WBR / design: lane LT distribution pipeline.

LC-6. Someone on your team challenged you to think differently. How did you respond? Outcome?

S10. Dotcom's lead challenged my "never promise in-transit" rule: "you've hidden sellable units from us." I asked for the two charts — sellable by channel, in-transit by lane — and they diverged. I brought it to the policy forum myself and replaced never with a per-lane window from lane data. Outcome: sellable units recovered; the policy process proved it could correct its author.

Probe "How did you respond in the moment?" — "Show me the two lines." Not a defense. **Probe** "What did it change in you?" — Defaults get a number on day one.

WBR / design: divergence chart as a standing input.

LC-7. Used external trends to improve your company's products or services. How did you keep up? How did you decide to apply? Result?

S6 with Amazon as the external example: I studied how Seller Fulfilled Prime turned an adoption problem into eligibility criteria with teeth, and applied the shape — published graduation criteria a wave must clear — to Flexport's onboarding. Keep up: public shareholder letters, carrier and port data standards, and the planners who live in the systems. Result: 68 customers two months early at \$0.03 a container. Also **S14:** LLM tool use applied to triage.

Probe "How did you decide it applied?" — Same structure: a program whose cost rises with uncontrolled admission.

Probe "What did you not copy?" — Penalties; gates were enough.

WBR / design: graduation criteria — the external pattern as a scorecard.

Ownership

OW-1. Took on something significant outside your area. Why important? Outcome?

S10. I was the fulfillment TPM; "what does available mean" sat between stores, finance, dotcom and the DC, and nobody owned it. Important because every oversell and every mis-shipped return traced to it. I wrote the stock-state model, took it to steering, designed the returns plant and the nightly ATS/OOB reconciliation. Outcome: zero non-sellable promises, legacy system off.

Probe "Who should have owned it?" — Everyone said someone else; that was the signal. **Probe** "Did anyone object to you taking it?" — Finance, briefly; the signature process gave them a veto, which is why they accepted it.

WBR / design: stock-state model with named owners — the seam now has an owner per state.

OW-2. Didn't think you'd meet a commitment. How did you identify and communicate the risk? Differently?

S8. The day I learned ticket content needed a security review, the cutover date was dead. I identified it from the data classification, communicated it the same day with the new date, the reason, and the fact that the parallel run kept

Monday alive on the manual path. Differently: "whose approval does this data need and what is its lead time" at kickoff — now a checklist item.

Probe "How did your manager react?" — Fine, because the business impact was zero and the date was already re-planned. **Probe** "What if the parallel run had not existed?" — Then a missed Monday, and a worse conversation.

WBR / design: approval lead times as RAID rows; parallel run as the safety net.

OW-3. An initiative that benefited the company but wasn't any group's responsibility.

S3. The seam between carrier brochure lead times and the inventory plan — carriers owned the brochure, planners owned the plan, nobody owned whether the number was true. I took the seam: measured two hundred lanes, replaced the parameter, built the drift flag. ~\$1.5M, 99%. Also **S8:** the review's data quality.

Probe "How did you get permission?" — Delivered the report I was asked for on time and pitched the parameter change with a three-lane proof. **Probe** "Who owns it now?" — Planning, with the drift flag as their trigger.

WBR / design: drift flags open as a planner-owned input.

OW-4. Sacrificed short-term gain for long-term value. Outcome? Differently?

S4. Three weeks of reconciliation before any planner saw an alert — a visible delay — for alerts that would be trusted for years. Outcome: adopted un-mandated, 70 → 85%, ~\$2M. Also **S1:** three weeks of robotics ROI for zero incidents through peak. Differently on S4: show the drift number falling daily from day one so the sacrifice looked like progress.

Probe "How did you know the long-term value was real?" — Alerts ignored cost more than alerts delayed; I had watched the first prototype get dismissed. **Probe** "Who absorbed the short-term pain?" — Planners, and I told them why in the first meeting.

WBR / design: recon drift published daily — the sacrifice made visible as a metric.

OW-5. Transitioned a project you owned to a new owner; and a time you got involved in one you had transitioned.

Transition: **S12** — the release gate to the QA lead: written criteria, exception path with reason codes, and two cycles of her defending it with me silent in the room before I left. It survived. Got involved again: **S3** — a quarter after handing the lane system to planning, a drift flag fired on a lane and sat two weeks; I stepped in to make the flag's ageing a red on the planner's line, not to decide the lane. Important because the mechanism, not the lane, was at risk.

Probe "How did you keep from taking it back?" — I changed the rule, not the decision. **Probe** "Signs a transition worked?" — Decisions made without you that you would have made.

WBR / design: owner named on the artifact; ageing of open flags as the health check.

OW-6. Saw a peer struggling and stepped in. Why? Actions?

S13. The engineering lead was carrying two go-lives and every night call. Not my program half, but my go-live chain. I stepped in by building the two-pass structure so the clock, not he, absorbed the load, and by taking the day-pass triage myself for the first two weeks. Both go-lives landed; no 24-hour days.

Probe "Did he want help?" — Not at first; the RAID log showing his name on 70% of rows made the case without a conversation about him. **Probe** "What did it cost you?" — Two weeks of early mornings.

WBR / design: rows per owner as a load input; people on both passes as the red.

Think Big

TB-1. Saw an opportunity to do something much bigger than the initial focus. Did you take it? Outcome?

S3. Asked for a report; saw that the plan ran on an unmeasured parameter across two hundred lanes. Took it — pitched leadership with mockups, delivered the report on time as proof of good faith. Outcome: 99%, ~\$1.5M, a drift flag that keeps it true. The ask was a report; the product was lane-level planning as a system.

Probe "What would you have lost by not taking it?" — A report read once; the brochure in the plan for another year.

Probe "What did taking it cost?" — Engineering time to productize, and a quarter of my own.

WBR / design: lane LT pipeline → parameter → drift flag.

TB-2. Changed the direction or view of a function and helped them embrace new thinking. Why? Outcome?

S7. Product and engineering argued roadmaps in features and dates; I moved the argument to cost-to-serve. After the flip, the two-column table became how the group compared options. Outcome: -66% effort, ~\$5M, and a function that priced sequences before debating them.

Probe "How do you know they embraced it?" — The next two roadmap decisions arrived at me as tables. **Probe** "Who resisted?" — Operators' advocates; the protected milestone kept them in.

WBR / design: CTS as the standing currency in roadmap review.

TB-3. Proposed a novel approach. Why novel? Successful?

S10 (returns plant and pick-reserve RTV). Novel because returns are usually a storage location, not a plant; making them a plant made "not promiseable until disposition" a system fact rather than a rule people had to remember, and pick-reserve on RTV meant finance could not credit ahead of the floor. Successful: zero returned units promised before grading; credit tied to the pick.

Probe "Why not a status flag?" — Flags get overridden; a plant does not. **Probe** "Cost of the heavier model?" — More STOs; cheaper than one damaged unit to a second customer.

WBR / design: returns-plant flow: receive → grade → restock / scrap / RTV; L&F and QI ageing as inputs.

TB-4. Drove adoption for your vision. How did you know it was adopted? How did you drive and track it? Differently?

S4. Vision: a supply-demand view planners trust enough to act on in S&OP. Drove adoption by earning it — recon at zero, reason lines, public false-alarm burn-down. Knew it was adopted when the alert was on the S&OP agenda without a mandate; tracked % of alerts actioned weekly. Differently: reason line in the first prototype.

Probe "What was adoption at week four?" — Low; dismissals were the data that fixed it. **Probe** "How is it tracked now?" — % alerts actioned and stockouts without prior alert.

WBR / design: % alerts actioned as the adoption input.

TB-5. An idea adopted by global stakeholders. How did you drive and track adoption? Differently?

S6. Graduation criteria for a global 4PL platform — customers across regions, data sources across carriers and ports. Drove adoption by publishing the six criteria to Sales and leadership and holding wave three publicly. Tracked by scorecard per wave and CTS per container. Also **S16:** a thousand supplier and carrier routes on one gate. Differently: kill switch first.

Probe "What differed by region?" — Data-source quality; waves were sequenced by it, not by geography. **Probe** "How did you know global stakeholders had adopted?" — Regional sales leads started quoting the criteria to customers.

WBR / design: graduation scorecard by wave, global.

TB-6. Thought differently to improve a process that was working. Assumptions questioned? How did you evaluate? Differently?

S8. The Monday review worked — on time, looked right. Assumption questioned: that "looks right" meant right. Measured: 60%. Evaluated the change by a 1% parallel run and by what the pipeline caught that the manual era could not — a 4% sync drop. Differently: security lead time on the checklist.

Probe "Why fix what works?" — A VP steering on 60% is a decision-quality problem waiting for the wrong week. **Probe** "What did you not change?" — The seven definitions; changing them would have hidden the pipeline's accuracy.

WBR / design: review accuracy; frozen definitions; parallel-run delta.

TB-7. (Manager) Encouraged a team member or organization to take a big risk. How did you balance it? Outcome? Learned?

S1. I encouraged the organization to go live with robotics inside a peak window — a big risk — and balanced it with the gate, the dial and the manual path. Outcome: 94% in week two handled as a Tuesday decision; +15%, zero incidents. Learned: the size of a risk you can take is set by the quality of the reversibility you build.

Probe "Who was most afraid?" — Store ops, of mis-shipped replenishment; the zone-level gate addressed their fear specifically. **Probe** "What risk did you refuse?" — Bots at 100% with a two-week "stabilization" that had no number.

WBR / design: bot share and zone accuracy — risk and its control on one page.

TB-8. (Manager) Established a vision where there wasn't one. Buy-in? Execution? Outcome? Differently?

S10. There was no vision for inventory truth across channels — each system had its own. Vision: one definition per stock state, one owner, one promising rule, proven nightly. Buy-in through workshops where each group named the customer harmed by its own definition, then signatures by a date. Execution as a program: RAID, weekly pack, 343-task runbook, numeric rollback. Outcome: zero non-sellable promises, MCS off. Differently: in-transit from lane data on day one.

Probe "How did you know it was a vision and not a config?" — Because the arguments were about words, not fields.

Probe "What did you say to get the first signature?" — "Here is the customer this state hurts if we promise it."

WBR / design: stock-state model + nightly ATS/OOB — vision as a table and a computation.

Strive to Be Earth's Best Employer

BE-1. Helped build a more inclusive working environment. Problems? What did you do? Reactions? How did it help?

S13. The offshore night team executed and the onshore day team decided — the night team was never in the room. I gave the night pass written decision rights for reversible actions and reviewed their decisions weekly without blame. Reaction: escalations fell by week three; the night lead started proposing fixes, not just applying them. It helped because issue age fell and because the people doing the work were making the calls.

Probe "What problem were you solving — inclusion or speed?" — Both; they were the same problem. **Probe** "How did you make the rights real?" — The log recorded night decisions as decisions, reviewed the same way as day ones.

WBR / design: decisions by pass as a visible input; escalations to morning trending down.

BE-2. Advocated for someone with less influence or seniority. Why? Outcome?

S12. The QA lead was being routed around. I advocated by giving her the load-bearing artifact and putting her name on it, and by telling engineering to argue the criteria, not around her. Outcome: she owned the gate; defects -50%; neighbors copied her model. Also MetricStream: offshore leads presenting their own work to the sponsor.

Probe "What did it cost you?" — Two cycles of friction with engineering. **Probe** "Why her and not just fix the process?" — The process was her; fixing it without her would have confirmed the exclusion.

WBR / design: owner names on scorecards — the advocacy is structural.

BE-3. Implemented a change to improve the work experience of yourself or coworkers. Goal? Why necessary? Outcome?

S13. Goal: no one works around the clock while issues move around the clock. Necessary because the same six people were on every call. Outcome: both go-lives on date, ~\$1M per DC, zero 24-hour days, and a structure I reuse. Also **S8:** removing two hours of my own Excel every Monday was a work-experience change for me and the review.

Probe "How did you enforce it?" — The log showed who touched what and when; anyone on both passes heard from me. **Probe** "Did anyone resist resting?" — Yes; the rule made resting a compliance item, which oddly helped.

WBR / design: hours per person and double-pass count as inputs.

BE-4. Promoted diversity, equity or inclusion. What forms of diversity? Approach?

Diversity of function and disposition in **S1** — skeptics write the criteria — and of geography and seniority in **S13** and MetricStream: offshore teams given decision rights and the floor with the sponsor. Approach: structural. I do not ask people to be inclusive; I put the under-heard person's name on the artifact and their decision in the log. Say plainly that this is how you have practiced it; do not claim programs you did not run.

Probe "A specific moment?" — The night lead's first unescalated decision being reviewed as a good call in front of the day team. **Probe** "What did you learn?" — Inclusion that is not written into decision rights evaporates under pressure.

WBR / design: decision rights and owners written down — inclusion as a mechanism.

BE-5. Showed compassion during a stressful situation. How did you recognize it mattered? What did you do? Outcomes?

S9 (RF guns). The track lead whose script failed was convinced he had sunk the program. I recognized it because he stopped speaking in the daily. I told leadership the miss was mine — I had scored it — and told him the recovery plan

was his to run. He ran it; training closed 1.5 weeks late and the program landed. Also **S13**: the two-pass rule was compassion with a schedule.

Probe "Was it really your miss?" — The scoring was; the script was his. Owning the part that was mine let him own his.

Probe "What did he do next?" — Led device readiness on the next DC.

WBR / design: misses attributed to the decision, not the person, in the RAID.

BE-6. Supported or empowered someone to achieve a goal. Your role? Impact on you?

S12. Role: I gave the authority and coached the form; she did the work. Impact on me: two cycles of more time, then much less, and a mechanism that outlived me. Also **S13**'s night lead: decision rights and a weekly review turned an executor into a decision-maker.

Probe "How did you know when to step back?" — When she defended the gate to engineering without looking at me.

Probe "What did you get?" — A release process that did not need me.

WBR / design: gate owner named; my name absent.

BE-7. Organized or participated in an initiative to foster an enjoyable work environment. Reactions?

S1: the 836-row issues log had a ritual — every Friday, rows closed that week were read out with the owner's name, across sixty stakeholders and three vendors. Small, cheap, and people started closing rows on Thursday. **S13**: hand-off notes between passes ended with one thing that went well; the night team kept it after the program. Say only what you actually did; do not inflate this.

Probe "Why does it matter?" — Fourteen months is a long program; a weekly moment of visible progress kept the log honest and the people in it. **Probe** "Reaction?" — Vendors asked to be included.

WBR / design: rows closed by owner, read weekly — recognition as part of the cadence.

BE-8. Noticed someone being excluded or treated unfairly. Could you help?

S12 and MetricStream talent reviews. **S12**: a QA lead routed around because she said no; I gave her the gate.

MetricStream: offshore leads reviewed by a sponsor who never saw them; I had them present their own work and scored everyone against written criteria before reading names. Both helped; both were structural.

Probe "How did you notice?" — She stopped speaking; they were never on the agenda. **Probe** "What did you not do?" — Make it a conversation about fairness; I made it a change to who held what.

WBR / design: criteria before names; owners on artifacts.

BE-9. (Manager) Created an environment where people felt comfortable speaking up with concerns, ideas or mistakes. How do you know it worked?

S9: mock one at 7% went up unvarnished with my name on it; **S10**: my own wrong in-transit rule corrected through the public policy forum; every weekly pack has a "what we got wrong" line, mine first. I know it worked because the RF-gun failure and the dotcom in-transit challenge both came to me early and directly. Matrix.

Probe "A time it did not work?" — The Flexport analyst who said late that he did not want the role; I had not asked.

Probe "How do you measure it?" — Time from a miss occurring to it appearing in the log.

WBR / design: "what we got wrong" as a standing pack section; miss-to-log latency.

BE-10. (Manager) Improved your team's work environment. Why needed? How did you decide steps? Impact?

S13. Needed because the RAID log showed six names on most rows and night calls on all of them. Steps decided from the log: one log, one daily, two passes, written night rights, no double passes. Impact: issue age down, both go-lives on date, no 24-hour days.

Probe "What did the team ask for?" — More people; the structure was faster and I said why. **Probe** "What did you measure?" — Hours per person, rows per owner, issue age by pass.

WBR / design: load per owner as a standing input.

BE-11. (Manager) Influenced a change that fostered DEI on your team or org. Indicators? Measures?

MetricStream and **S13**. Indicators: offshore leads absent from sponsor reviews; night team's decisions recorded as escalations, not decisions. Measures: written criteria scored before names; offshore leads presenting their own work; night decision rights in writing with weekly review. Outcome: decisions and recognition distributed by contribution, visible in the log. Honest framing: these were my teams' practices, not an org-wide program.

Probe "How did you know the indicators were real?" — The agenda and the log; both were countable. **Probe** "What would you scale?" — Criteria-before-names into every review I influence.

WBR / design: who decides and who presents, counted.

Success and Scale Bring Broad Responsibility

SS-1. A project with impact beyond your immediate customer or organization. How did you approach downstream impact? When did you start thinking about it? Risks?

S6. Scaling a visibility platform 10× affects every customer's downstream planners and every carrier whose data we score. I thought about it at wave one: the gates were designed so growth did not dump cost on the support team or bad data on customers. Risk: a wave boarded on bad data would have propagated wrong ETAs into sixty customers' plans. Response 72 → 90% at 5× is the proof the downstream was protected.

Probe "Who outside Flexport felt it?" — Carriers, who got source-quality scorecards and in some cases a kill switch.

Probe "What did you not foresee?" — The need for the kill switch itself.

WBR / design: per-source quality and kill-switch activations — the downstream made visible.

SS-2. Changed a decision because of unintended consequences for customers. Impact? Trade-offs? How did the new solution serve better?

S10 (in-transit). Unintended: dotcom customers saw fewer units for sale than physically existed. I changed "never" to a per-lane window. Trade-off: a small risk of promising a unit on a late truck, priced by lane. Better: sellable units recovered without re-opening the policy's integrity.

Probe "How did you find out?" — Dotcom's lead and two diverging lines. **Probe** "How did you bound the new risk?" — Window set from measured lane transit, same method as the 200 lanes.

WBR / design: in-transit outside window as an input; sellable by channel as the output.

SS-3. Changed course because of negative impact to a third party.

S16. The original plan cut off suppliers whose ASNs failed validation — which would have stopped their shipments and their cash. I changed course to cohort gates with supplier scorecards and reason codes, so a supplier failing got a specific list, not a cutoff. Completeness 60 → 95% without a supplier losing a route. Also **S7**: operators' date protected.

Probe "Did it slow you down?" — A little per cohort; faster overall because suppliers fixed their own data. **Probe** "Who was the third party in S7?" — Operators, inside the company but outside the decision.

WBR / design: supplier scorecard by cohort.

SS-4. Thought carefully about social responsibility in decision-making. Why? How did it affect the decision?

S5. The routing engine's cheapest answer on some lanes was air expedite. Measured properly, the data also showed which lanes could consolidate to ground without a service miss, and I weighted the engine toward consolidation where service held. It affected the decision by making fewer expedited miles a scored outcome, not a slogan. Say this only to the extent you actually did it; otherwise use **S10**: the returns plant's grade step means restock before scrap.

Probe "Did it cost savings?" — Marginally on a few lanes; the \$8M held. **Probe** "How did you measure it?" — Expedite share by lane, before and after.

WBR / design: expedite share and restock-vs-scrap ratio as inputs.

SS-5. Considered everyone affected by your work, not just direct customers.

S10. Direct customers: dotcom shoppers and stores. Also affected: finance's auditors (credit timing), the returns team (grading labor), vendors (RTV timing), and the DC's night shift (cutover weekend). Each had a seat in the stock-state workshops and a row in the runbook. The rollback written as a number was for the night shift specifically — so a tired bridge did not carry a judgment call.

Probe "Who did you almost miss?" — Vendors on RTV timing; the pick-reserve model changed when they got credit notices. **Probe** "How did you include them?" — A vendor-facing row in the runbook and a communication before the weekend.

WBR / design: stakeholders as rows in the runbook with owners.

SS-6. Balanced growth or progress with environmental or societal impacts. Impacts? Steps?

S1. Consolidating two warehouses into one removed inter-DC transfers — fewer truck miles — and duplicated stock. I did not run it as an environmental program and will not claim I did; the step I took was to keep transfer elimination as a tracked outcome of the consolidation rather than a side effect. Also **S5** expedite share. Be honest about scope here.

Probe "Did you measure emissions?" — No; I measured transfers and miles, which are the input. **Probe** "What would you do at Amazon?" — Placement decisions already carry a distance term; I would want it on the cost page, not a separate one.

WBR / design: transfer volume and miles as inputs alongside cost.

SS-7. Realized a project would have unintended consequences in the future; and one where you didn't realize until complete.

Foresaw: **S6** — scaling without gates would have buried support; built the gates before wave one. Did not: **S10** — the in-transit rule's effect on dotcom only showed in production. Both now shape how I write defaults: every rule gets a measured number and a divergence chart from day one.

Probe "What did S10 teach about foresight?" — Conservative defaults are decisions too; price them. **Probe** "What else did you miss on S6?" — Kill switch.

WBR / design: divergence charts on every new policy for its first quarter.

SS-8. Left something better than you found it. How did your actions influence things for the better?

S3: a planning parameter that measures itself and a drift flag planning still uses. **S12:** a release gate owned by QA that neighbors copied. **S10:** a stock-state model and a nightly reconciliation that run without me. The test I apply: is the mechanism still firing a year after I left, with my name nowhere on it?

Probe "How do you know they're still running?" — Former colleagues; the drift flag and the gate were both referenced in later programs. **Probe** "What did you leave worse?" — Two weeks of security-review scar tissue on the infrastructure team; I fixed the checklist.

WBR / design: mechanisms with owners that are not me.

SS-9. (Manager) Influenced an organizational change toward social awareness and responsibility. Activities? Outcome?

Honest answer: I have not run a social-responsibility program. What I have done is make downstream parties visible in decisions — suppliers as scorecards rather than cutoffs (**S16**), operators' dates protected in a flip that cost them (**S7**), vendors and night shifts as rows in a cutover runbook (**S10**). The activity was structural: a row, an owner, a metric. Outcome: decisions that counted who they affected. Say that, and stop.

Probe "Is that social responsibility?" — It is the version of it a program manager controls. **Probe** "What would you add at Amazon's scale?" — The downstream row on every capacity decision: delivery stations, DSPs, communities around a new building.

WBR / design: a "who else is affected" row in every decision document.

7A. How Amazon manages capacity — the unit, the flow, the money, the review

This is the depth section. You will not recite it. It exists so that when the interviewer says "walk me through how you would think about capacity in a region," every step, metric and dollar you name is attached to a unit moving through a building. **Three kinds of statements appear here.** (1) Structure that Amazon has described publicly — FC types, inbound and outbound process paths, sort centers and delivery stations, CPT. Use freely. (2) Standard capacity-planning economics — utilization ceilings, fixed-cost dilution, peak factors, labor math. Use freely; this is the job. (3) Numbers. Every number below is **ILLUSTRATIVE**, chosen so the arithmetic can be checked on paper. Say so if you use one, and say what replaces it on day one: the S&OP demand file, Ops Engineering design rates, Finance's cost-per-unit ledger.

7A.0 Supply, demand, capacity — three words, four equations

Say these three words in this order, with these meanings, and the interviewer knows you are speaking the team's language. **Demand** is units customers will order, by week, by region, by building type, by speed tier — the forecast, with its error. **Supply** is units arriving to be stowed — vendor inbound, FBA inbound, transfers, returns — and the inventory already on hand. **Capacity** is what a building can do with those units: **store** them (cube) and **move** them (hours through each process path). Demand and supply are flows; inventory is the stock between them; capacity is the pipe and the tank. The plan is the statement that for every building and every week the four equations below hold.

- (1) INVENTORY BALANCE
$$I(t) = I(t-1) + \text{Supply_in}(t) - \text{Demand_out}(t) - \text{Shrink}(t)$$

what you stow this week minus what you ship is the change in what you hold
- (2) STORAGE CONSTRAINT (Ceiling_storage $\approx$ 0.85–0.90)
$$I(t) \times \text{cube_per_unit}(\text{mix}) \leq \text{UsableCube} \times \text{Ceiling_storage}$$

what you hold must fit — at the ceiling, not at 100%
- (3) INBOUND THROUGHPUT (steps: unload, receive, decant, stow)
$$\text{Supply_in}(t) \leq \min \text{ over steps } s \text{ of } [\text{Hours}_s(t) \times \text{Rate}_s]$$

you can only stow what every inbound step can push in the hours you have
- (4) OUTBOUND THROUGHPUT (steps: pick, rebin, pack, SLAM, ship-sort)
$$\text{Demand_out}(t, \text{CPT}) \leq \min \text{ over steps } s \text{ of } [\text{Hours}_s(\text{window before CPT}) \times \text{Rate}_s]$$

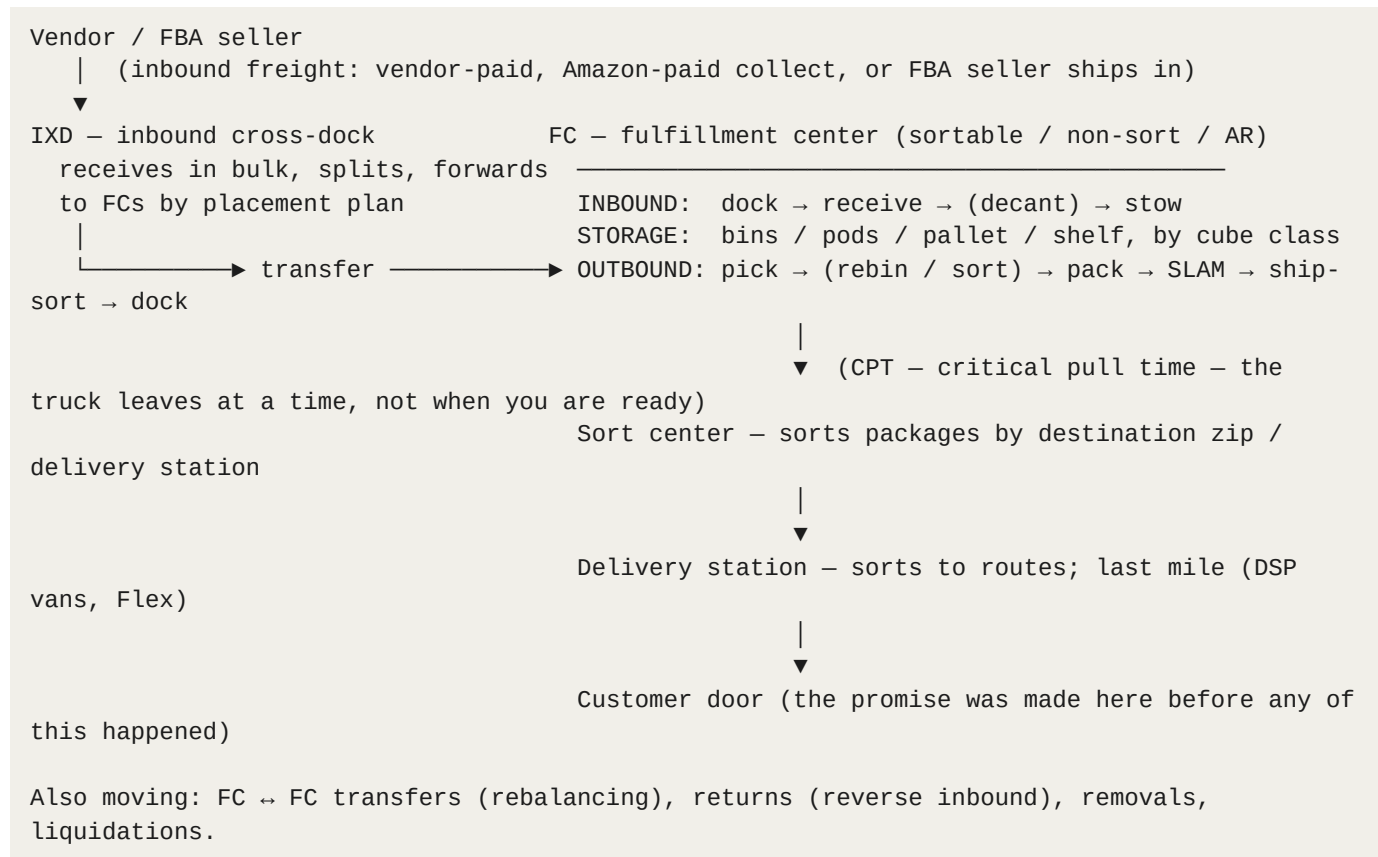
you can only ship what every outbound step can push BEFORE EACH TRUCK LEAVES
- Rate_s is not a constant:
$$\text{Rate}_s = \text{DesignRate}_s \times f(\text{utilization}) \times g(\text{new-hire share}) \times h(\text{flow evenness})$$

f falls past ~85–90% storage utilization (walk, hunt, congestion)
 $g \approx 0.6 \rightarrow 1.0$ over ~2 weeks of ramp; $h < 1$ when work arrives in bursts

How to read a gap with these. If (2) fails, you have a *storage* gap: lower $I(t)$ (days of cover, slow movers out, inbound pushed back) or raise UsableCube (overflow, retrofit, new build). If (3) fails, *inbound* gap: hours, appointment shaping, IXD absorbs. If (4) fails, *outbound* gap: hours, placement to buildings with slack, CPT structure. If (1) is drifting up week over week with (2) approaching the ceiling, you have a *placement or forecast* problem wearing a capacity costume — supply is being sent to a building whose demand does not justify it. Name the equation that failed before you name a lever. That is the whole discipline.

The one sentence: "Demand tells me how many units must leave each building each week; supply tells me how many must arrive; inventory is the difference; capacity is whether the difference fits in the cube and whether the flows fit in the hours — before each truck leaves."

7A.1 The network one unit travels



The sentence that organizes everything below: capacity is planned in two currencies — **cube** (can the unit be stored) and **hours** (can the unit be moved through each step before its truck leaves) — and the plan is honest only if both are checked at the step where each binds. A building can have empty shelves and still miss CPT because pack is short of hours. A building can have hours and still refuse inbound because it is 97% full.

7A.2 Inbound — step by step, what the unit does, how it is measured, what it costs

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Inbound transportation	Trailer or container travels vendor → IXD/FC. Appointment booked; trailer arrives at a dock door.	Trailers/day vs appointments; on-time arrival; units per trailer (cube fill); dwell in yard (hours).	Dock doors × turns/day; yard spots; appointment slots.	Variable: linehaul \$/mile or \$/trailer, drayage, fuel; semi-fixed: yard jockeys, gate staff. Paid by Amazon on collect freight; embedded in cost of goods on prepaid.
Dock / unload	Pallets or fluid-loaded cartons come off. Cartons scanned to a receive queue.	Cartons unloaded/hour; door dwell (hours door is occupied per trailer); unload backlog (trailers waiting).	Doors; unload labor hours; forklift/equipment.	Variable labor: unload hours × rate. Fixed: doors, levelers, forklifts (capex / lease).

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Receive	Carton opened, ASIN identified, quantity counted, condition checked, unit now exists in inventory as "received, not yet stowed."	Units received/hour per associate (receive rate); receive backlog (units or hours of work waiting); vendor compliance defects (mislabeled, wrong qty) per 1,000 cartons.	Receive hours; problem-solve queue for exceptions.	Variable labor: receive hours. Exceptions cost 5–10× a clean carton (illustrative) — the compliance metric is a cost metric.
Decant / prep	Units pulled from vendor cartons into totes or prepped (poly-bag, label) so they can be stowed.	Units/hour; prep required %; tote fill.	Prep hours; consumables.	Variable: labor + packaging consumables.
Stow	Unit placed into a bin (shelf, pod in AR, pallet location). Inventory becomes available to promise at this scan.	Units stowed/hour (stow rate); dock-to-stow hours (the headline inbound flow metric: carton arrival to unit available); stow backlog; bin-fullness at stow.	Stow hours; empty cube — if bins are full, stow slows or stops; in AR, pod availability and robot drive-unit count.	Variable labor: stow hours. Stow rate falls as the building fills (walk time, hunting for empty bins) — this is the first place the utilization curve costs money.

Why dock-to-stow is the inbound metric that matters. Until the unit is stowed, the customer cannot buy it against this building's promise, the vendor's trailer is still at your door or in your yard, and the carton is occupying floor space that is not a bin. A dock-to-stow target (say ≤ 24 hours, illustrative) converts inbound capacity into a flow commitment. Miss it and you have an inbound backlog, which shows up as refused appointments, which shows up as inventory stuck upstream, which shows up as a speed miss three weeks later.

7A.3 Storage — the cube currency

QUANTITY	HOW TO COMPUTE IT (ILLUSTRATIVE)	WHAT IT TELLS YOU
Gross storage cube	Bins $\times$ cubic feet per bin, by bin type (shelf, pod, pallet, bulk floor).	The building's theoretical ceiling. Never plan to it.
Usable cube	Gross $\times$ bin-fill efficiency (air inside a bin because units are odd shapes; ~60–75% illustrative) $\times$ operating ceiling (85–90%).	The number you actually plan storage to.
Units on hand	Daily outbound units $\times$ days of cover (inventory policy, by ASIN velocity class).	Required storage. Slow movers are the cube problem; fast movers are the hours problem.
Storage utilization	Cube occupied $\div$ usable cube, weekly.	Above the ceiling: stow rate falls, inbound gets refused, pick paths lengthen. This is the capacity-plan red light.
Inventory health	% of cube held by units with $> N$ weeks of cover; aged inventory; unsellable (damaged, quality hold).	Cube you are paying for that cannot ship. Your S/4HANA story: quality-hold counted as available overstates capacity.

Illustrative storage math. A building with 2.0M usable cubic feet at a 90% ceiling has 1.8M plannable. If its mix averages 0.25 cu ft per unit and the inventory policy is 21 days of cover on 120K daily outbound units, it needs $2.52\text{M units} \times 0.25 = 630\text{K cu ft}$ — 35% utilized, lots of room. Push the same building to 20 days of cover on a 300K/day peak outbound and it needs 1.5M cu ft — 83%, at the edge. The same shelves went from comfortable to critical because *days of cover* and *daily outbound* moved, not the building. That is why capacity planning owns the inventory policy conversation, not just the real-estate one.

7A.4 Outbound — step by step

STEP	WHAT HAPPENS TO THE UNIT	HOW FLOW IS MEASURED	WHAT BINDS	COST ATTRIBUTES
Order drop / batching	Customer order is assigned to this building (placement + promise decision). Units are batched into pick work by CPT.	Units dropped per hour by CPT window; work-in-queue hours ahead of each CPT.	Planning logic, not labor. Over-drop and pack drowns; under-drop and pick idles.	No direct cost; it decides every downstream cost. Wrong-building placement adds transportation and transfer cost.
Pick	Associate (or robot brings pod to associate) retrieves unit into a tote.	Units picked/hour (pick rate); picks per tote; pick backlog vs CPT; mispicks per 1,000.	Pick hours; in AR, drive-unit and pod availability; pick-station count.	Variable labor: largest single labor line in most buildings. Rate drops as storage fills (path length) and at peak with new hires on a learning curve.
Rebin / sort (multi-unit orders)	Totes from many pickers are consolidated so one order's units land together.	Units sorted/hour; wall utilization; orders waiting on last unit.	Sort-wall slots; hours.	Variable labor; fixed sort walls / sorters.
Pack	Units go into a box or mailer; box size chosen; dunnage added.	Units or packages packed/hour; packages per hour per station; box-size accuracy (cube out — shipping air is a transportation cost).	Pack stations and hours. Pack is frequently the outbound bottleneck at peak.	Variable labor; variable packaging (corrugate, mailers, tape, dunnage — \$/package, mix-dependent).
SLAM	Scan, Label, Apply, Manifest — package weighed, label printed and applied, manifested to a carrier and a truck.	Packages/hour; weight/dimension mismatches; label errors.	SLAM lines (machines) — a fixed-capacity step.	Fixed equipment; small variable labor and labels.
Ship sort / dock	Package routed to the right outbound door for its carrier, sort center, and CPT; loaded into a trailer.	Packages sorted/hour; CPT attainment (% of planned packages on the truck before it leaves — the headline outbound flow metric); trailer cube fill; late trailer departures.	Sorter capacity; outbound doors; trailer availability.	Fixed sorter; variable load labor; variable transportation — a half-full trailer costs the same as a full one.

Why CPT attainment is the outbound metric that matters. A critical pull time is the moment the promise stops being a plan and becomes a truck. A package that misses its CPT either rolls to the next truck (speed downgrade, possibly a missed promise) or goes on a more expensive mode (premium transportation). Every upstream step — pick, pack, SLAM — is measured against hours-ahead-of-CPT, not against the shift. That is what "flow" means inside a building: work arriving at each step early enough that the last step finishes before the door closes.

7A.5 Downstream — sort center, delivery station, last mile

- **Sort center.** Receives trailers from many FCs, sorts packages by destination delivery station or carrier injection point. Capacity in packages/hour through the sorter and outbound doors by destination. Metrics: sort rate, missort rate, trailer utilization, sort-center CPT attainment to delivery stations.
- **Delivery station.** Receives sorted packages overnight, sorts to routes, dispatches to delivery partners. Capacity in packages/day by route count × packages/route. Metrics: packages per route, dispatch on time, delivery success rate.
- **Why NACF capacity planning cares.** FC outbound capacity is only useful if there is downstream sort and delivery capacity on the same lanes on the same night. A region can have FC slack and still miss speed because the delivery stations near the demand are full. Say "FC capacity and downstream capacity have to be planned on the same page" — it is the regional version of "storage and throughput both."

7A.6 A worked example — demand meets capacity in one region (ILLUSTRATIVE)

Three FCs serve a region. Numbers invented so you can check them.

BUILDING TYPE		USABLE STORAGE (M CU FT, AT CEILING)	OUTBOUND DESIGN RATE (K UNITS/WK)	PLANNABLE OUTBOUND AT 90% (K/WK)	AVG WEEK DEMAND PLACED (K/WK)	PEAK WEEK (×1.6)
FC-A	Sortable AR	1.8	1,400	1,260	1,000	1,600
FC-B	Sortable (manual)	1.2	900	810	700	1,120
FC-C	Non-sort	2.5	500	450	300	480
Region		5.5	2,800	2,520	2,000	3,200

Read it. Average week: 2,000 against 2,520 plannable — 79% utilized, fine. Peak week: 3,200 against 2,520 — a **680K-unit gap**, and it is not evenly distributed: FC-A is 340K over, FC-B 310K over, FC-C is 30K over. "The region is 27% short" is the wrong sentence. "Two sortable buildings are each about 35% short in peak week; non-sort is fine" is the right one.

Check the other currency. FC-A peak inventory: $1.6\text{M/wk} \div 7 = 229\text{K units/day} \times 18 \text{ days of cover} \times 0.25 \text{ cu ft} = 1.03\text{M cu ft}$ against 1.8M usable — 57%, storage is not the problem. FC-B: $160\text{K/day} \times 18 \times 0.30 = 0.86\text{M}$ against 1.2M — 72%, getting tight but not binding. So this is a **throughput** gap in the two sortable buildings, concentrated in pick and pack hours. The lever stack is a labor and flow conversation first, a steel conversation never (for this peak).

Lever stack with illustrative costs for the 650K sortable gap, cheapest and most reversible first:

LEVER	CAPACITY IT BUYS	ILLUSTRATIVE COST	TRIGGER AND OWNER
1. Placement shift — move sortable-eligible demand to FC-C's slack and to the next region's sortable buildings	~150K	Extra outbound transportation ~\$0.40/unit on shifted volume (longer zone) → ~\$60K/wk	Fires when week-8 forecast still shows >10% gap. Owner: placement / SCOT partner.
2. Hours — add a shift, extend to 6 days, overtime	~250K (FC-A 150K, FC-B 100K)	Overtime premium 1.5× on ~180K incremental hours-equivalent... illustratively +\$1.2M for the peak period	Labor plan locked at week-10. Owner: FC Ops + regional labor planning.

LEVER	CAPACITY IT BUYS	ILLUSTRATIVE COST	TRIGGER AND OWNER
3. Peak hiring with learning-curve haircut	~200K	New hires at ~60% rate for 2 weeks; training and attrition cost; ~\$2M for the period	Requisitions open week-12; ramp complete week-2. Owner: FC Ops / HR.
4. Inbound shaping — pull fast-mover inbound forward to pre-peak weeks, push slow-mover inbound to post-peak	Frees pick/stow hours in peak week (~50K equivalent)	Carrying cost on earlier inventory (small); vendor negotiation effort	Inbound appointment caps by week published week-10. Owner: inbound planning.
5. Overflow site / 3P	Residual	\$1.50–3.00/unit all-in	Only for what 1–4 do not cover. Contract trigger at week-8.
6. Retrofit / automation	+15–30% building throughput	Capex, 2–4 quarters	Not this peak. Goes into the OP cycle for next year.
7. Demand shaping — speed tier, promise	Last resort	Conversion cost (unpriced trust)	Only when a customer-facing miss is the alternative.

What you say at the end of the example. "The plan closes 650K with placement, hours, hiring and inbound shaping, at roughly three to four million dollars of incremental cost for the peak period, and leaves overflow as a priced option with a trigger date. Every lever has an owner and a week it fires. The two things I will not do: spread a pad across all three buildings, which hides that FC-C is fine; and plan either sortable building above 90% of design rate, because congestion arrives before the ceiling does."

7A.7 The labor model — how hours are computed

Required hours (step, week) = units through the step ÷ planned rate (units/hour) for that step

Planned rate = engineered standard × building-specific factor × learning-curve factor × congestion factor

Headcount (step) = required hours ÷ (hours per shift × shifts per week × attendance %)

Total direct hours (week) = Σ over steps (receive, stow, pick, rebin, pack, ship-sort, dock, problem-solve)

Indirect hours = supervision, training, quality, ICQA (inventory counts), safety, maintenance labor

ILLUSTRATIVE FC-A peak week: 1.6M units

pick	$1.6M \div 120/hr = 13,300 \text{ hrs}$	stow	$1.5M \text{ inbound} \div 110/hr = 13,600 \text{ hrs}$
pack	$1.6M \div 90/hr = 17,800 \text{ hrs}$	receive	$1.5M \div 150/hr = 10,000 \text{ hrs}$
rebin	$0.8M \div 250/hr = 3,200 \text{ hrs}$	ship/dock	$\approx 4,000 \text{ hrs}$

direct $\approx 62,000 \text{ hrs}$; at 36 productive hrs/associate/week and 92% attendance $\rightarrow \sim 1,870$ associates

average week $\approx 1,170 \rightarrow$ peak requires ~ 700 incremental associates, ~ 2 weeks of ramp

Two things the interviewer wants to hear you know: **rates are not constants** — they fall when storage is full, when the building is on new hires, and when flow is uneven (idle pack then flooded pack) — and **pack, not pick, is often the peak bottleneck**, because pack rate per unit is lower and pack stations are fixed.

7A.8 The cost model — every dollar a unit picks up

Think of cost per unit as the sum of what the unit touched, divided by how many units shared the building that week. Fixed costs make the denominator matter; variable costs make the process paths matter; congestion makes both worse at the same time.

COST FAMILY	LINES	FIXED / VARIABLE	HOW IT BEHAVES WITH VOLUME AND UTILIZATION
Facility	Lease or depreciation on the building; property tax; insurance; base utilities; security	Fixed	Per-unit cost falls as volume rises — <i>fixed-cost dilution</i> . This is why under-utilized buildings are expensive per unit and why Finance resists new steel until the existing floor is used.
Equipment	Conveyors, sorters, SLAM lines, AR drive units and pods, racking, forklifts, IT/scanners	Fixed (capex, depreciated) + maintenance (semi-variable)	Same dilution. Automation shifts cost from variable labor to fixed equipment — good at high utilization, bad at low.
Direct labor	Receive, stow, pick, rebin, pack, ship-sort, dock, problem-solve	Variable (hours × wage), with step-ups	Roughly linear until the building is full or on new hires; then rate falls and cost per unit rises. Overtime at 1.5×; peak new-hire premium; shift differentials.
Indirect labor	Area managers, process assistants, learning, ICQA, safety, HR, facilities	Semi-fixed (steps with shifts and headcount bands)	Moves in steps when you add a shift; otherwise flat.
Packaging	Corrugate, mailers, tape, dunnage, labels	Variable per package	Mix-driven. Box-size accuracy also drives outbound transportation (dimensional weight).
Inbound transportation	Linehaul vendor → IXD/FC; IXD → FC transfers; drayage	Variable per trailer / per unit	Driven by placement: a unit stowed far from its vendor and far from its demand pays twice. Trailer cube fill is the efficiency metric.
Outbound transportation	FC → sort center linehaul; sort center → delivery station; last mile; carrier injection	Variable per package, zone and weight dependent	Usually the largest cost family after labor. Driven by placement (distance to customer), speed tier (air vs ground), package cube, and trailer fill. A placement shift to close a capacity gap is paid here.
Transfers and rebalancing	FC ↔ FC moves to fix placement after the fact	Variable	A tax on bad placement or bad forecasting. Track it as its own line so it cannot hide in inbound.
Inventory carrying	Cost of capital on units on hand; shrink; damage; obsolescence/markdown	Variable with inventory level	Rises with days of cover. Inbound shaping (pull-forward) spends this to save labor in peak week — price the trade.
Quality and defects	Mispicks, mis-ships, damages, concessions, returns processing	Variable per defect	Rises when rates are pushed past standard. A building running hot pays here two weeks later.
Overflow / 3P	Short-term space and labor contracts	Variable (all-in per unit) with minimums	Expensive per unit but no fixed commitment — the right lever for a short peak, the wrong one for a structural gap.

Cost per unit, illustrated at three utilizations (same building, one week, invented numbers):

	60% utilized	85% utilized	97% utilized ("full")	
Units shipped	840K	1,190K	1,360K	
Fixed (facility+equipment)	\$1.70M → \$2.02/u	\$1.70M → \$1.43/u	\$1.70M → \$1.25/u	
Direct labor	\$2.0M → \$2.38/u	\$2.7M → \$2.27/u	\$3.6M → \$2.65/u	← rates
fall, OT rises				
Indirect labor	\$0.45M → \$0.54/u	\$0.50M → \$0.42/u	\$0.55M → \$0.40/u	
Packaging	\$0.55/u	\$0.55/u	\$0.55/u	
Outbound transport	\$3.20/u	\$3.20/u	\$3.45/u	← CPT
misses → premium mode				
Defects / concessions	\$0.06/u	\$0.08/u	\$0.18/u	
Total	≈ \$8.75/u	≈ \$7.95/u	≈ \$8.48/u	

Read it. Cost per unit is U-shaped in utilization. Too empty and fixed cost is undiluted; too full and labor, transport and defects spike. The economic optimum sits near the operating ceiling, not at 100%. That is the whole argument for planning to a ceiling, and it is the argument you make to Finance in their own currency: "the last 7% of utilization costs more per unit than the first 85% saved."

Inbound vs outbound cost, said plainly. Inbound cost per unit (freight + receive + stow) is small per unit but front-loaded and placement-sensitive. Outbound cost per unit (pick + pack + packaging + transportation) is larger and speed-sensitive. Placement is the one decision that moves both: put the unit near its demand and you pay more inbound and less outbound; the capacity plan has to know which side of that trade a region is on.

7A.9 The WBR this role would run

Outputs are what the business feels. Inputs are levers a named owner can move this week. Every line has a grain, a target, an owner, and a red that arrives pre-decomposed: which input moved, whose lever, what correction, verified by the same number next week. Illustrative targets.

	METRIC	GRAIN	ILLUSTRATIVE TARGET	OWNER / WHY IT IS ON PAGE ONE
Outputs				
O1	Capacity delivered vs plan (units/week the network could process)	Region × building type × week	≥ 100% of committed plan	You. The headline: did the capacity exist when the plan said it would.
O2	Throughput utilization vs ceiling	Building × week (peak: × day)	75–90% band	You with FC Ops. Below band = cost; above = speed and defects.
O3	Storage utilization vs ceiling	Building × week	≤ 90%	You with inventory planning. The second currency.
O4	Customer speed attainment (promised speed met)	Region × speed tier × week	≥ target by tier	Shared with S&OP / transportation. The customer-facing reason the plan exists.
O5	Cost per unit vs plan (fully loaded, and by family)	Building × week; region × month	≤ plan	Finance partner. The second half of the charter.
O6	CPT attainment	Building × CPT × day	≥ 98%	FC Ops. Outbound flow health; leading indicator of O4.
O7	Dock-to-stow hours	Building × day	≤ 24 h	FC Ops / inbound. Inbound flow health; leading indicator of refused appointments.

	METRIC	GRAIN	ILLUSTRATIVE TARGET	OWNER / WHY IT IS ON PAGE ONE
Inputs				
I1	Forecast accuracy at planning grain (and bias)	Building type × region × week, 4- and 8-week horizons	MAPE within band; bias ≈ 0	S&OP / SCOT. A biased forecast is a capacity error wearing a demand costume.
I2	Placement compliance (units landed in the planned building)	Region × week	≥ 95%	Placement partner. Off-plan placement is the silent capacity thief.
I3	Labor plan attainment (hours staffed ÷ hours required, by step)	Building × step × week	≥ 97%	FC Ops / labor planning.
I4	Rate attainment vs planned rate, by step	Building × step × week	≥ 95% of plan	FC Ops. Where congestion and learning curve show up first.
I5	Launch and retrofit milestones on dated gates	Project × gate	Green/red vs date	Launch / Engineering / Real Estate. New capacity that is late is a gap you planned on.
I6	Inbound appointment adherence and vendor compliance defects	Building × week	≥ 95%; defects ≤ x per 1,000	Inbound planning / vendor management.
I7	Inventory health: days of cover by velocity class; % cube in aged / unsellable	Building × week	Within policy; aged ≤ x%	Inventory planning. Cube you cannot sell is capacity you do not have.
I8	Trailer cube fill, inbound and outbound	Lane × week	≥ target	Transportation. The cheapest transportation dollar.
I9	Transfers and rebalancing volume and cost	Region × week	Trending down	You. The tax on bad placement, kept visible.
I10	Lever status: which planned levers have fired, on which trigger, at what cost	Lever × week	All triggered levers fired on time	You. The plan is a list of triggers; this line proves it is alive.

Page one vs annex. Page one: O1–O7, I1–I10, RAG with week-over-week deltas, five callouts, and the action log with owner and date. Annex: forecast histograms, rate curves by building, placement heatmaps, cost bridges, launch Gantt. **Correction of error** is written when O1 misses plan by more than a published threshold, when O2 or O3 exceeds ceiling for two consecutive weeks, or when a lever that should have fired did not. It ends in a mechanism change — a trigger moved earlier, a ceiling re-measured, an owner named — never in "try harder."

7A.10 How you would model it

Four layers, each feeding the next. Build the simplest version of each first; add fidelity where the plan is actually sensitive.

LAYER 1 – DEMAND	units by week × region × building type × speed tier, from S&OP peak factors by week; a range (P10 / P50 / P90), not a point – the plan is sized to a quantile you choose on purpose
LAYER 2 – PLACEMENT	which building absorbs which demand: minimize inbound + outbound transport + transfer cost, subject to each building's storage and throughput ceilings → units/week per building. The network LP lives here; SCOT owns the engine, you own the constraints and whether the answer is plannable

LAYER 3 – BUILDING	per building per week: required hours by step (7A.7) and required cube (7A.3) vs available; gap by currency by step; rate curves as a function of utilization and new-hire share; CPT structure (how many trucks, when) – flow inside the week, not just the weekly total
LAYER 4 – COST	per building per week: fixed + labor + packaging + transport + carrying + defects (7A.8); cost per unit; incremental cost of each lever; the U-curve
OUTPUTS	gap by week × building × currency; ranked lever stack with cost and lead time; triggers; sensitivity – which forecast error, rate miss or launch slip breaks the plan
VALIDATION	backtest on last year's weeks: did the model's "available" match what the buildings shipped? Where it missed, which layer was wrong – demand, placement, rate, or ceiling?

Three modeling judgments to say out loud. (1) Plan to a quantile, not a point: if you size to P50 you will be short half the time. Decide with Finance which quantile the cost of shortage justifies. (2) Rates are curves, not constants: a model with fixed rates will tell you a 97%-utilized building is fine. It is not. (3) Weekly totals hide CPT misses: a building can hit its weekly number and miss every 6 PM truck. Model at least the daily CPT structure for peak weeks.

What you would ask on day one. What is the current ceiling by building type, who set it, and when was it last re-measured against actuals? Does placement see the throughput ceiling or only storage? Which lever fired last peak that was not in the plan? Where does forecast bias live by region? And: which of these numbers does Finance accept, and which does it dispute?

7A.11 Whiteboard math — the ten computations you must be able to do cold

Each one: the formula, why it is the right formula, a worked number (ILLUSTRATIVE), and what it tells a capacity planner. Do them on paper once before Friday. In the room, say the formula first, then the numbers, then the implication.

1. Peak sizing from a forecast with error — plan to a quantile, not a point

Required capacity at service level α = Forecast_P50 × (1 + z_{α} × CV)
 CV = forecast error ($\sigma \div$ mean) at the planning grain and horizon
 $z_{0.80} \approx 0.84$ $z_{0.90} \approx 1.28$ $z_{0.95} \approx 1.65$

Worked: peak-week P50 = 1.6M units at FC-A; 8-week-ahead CV = 12%
 $P80 = 1.6M \times (1 + 0.84 \times 0.12) = 1.76M$
 $P90 = 1.6M \times (1 + 1.28 \times 0.12) = 1.85M$
 $P95 = 1.6M \times (1 + 1.65 \times 0.12) = 1.92M$

Implication. Sizing to P50 is short half the time. The quantile is a *business decision*: the cost of being short (premium transport, speed misses, overflow at spot rates) vs the cost of being long (undiluted fixed cost, idle hours). Choose it with Finance, write it down, and size every lever to it. If CV is 12% at 8 weeks and 6% at 2 weeks, the plan can hold reversible levers (hours, placement) at P90 and commit irreversible ones (overflow contracts) at P80, then upgrade as the forecast tightens.

2. Storage required and utilization — the cube currency

Inventory units = DailyOutbound × DaysOfCover
 Cube needed = Inventory units × cube_per_unit(mix) ÷ BinFillEfficiency
 Utilization = Cube needed ÷ (GrossCube × Ceiling)

Worked (FC-A, same as 7A.6):
 Inventory = 229K units/day × 18 days = 4.12M units; × 0.25 cu ft = 1.03M cu ft of product

Usable = 2.86M gross bin cu ft × 0.70 bin fill × 0.90 ceiling = 1.80M cu ft
 Utilization = 1.03 ÷ 1.80 = 57%
 Sensitivity: +1 day of cover = 229K × 0.25 = 57K cu ft ≈ 3.2 points; +10 days ≈ 32 points → 89%,
 at the ceiling

Implication. Days of cover is the biggest storage lever and it is an inventory-policy decision, not a real-estate one. Three days of cover is ten points of utilization — the difference between comfortable and congested. The capacity planner must own the days-of-cover conversation with inventory planning.

3. Throughput required vs available — the hours currency

Required hours_s = Units_s ÷ Rate_s
 Available hours_s = Heads_s × ShiftHrs × Shifts/wk × Attendance
 Gap_s = Required - Available ; binding step = largest positive gap

 Worked (FC-A peak week, 1.6M outbound):
 pack: 1.6M ÷ 90/hr = 17,800 hrs needed; staffed 420 heads × 8 h × 5 days × 0.92 = 15,460 hrs
 gap = 2,340 hrs ≈ 210K units at pack → pack binds even though pick (13,300 hrs needed) is
 staffed to 14,000
 Fixes priced in hours:
 +1 shift-day/wk on pack = 420 × 8 × 0.92 = +3,090 hrs at 1.0× wage
 OT on existing heads = 2,340 hrs at 1.5× wage
 move 210K units to FC-C = transport Δ (7A.6: ~\$0.40/unit)

Implication. A building's throughput is its slowest step *at the hour the truck leaves*. Always compute the gap per step; the binding step is where the money goes. Pack binds more often than pick because its rate is lower and stations are fixed.

4. Little's Law — turning a flow metric into a backlog and back

WIP = Throughput × CycleTime ⇔ CycleTime = WIP ÷ Throughput

 Inbound: 215K units/day arriving, dock-to-stow target 24 h
 → WIP allowed on the inbound floor = 215K × 1.0 day = 215K units
 if the floor holds 430K units, dock-to-stow is running 48 h, whatever the dashboard says
 Outbound: a CPT with 180K units due in 6 h needs 30K units/h through pack
 at 90 units/h that is 333 pack heads working that CPT

Implication. You can measure cycle time by counting what is on the floor. Dock-to-stow hours, backlog units and receive rate are one relationship, not three metrics; if two are green and one is red, one of the greens is wrong.

5. Congestion — why cost per unit turns up past the ceiling

Queueing approximation: Wait ∝ $\rho \div (1 - \rho)$, ρ = utilization
 $\rho = 0.70 \rightarrow 2.3$ $0.85 \rightarrow 5.7$ $0.90 \rightarrow 9.0$ $0.95 \rightarrow 19$ $0.97 \rightarrow 32$ (relative wait)

 Storage analog — stow rate falls as empty bins get scarce (illustrative curve):
 Rate(ρ) = DesignRate × $[1 - k \times \max(0, \rho - 0.85)^2 \div 0.0225]$, $k \approx 0.3$
 $\rho = 0.85 \rightarrow 100\%$ of design $0.90 \rightarrow 97\%$ $0.95 \rightarrow 87\%$ $1.00 \rightarrow 70\%$

Implication. Going from 90% to 95% utilization doubles waiting; going to 97% triples it again. That is the arithmetic behind "plan to a ceiling, not to 100%." It is also why the last five points of utilization cost more per unit than they save in fixed-cost dilution (7A.8).

6. Fixed-cost dilution vs variable-cost inflation — the U-curve, derived

CPU(ρ) = Fixed ÷ Units(ρ) + VarLabor(ρ) + Packaging + Transport(ρ) + Defects(ρ)
 Fixed ÷ Units falls like 1/ ρ
 VarLabor, Defects rise past the ceiling (Rate(ρ) falls, queueing grows)
 Transport rises when CPT misses force premium modes

Worked (7A.8 numbers): CPU at 60% = \$8.75 at 85% = \$7.95 at 97% = \$8.48
 60 → 85 dilution: fixed/unit \$2.02 → \$1.43 (-\$0.59); labor/unit \$2.38 → \$2.27 (-\$0.11)
 85 → 97 inflation: fixed/unit \$1.43 → \$1.25 (-\$0.18); labor +\$0.38, transport +\$0.25, defects +\$0.10 (+\$0.73 total) → net +\$0.53 per unit

Implication. The minimum of the curve is the economic ceiling, and it is *below* the physical ceiling. When Finance asks why you will not plan to 100%, this is the table you show: the last 12 points cost 53 cents a unit.

7. Lever cost per unit of capacity — ranking the stack

Lever cost (\$ per unit of gap closed) = Incremental cost ÷ Units of gap it closes
 Rank ascending, subject to lead time ≤ weeks remaining to peak

Worked (7A.6 lever stack, 650K/wk gap over a 4-week peak):

placement shift	\$60K/wk × 4 ÷ (150K × 4) ≈ \$0.40/unit	lead 1-2 wks	reversible
inbound shaping	carrying cost on pulled-forward stock ≈ \$0.30/unit	lead 6-8 wks	
reversible			
OT / extra shift	\$1.2M ÷ (250K × 4) ≈ \$1.20/unit	lead 4-6 wks	reversible
peak hiring	\$2.0M ÷ (200K × 4) ≈ \$2.50/unit	lead 10-12 wks	semi-reversible
overflow / 3P	\$1.50-3.00/unit all-in	lead ~8 wks	contractual
demand shaping	conversion loss — unpriced, so last		

Implication. Flow before hours, hours before hiring, hiring before overflow, overflow before the promise — but the ranking is an output of this arithmetic, not a slogan. If placement shift costs \$0.40 and the next region has no slack, it drops off the list and hours move up.

8. Build vs no-build — the break-even forecast

Build (annualized) = Capex ÷ Life + Lease/fixed opex + Fixed labor ← paid regardless of volume
 No-build (annual) = $\sum_{\text{weeks}} [\text{Overflow\$} + \text{Transfers\$} + \text{PremiumTransport\$} + \text{SpeedDowngrade} \times \text{ConversionCost} + \text{P(peak miss)} \times \text{PeakMiss\$}]$
 Break-even demand D* = the demand at which the two are equal; report it as a forecast quantile

Worked (illustrative): build = \$28M/yr all-in

no-build cost at P50 demand = \$19M/yr; at P75 = \$31M/yr; at P90 = \$44M/yr → D* ≈ P68

sizing to P80: the build pays. sizing to P50: it does not, and overflow is the plan.

Implication. Do not argue "we need the building." Show D*, show where the planning quantile sits, and let the decision fall out. Also show the middle path — retrofit for +20% throughput at a third of the capex — on the same axis.

9. Learning curve and attrition — why peak heads are not peak hours

Effective hours (wk) = $\sum_{\text{cohorts}} \text{Heads}_c \times \text{ShiftHrs} \times \text{Attendance} \times \text{Productivity}_c(\text{weeks since hire})$

Productivity ≈ 0.60 (wk 1) 0.80 (wk 2) 0.95 (wk 3) 1.00 (wk 4+) [illustrative]

Ramp attrition ≈ 10-15% of a cohort → hire ~1.15× the heads you need

Worked: need +700 effective associates in peak week 0

hire 300 in wk -3 and 500 in wk -2 (800 gross)

effective in wk 0 ≈ 300 × 0.95 + 500 × 0.80 = 685 → 15 short, borderline → add 50 to the wk -3 cohort

Implication. A labor plan in heads is wrong by the learning curve. Plan in effective hours by cohort, and the hiring trigger moves earlier than the heads math suggests.

10. Speed — the CPT cliff and what a late unit costs

A unit that misses its CPT by 1 minute loses a full truck cycle — often a calendar day of promise.
Speed loss (units/day) = $\sum_{\text{CPT}} \max(0, \text{Units due at CPT} - \text{Pack capacity in the window before CPT})$
Cost per missed unit $\approx$ premium-transport uplift (if upgraded)
or $P(\text{cancel} \mid \text{late}) \times \text{margin} + \text{trust (unpriced)}$

Worked: 18:00 CPT has 180K due; pack capacity 14:00–18:00 = 4 h $\times$ 36K/h = 144K $\rightarrow$ 36K miss,
daily
60% upgraded to premium at +\$2.50 $\rightarrow$ 21.6K $\times$ \$2.50 = \$54K/day in transport alone, before cancels

Implication. Weekly utilization can be 80% and the 18:00 CPT can miss every day. Capacity must be planned at the CPT window grain for peak weeks. This is Case C in Section 7, in numbers.

7A.12 Lines for this section

- "Capacity is two currencies — cube and hours — and the plan is honest only if both are checked at the step where each binds."
- "Dock-to-stow is inbound flow. CPT attainment is outbound flow. Everything else is a component."
- "Cost per unit is U-shaped in utilization. The optimum is near the ceiling, not at a hundred percent."
- "Placement is the one decision that moves inbound and outbound cost at the same time."
- "A biased forecast is a capacity error wearing a demand costume."
- "Inbound shaping spends carrying cost to buy peak-week labor. Price the trade; don't just do it."
- "Pack, not pick, is usually where peak breaks."

7. The functional block — capacity planning

You will get at least one of these. Format they expect: clarify out loud (60–90 seconds) $\rightarrow$ structure (60 seconds) $\rightarrow$ a math checkpoint $\rightarrow$ recommendation with owners, triggers and risks. Say "both constraints — storage and throughput" early in every answer. Label every number you make up as illustrative, and say what you would replace it with on day one.

Primer: the shape of a capacity plan

Demand side	units by week × building type × speed tier, with a peak factor (forecast owner: S&OP / SCOT; your job is to interrogate its grain and mix)
Required capacity needed	storage: units on hand ≈ daily outbound units × days of cover → cube throughput: units/week each process step must push (inbound, stow, pick, pack, ship) both, by building type; the binding one is the one you plan to
Available capacity before 100%)	buildings × design rate × a utilization ceiling (congestion starts before 100%) minus planned downtime, retrofits, launches still ramping
Gap by week	required – available, by region, by type; where and when it binds
Levers, cheapest and fastest first	
1. flow: move units – placement / allocation to buildings with slack (days to weeks)	
2. hours: shifts, overtime, labor plan (weeks)	
3. inventory policy: pull inbound forward or push it back; trim slow movers (weeks)	
4. overflow: 3P / temporary space (weeks to months)	
5. retrofit / automation: more throughput in the same cube (quarters)	
6. expansion / new build: more cube (OP cycle; years)	
7. demand shaping: speed tier, promise, selection (the lever nobody wants to admit)	
Plan	named owner per lever, trigger that fires it, review cadence, and what you will not do

Case A — "Build next year's capacity plan for a region."

Clarify (say these out loud). Which building types are in scope? Is the demand forecast handed to me or do I own it? What is the peak factor and what week does it peak? What are the policy ceilings on utilization? Which launches and retrofits are already committed? What is the planning grain — week, building, type?

Structure. The primer, top to bottom. Spend the most time on the gap-by-week and the lever ranking.

Math checkpoint (illustrative). "If a building ships 1.0M units a week at average and the peak factor is 1.6, peak week is 1.6M. If its throughput design rate is 1.4M and the ceiling I am allowed to plan to is 90%, usable is 1.26M. The gap is 340K units in peak week. Flow first: how much of that can neighboring buildings absorb inside their own ceilings? Then hours. Overflow only for what is left. I would replace every one of those numbers with the S&OP demand file and the engineering design rates on day one."

Recommendation. A one-page plan: gap by week by region, the lever stack with owners, the triggers that promote a lever (e.g., "if the week-8 forecast is still 5% above plan, overflow contracts sign"), and the two things you will not do — plan to 100% of design rate, or pad every building's forecast to hide a placement problem.

Risk: forecast mix shifts between types; you plan by type, not by total.

CASE A — THE PROBES THEY WILL RUN INSIDE THE CASE, AND THE ANSWERS

"Where does your demand number come from, and how much do you trust it?" From S&OP/SCOT at week × region × building type × speed tier. I trust it as a distribution, not a point: I ask for the error (CV) at the 8-week and 2-week horizons and the bias by region, and I size reversible levers at P90 and irreversible ones at P80 until the forecast tightens (7A.11 #1).

"Which constraint binds first in your example — storage or throughput?" Throughput, in the two sortable buildings, specifically pack hours before the evening CPTs. Storage is at 57% and 72%. I know that because I ran equations (2) and (4) separately per building; the network total would have hidden it.

"Why 90% and not 95%?" Queueing: waiting roughly doubles between 90 and 95 (7A.11 #5), stow and pick rates fall as bins fill, and the cost-per-unit curve turns up past the economic ceiling (7A.11 #6). If the building's own measured rate curve says the knee is at 93%, I plan to 93%. The 90 is a placeholder for a measured number.

"Your placement shift costs transport. How much, and who pays?" Illustratively \$0.40 a unit on shifted volume — longer zone. Transportation's budget pays; the capacity plan's cost line shows it. I would not hide a capacity fix inside someone else's cost center; the WBR line I9 (transfers and rebalancing) exists so this stays visible.

"What if the forecast is wrong by 15% the other way — you over-hired?" The levers are sequenced by reversibility for exactly this: placement and OT unwind in a week; hiring is the first semi-irreversible lever and it fires at week -12 only if the P80 still shows the gap. If demand comes in low, the cost is undiluted fixed cost and some idle hours — I would show that number next to the cost of being short, and that comparison is how we chose the quantile in the first place.

"How do you know the lever actually fired and worked?" WBR line I10: each lever has a trigger date, an owner, and a 'fired / not fired / effect measured' state. Effect is measured in the same currency as the gap — units of pack capacity in the week it was needed — not in 'we added a shift.'

"What would you not do?" Spread a pad across all three buildings — it hides that FC-C is fine and spends everywhere. Plan either sortable building above its ceiling. Pull the promise before the priced levers are exhausted. And I would not present a plan without the quantile written on it.

"How would you present this to Finance?" Two columns, same rows (S7): cost of the plan by lever versus cost of the shortfall it prevents — overflow at spot, premium transport, speed downgrades at conversion cost, risk-adjusted peak miss. Break-even demand marked on the forecast distribution (7A.11 #8). Six pages, the table on page one.

"Walk me through the data pull." Demand: forecast file at week × building type × region × speed tier with error by horizon. Capacity: design rates by process path by building from Ops Engineering; measured rates by week for the last 52 weeks to fit the utilization curve; usable cube and bin fill by building. Labor: heads, attendance, cohort start dates. CPT structure: trucks and due units by hour. Costs: wage rates, OT premium, lease and depreciation by building, transport \$/unit by lane, overflow quotes. Join everything at building × week; keep CPT × day for peak weeks.

If they ask for it as a STAR ("tell me about a time you balanced capacity and demand"): S2 is the lead — demand exceeded the apparent capacity of the network at 4 AM; the task was 99% fill; the actions were promise-at-risk ranking and rebalancing to nodes with hours; the result was 99% held and \$8M protected — and the capacity turned out to be a planning-system failure, which is the lesson. S1 is the backup: consolidation as a capacity decision, throttled against a quality gate.

Case B — "Peak forecast just came in 10% above capacity. Go."

1. **Verify the number before you spend on it.** Grain and mix: is the 10% real demand, or a type-mix shift that makes one building type bind while another has slack? Same cohort weights as last forecast?
2. **Find where it binds.** Storage or throughput? Inbound dock, stow, pick, pack, or outbound? One region or network? "A network 10% is usually three buildings at 25% and the rest fine."
3. **Cheapest lever first, with a trigger.** Placement and allocation shifts this week. Labor plan next. Pull slow-moving inbound back; pull fast-moving inbound forward. Overflow for the residual. Speed tier only if the customer promise is the last protection.
4. **Publish the plan with a date on every lever** and a freeze on non-essential change into peak — the UDC October-to-December freeze.
5. **What you will not do.** Spread a national pad across every building. That hides the cell that is actually breaking and spends cost everywhere.

Story to attach: S2 — rank by promise-at-risk, rebalance across nodes, timestamped log, then a mechanism so you never discover the regime in November again.

CASE B — PROBES AND ANSWERS

"You have one hour. What's the first number you pull?" The gap by building by step, not the network 10%. Equation (4) per building: units due at each CPT vs pack-and-ship hours in the window. Then (2): any building above 85% storage. Ten percent on the network is usually three buildings at 25–35% and the rest fine (7A.6).

"Is it real demand?" Three checks before spending: same cohort weights and promotions as last forecast; mix shift between building types (a 10% rise in non-sort with sortable flat is a different plan); and bias — if this region's 8-week forecast has run 6% high for a year, the 10% is 4%.

"What fires this week?" Placement to buildings with slack, with the transport delta priced (19). OT authorization on the binding step in the binding buildings. Inbound appointment caps on slow movers into those buildings, pulling fast movers forward. All reversible.

"What fires only on a trigger?" Hiring at week –12 if P80 still shows the gap. Overflow contract at week –8 for the residual above what hours and placement close. The promise lever only when the priced levers are exhausted — and I would write the dollar value of a one-day speed downgrade next to it so nobody pulls it casually.

"How do you communicate it?" One page by Friday: gap by building by week by currency, the lever stack with cost per unit closed, triggers with dates, owners, what I will not do, and the number I will report next week to show whether the first levers worked.

Case C — "Utilization looks fine but speed is missing."

Utilization is a weekly average. Speed misses live at the hour, the dock door, the sort type, the cutoff. Change the grain: by building by hour by process step. Check mix: did fast-promise demand land in buildings far from the customer? Check cutoff cliffs: units arriving an hour late lose a calendar day. **Story:** S5 — the reports were wrong in both directions because they measured the wrong grain. **Line to say:** "If I am still talking about network utilization after two follow-ups, I am failing the question."

CASE C — PROBES AND ANSWERS

"Give me the math." 7A.11 #10: a CPT with 180K due and 144K of pack capacity in its window misses 36K units every day while weekly utilization reads 80%. Speed is lost at the CPT window, not the week.

"What else could it be?" Placement — fast-promise demand landing in buildings far from the customer, so the unit ships on time and still arrives late; downstream — sort center or delivery station capacity on that lane; or a dock-to-stow backlog making units unavailable to promise at order time (Little's Law, 7A.11 #4).

"Which one first?" CPT attainment by building by CPT by day (O6) splits it: if CPT attainment is high and speed is low, the problem is placement or downstream; if CPT attainment is low, it is inside the building, and the step with the largest hours gap in the window is the cause.

Case D — "Finance says the new building is not justified."

Do not argue. Price both columns. Cost of the build, annualized. Cost of not having it: overflow contracts, transfers and split shipments, speed downgrades and their conversion cost, labor premiums, the risk-adjusted cost of a peak miss. Show the sensitivity: at what forecast does the build break even? Offer the middle path — retrofit or flow redesign — priced in the same table. **Story:** S7 — the two-column table that flipped an eighteen-month plan; Finance attacked it and it survived. **Line:** "I would rather lose this argument to a better number than win it with a louder voice."

CASE D — PROBES AND ANSWERS

"What's in the no-build column?" Overflow at contract or spot rates for the shortfall weeks; transfers and rebalancing; premium transportation for CPT misses; speed downgrade at conversion cost; labor premium from

running existing buildings past the economic ceiling (7A.11 #6); and a risk-adjusted cost of a peak miss — probability × cost — which I state as an assumption and let Finance change.

"Where's the break-even?" 7A.11 #8: the demand level D^* where annualized build cost equals annual no-build cost, reported as a quantile of the forecast — "the build pays at P68 and above." Then the question is not "is the building justified" but "what quantile are we planning to" — a decision Finance owns with us.

"What if they still say no?" Commit. Then the no-build plan becomes the plan: overflow contracts signed by week -8, the transport and labor premiums budgeted, and the speed exposure written down. I bring the realized no-build cost back to the OP cycle next year as data, not as an argument.

Case E — "Define the metrics for a network capacity program."

Outputs (what the business feels): capacity delivered vs plan by week; utilization vs ceiling by type; cost per unit; customer speed attainment. Inputs (what a named owner can move this week): launch and retrofit milestones on dated gates; placement compliance; forecast accuracy at the planning grain; labor plan attainment; inbound ASN accuracy. Mechanism: a weekly review where reds arrive pre-decomposed — which input moved, whose lever, what correction, verified by the same number next week — and a correction-of-error write-up for any miss above a published threshold that ends in a mechanism change, not "try harder." **Line:** "Inputs without outputs are a hobby. Outputs without inputs are weather."

CASE E — PROBES AND ANSWERS

"Pick the one output you'd be judged on." Capacity delivered vs plan (O1), because it is the promise this team makes to everyone else. Cost per unit (O5) is the constraint on how I deliver it.

"Pick the one input you'd watch hardest." Forecast bias by region (I1), because a biased forecast makes every other metric lie in the same direction, and placement compliance (I2) a close second, because off-plan placement is the silent thief.

"How do you stop the WBR from becoming a status meeting?" Reds arrive pre-decomposed — which input moved, whose lever, what correction, verified by the same number next week. Definitions frozen (S8). A correction of error above a published threshold that ends in a mechanism change. And a line (I10) that shows whether triggered levers actually fired.

"What's the full list?" 7A.9: seven outputs, ten inputs, each with grain, target and owner.

Fermi, if they throw one

- State assumptions. Compute out loud. Label it illustrative. Say what you would replace it with.
- *"How many units does a large sortable FC ship in a week?"* Structure: shifts × hours × headcount × units per hour, cross-checked against outbound dock capacity. Do not quote an Amazon number you have not seen. "I would take the design rate from Ops Engineering on day one."
- *"How much storage for a week of demand?"* Daily outbound × days of cover ÷ units per cubic foot ÷ bin utilization. Say the structure, pick round numbers, flag the two you are least sure of.
- *"Cost of a 4-point speed miss at peak?"* Volume × delta × cost per late unit, plus the unpriced trust cost. Name the grain. Never say "huge."

Lines that read as judgment, not worship

- "I plan to a ceiling, not to design rate. Congestion starts before the building is full."
- "A network average hides the three buildings that are actually breaking."
- "Flow is the cheapest capacity in the building. Steel is the most expensive. I spend them in that order."
- "The forecast is an input I interrogate, not a fact I plan to."
- "Every lever gets a trigger and an owner before peak, or it is a wish."

8. The hard questions

"You have never built an Amazon FC capacity plan. Why you?"

"Correct, and I will not pretend otherwise. An internal transfer brings Amazon scar tissue. I bring the three things a capacity plan is made of, each proven: allocating demand across a network when the plan did not fit — 99% fill through peak; consolidating buildings and running a launch behind gates — plus fifteen percent, zero incidents; and turning measured data into planning parameters — two hundred lanes, a million and a half released. I also bring the review grammar this team already runs — input levers, output metrics, pre-wired reds. Days one to thirty I learn your numbers, your building types, and the five decisions last peak that nobody wants to talk about. I will not arrive trying to rebuild it. I will arrive with the mechanism."

"This role is in Bellevue."

Decide before Friday and say it in one sentence. Do not improvise a relocation answer in the room. [Your line here.]

"Tell me about a failure."

Pick one, own it in the first sentence, name the mechanism it built, stop. Three ready:

- **UAT that crashed under load (Elementum).** Five users, ten scenarios; I sent screenshots instead of digging. Concurrent load crashed it; sign-off missed; sponsor escalation. The implicit requirement — performance at the customer's shape — was not in the entrance gate. Every critical UAT since has numeric load criteria; Flexport's launch proof was 500 containers in 20 minutes at 99.6% connectivity before a customer touched it. "I was the mechanism that failed; I wrote the mechanism that replaced me."
- **RF devices, 1.5-week training slip (Sephora JDA).** In S9. Single-point dependency in front of an exit criterion now gets a what-if plan regardless of size.
- **Security review, two weeks (Flexport Snowflake).** In S8. "Whose approval does this data need, and what is its lead time" is now a kickoff question.

"Tell me about a time you had to influence without authority, two levels up."

S7. Then one sentence: "I set the instrument, not the person — a named line, a numeric target, a published grain, a date. Next Monday's number is the appeal."

"How do you set goals for people who don't report to you?"

"I set the instrument, not the person. A named input line, a numeric target, a published grain, and a date the number gets checked. On UDC the issues log made it unemotional: an owner's row either moved or it did not. I have never won a goal argument with a title; I have won them with a line on the page everyone reads."

"What would you do in your first 90 days?"

"Thirty days: learn the plan as it actually exists — the demand file, the design rates, the ceilings, which lever fired last peak and why. Sixty: take over the weekly review's output block and pre-wire the reds. Ninety: one output metric moved by a mechanism the review can audit — a trigger that fires a lever before the week it is needed, or a placement change that bought capacity without steel. Not a new dashboard."

"Are you an L6 or an L7?"

If asked, do not oversell. "L6 is running the plan and the review well, independently, on complex problems with real trade-offs. That is what I have done. I have also flipped a socialized roadmap with a table, published

tolerances that stopped go-lives, and throttled my own launch. I have not held a large direct-report org, and I will not pretend that is my differentiator."

9. Your questions — pick two

1. "When the capacity plan was wrong last peak, how long did it take to know which input was wrong — the forecast, the design rate, or the placement?" (*Shows you think in decomposition. Whatever they name, you have the matching story.*)
2. "Where does the storage-versus-throughput call actually get made today — in the model, in a planning table, or in a negotiation between S&OP and Ops?" (*The answer is almost always "it's complicated," which is your opening.*)
3. "What does PPT own relative to this team, and where do the two of you most often disagree?"
4. "Which lever is the team least willing to pull — flow, hours, overflow, or the promise — and why?"
5. "What does a good first ninety days look like to you, in one output metric?"

10. Close — fifteen seconds

"I want this work. I have not built an Amazon capacity plan, and I have said so. I have allocated a network through peak, consolidated buildings behind gates, and turned measured data into the parameters a plan runs on. In ninety days I will move one output with a mechanism your review can audit. If that is the job, I am here for it."

11. Numbers worth memorizing

All on your resume or in your war book. If you cannot say what sits under one, leave it out.

NUMBER	WHAT IT IS
3 mega-DCs, ~100 stores, 99% fill, ~\$8M, ~20% stockouts	Manhattan allocation and promising through peak; Cyber Monday 4 AM
200 lanes, p99, 99% LT accuracy, 580K units, ~\$1.5M	Elementum lead-time distributions replacing brochure numbers
70% → 85%, ~\$2M, zero recon drift	Elementum forecast accuracy and safety stock; MD04/MB52 daily recon
40% / 55% / 99%, ~\$8M, 25 locations	Austria: reported vs true data quality; freight savings
~1,000 routes, ≥95% valid ASN, 60% → 95%	EDI onboarding gate; shipment completeness
7 → 68, \$0.66 → \$0.03, 72% → 90% at 5×, 2 months early	Flexport visibility platform; six graduation criteria
~50% vs ~25%, 66%, \$5M, 6 vs 18 months	Roadmap flip on cost-to-serve
4 → 1, 60% → 90%, >\$100K/yr, 1% parallel run, 4% first catch	Snowflake WBR pipeline
8 months, 7% → 1.8% → 0.3% vs 0.5%, ~3% under \$1.8M, \$20M	Dual-DC JDA migration; mocks; pipeline unblocked
15+ tracks, 60+ stakeholders, 836 rows, Nov 22 / Dec 20 / Jan 30	UDC omni DC program; test gates; Oct–Dec freeze

NUMBER	WHAT IT IS
343 tasks; sync < 99% × 4 h or p95 > 2× × 1 h; never fired; MCS off; 0 incidents	S/4HANA ↔ ISM cutover runbook and rollback triggers; legacy inventory system retired
Returns plant; RTV STOs as pick-reserve; receive → grade → restock / scrap / RTV	S/4HANA stock-state model: nothing promiseable until disposition; credit never ahead of the physical unit
94% vs 98%, ~30%, week 3, +15%, zero incidents	Locus robots throttled; processing efficiency
6-week → 4-week in 6 weeks, defects ~50%	Agile conversion; QA owned the gate
~\$2.5M, a couple hundred orders	OMS–WMS discrepancy disclosed, not hidden
~\$200K/month per DC, ~\$1M per DC	Two compressed go-lives on one team
\$25 → \$0.40, 98%, 100% citations	LLM triage agent (side build)
11 partners, 10 countries, 48 of 53 in a month	FF App export MVP, April 2022

Leave "\$27M+ cost savings" off the table unless you can show the addition live.

12. Traps

Do not quote an Amazon internal number. You do not know their design rates, ceilings or peak factors. Label every number illustrative and say what replaces it on day one.

Do not say "we" for a decision. "I published the gate. I made the throttle call. I wrote the tolerance."

Do not stack stories. One question, one story, one number. Offer depth; do not volunteer it.

Do not let Situation eat the clock. Fifteen seconds of context. Seventy percent of the story is what you did and what it measured.

Do not open with SAP governance. This room is network and capacity. S/4HANA (S10) is a strong answer to inventory-flow, returns, cutover, and being-wrong questions, and a diluted answer to "tell me about yourself." Use it when they ask about inventory; do not lead with it when they ask about capacity.

Do not invent a headcount. Matrix leadership, said plainly, is fine at L6. A fake org chart is not.

Do not solve the case before clarifying. Sixty seconds of questions out loud is scored, not tolerated.

"What would you do differently" needs a real answer. You have five: shadow one zone first; scheduler heartbeat; measurement audit in week one; dual-path devices; security lead time on the kickoff checklist.

Decide the Bellevue answer before Friday.

13. The morning-of card

1. Read "tell me about yourself" and "why Amazon" (section 3) out loud once. End on the gap sentence.
2. Tell S1 and S2 to a wall in STAR — problem, goal, metric, S, T, A, R, did it move — in under two minutes each. Then say the supply/demand/capacity box for each.
3. Walk Case A on paper: demand → two currencies (cube, hours) → gap by week by building → lever stack with cost → owners and triggers. Re-read the 7A.6 worked example and the 7A.8 U-curve.
4. Do whiteboard math #1, #2, #3 and #6 (7A.11) on paper. Say the five judgment lines in section 7 once.

5. Pick your two questions. Pick your Bellevue sentence.